article	gonzalez-hoelling_effects_2021
title	Effects of a {Music}-{Based} {Rhythmic} {Auditory} {Stimulation} on {Gait} and {Balance} in {Subacute} {Stroke}
volume	18
copyright	http://creativecommons.org/licenses/by/3.0/
issn	1660-4601
url	https://www.mdpi.com/1660-4601/18/4/2032
doi	10.3390/ijerph18042032
abstract	Gait and balance impairments are common after stroke. This study aimed to evaluate the effect of a music-based rhythmic auditory stimulation (RAS) in cor conventional physiotherapy on gait parameters and walking ability in subacute stroke. This single-blind, historical controlled trial, included 55 patients who within the three weeks prior to enrolment. Patients from 2018 (n = 27) were assigned as the historical control group whereas 2019 patients (n = 28) received times a week. Both groups received 11 h of conventional physiotherapy per week during hospitalization. Primary outcomes were gait and balance parameter Timed Up\&Go test) and walking ability (Functional Ambulation Category scale). Secondary outcomes were trunk control, assistive devices, functional inde Independence Measure, Barthel index), and stroke severity and disability (modified Rankin scale, National Institutes of Health Stroke Scale). Results: No be differences were identified for gait and balance parameters nor for secondary outcomes. Significant between-group differences were observed in the Functio Category: the intervention group (Δmean ± SD; 3.43 ± 1.17) showed greater improvement (p = 0.002) than the control group (Δmean ± SD; 2.48 ± 1.09). Co conventional physiotherapy alone, our results suggest that the walking ability of subacute stroke patients might be improved with music-based RAS combine physiotherapy, but this treatment is not more effective than conventional physiotherapy in obtaining gait and balance gains.
language	en
number	4
urldate	2024-04-03
journal	International Journal of Environmental Research and Public Health
author	Gonzalez-Hoelling, Samira and Bertran-Noguer, Carme and Reig-Garcia, Gloria and Suñer-Soler, Rosa
month	jan, year = {2021
note	Number: 4 Publisher: Multidisciplinary Digital Publishing Institute
keywords	musical therapy, physical therapy, rehabilitation, rhythmical auditory stimulation, stroke
pages	2032
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\QSSKP7HT\Gonzalez-Hoelling et al. - 2021 - Effects of a Music-Based Rhythmic Auditory Stimula.pdf:appli
article	**lee_gait_2018**
title	Gait {Training} with {Bilateral} {Rhythmic} {Auditory} {Stimulation} in {Stroke} {Patients}: A {Randomized} {Controlled} {Trial}
volume	8
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2076-3425

shorttitle	Gait {Training} with {Bilateral} {Rhythmic} {Auditory} {Stimulation} in {Stroke} {Patients}
url	https://www.mdpi.com/2076-3425/8/9/164
doi	10.3390/brainsci8090164
abstract	The aim of this study was to investigate the effect of gait training with bilateral rhythmic auditory stimulation (RAS) on lower extremity rehabilitation in str participants (<6 months after stroke) were randomly allocated to the gait training with bilateral rhythmic auditory stimulation (GTBR) group (n = 23) and th 21). The GTBR group had gait training with bilateral RAS for 30 min a day, 5 days a week, in a 6-week period, in addition to conventional therapy. The con training without RAS, and conventional therapy. Outcome measures included gait symmetry, gait ability, balance ability, and lower extremity function. Gait showed significant improvements compared to baseline ($p < 0.05$) in the GTBR group, but not in the control group. Gait ability was significantly improved i to baseline values ($p < 0.05$), and the GTBR group showed significantly greater improvement in comparison to the control group ($p < 0.05$). Both groups sho improvements in the Timed Up and Go test (TUG), Berg Balance Scale (BBS), and Fugl–Meyer Assessment (FMA) compared to baseline ($p < 0.05$). GTBR therapeutic method of improving symmetric gait in stroke rehabilitation. Moreover, we found that GTBR beat frequency matching fast step time might be ev improving gait symmetry. Future studies may develop a method of applying RAS on step time and length for improvement of gait symmetry in stroke patien
language	en
number	9
urldate	2024-04-03
journal	Brain Sciences
author	Lee, Soonhyun and Lee, Kyeongjin and Song, Changho
month	sep, year = {2018
note	Number: 9 Publisher: Multidisciplinary Digital Publishing Institute
keywords	stroke, gait, rhythmic auditory stimulation, symmetry
pages	164
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\D64W3AFL\Lee et al. - 2018 - Gait Training with Bilateral Rhythmic Auditory Sti.pdf:application/pdf
article	**ahmed_efficacy_2023**
title	Efficacy of rhythmic auditory stimulation on gait parameters in hemiplegic stroke patients: a randomized controlled trial
volume	59
issn	1687-8329
shorttitle	Efficacy of rhythmic auditory stimulation on gait parameters in hemiplegic stroke patients
url	https://doi.org/10.1186/s41983-023-00606-w
doi	10.1186/s41983-023-00606-w
abstract	Gait disorders which caused by stroke are one of the most common causes of functional disabilities in hemiparetic stroke patients that leads to significant im and financial cost. In this study, we focused on how certain gait metrics in stroke patients were affected by rhythmic auditory stimulation and treadmill train
number	1

urldate	2024-04-03
journal	The Egyptian Journal of Neurology, Psychiatry and Neurosurgery
author	Ahmed, Gehan M. and Fahmy, Ebtesam M. and Ibrahim, Mohamed F. and Nassief, Ayman A. and Elshebawy, Haidy and Mahfouz, Marwa M. and Elzanaty
month	jan, year = {2023
keywords	Biodex gait trainer 2TM, Rhythmic auditory stimulation, Stroke
pages	8
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\7L8Y6E85\Ahmed et al. - 2023 - Efficacy of rhythmic auditory stimulation on gait .pdf:application/pdf
article	**suh_effect_2014**
title	Effect of rhythmic auditory stimulation on gait and balance in hemiplegic stroke patients
volume	34
issn	1053-8135
url	https://content.iospress.com/articles/neurorehabilitation/nre1008
doi	10.3233/NRE-131008
abstract	Background: Rhythmic auditory stimulation (RAS) improves gait velocity, cadence, and stride length in hemiplegic stroke patients, yet little is known about RAS on standing balance. Objectives: To examine and compare the effective
language	en
number	1
urldate	2024-04-03
journal	NeuroRehabilitation
author	Suh, Jee Hyun and Han, Soo Jeong and Jeon, Seo Young and Kim, Ho Jeong and Lee, Jeong Eun and Yoon, Tae Sik and Chong, Hyun Ju
month	jan, year = {2014
note	Publisher: IOS Press
pages	193--199
article	**bood_power_2013**
title	The {Power} of {Auditory}-{Motor} {Synchronization} in {Sports}: {Enhancing} {Running} {Performance} by {Coupling} {Cadence} with the {Right}
volume	8
issn	1932-6203
shorttitle	The {Power} of {Auditory}-{Motor} {Synchronization} in {Sports}
url	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0070758
doi	10.1371/journal.pone.0070758

abstract	Acoustic stimuli, like music and metronomes, are often used in sports. Adjusting movement tempo to acoustic stimuli (i.e., auditory-motor synchronization) sports performance. However, music also possesses motivational qualities that may further enhance performance. Our objective was to examine the relative motor synchronization and the motivational impact of acoustic stimuli on running performance. To this end, 19 participants ran to exhaustion on a treadmill without acoustic stimuli, 2) a metronome condition with a sequence of beeps matching participants' cadence (synchronization), and 3) a music condition wit motivational music matched to participants' cadence (synchronization+motivation). Conditions were counterbalanced and measurements were taken on sepa time to exhaustion was significantly longer with acoustic stimuli than without. Unexpectedly, however, time to exhaustion did not differ between metronom music conditions, despite differences in motivational quality. Motivational music slightly reduced perceived exertion of sub-maximal running intensity and l)maximal running intensity. The beat of the stimuli –which was most salient during the metronome condition– helped runners to maintain a consistent pace l the prescribed tempo. Thus, acoustic stimuli may have enhanced running performance because runners worked harder as a result of motivational aspects (mo motivational music) and more efficiently as a result of auditory-motor synchronization (most notable with metronome beeps). These findings imply that runn music with a very prominent and consistent beat matched to the runner's cadence will likely yield optimal effects because it helps to elevate physiological ef exertion, whereas the consistent and correct cadence induced by auditory-motor synchronization helps to optimize running economy.
language	en
number	8
urldate	2024-04-03
journal	PLOS ONE
author	Bood, Robert Jan and Nijssen, Marijn and Kamp, John van der and Roerdink, Melvyn
month	aug, year = {2013
note	Publisher: Public Library of Science
keywords	Acoustics, Exercise, Heart rate, Human performance, Music perception, Psychophysics, Running, Sports
pages	e70758
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\VX2DAHGB\Bood et al. - 2013 - The Power of Auditory-Motor Synchronization in Spo.pdf:application/pdf
article	**ctrl-labs_at_reality_labs_generic_2024**
title	A generic noninvasive neuromotor interface for human-computer interaction
url	http://biorxiv.org/lookup/doi/10.1101/2024.02.23.581779
doi	10.1101/2024.02.23.581779
abstract	Abstract Since the advent of computing, humans have sought computer input technologies that are expressive, intuitive, and universal. While diverse modali developed, including keyboards, mice, and touchscreens, they require interaction with an intermediary device that can be limiting, especially in mobile scena systems utilize cameras or inertial sensors to avoid an intermediary device, but they tend to perform well only for unobscured or overt movements. Brain co (BCIs) have been imagined for decades to solve the interface problem by allowing for input to computers via thought alone. However high-bandwidth comm been demonstrated using invasive BCIs with decoders designed for single individuals, and so cannot scale to the general public. In contrast, neuromotor sign offer access to subtle gestures and force information. Here we describe the development of a noninvasive neuromotor interface that allows for computer inpu electromyography (sEMG). We developed a highly-sensitive and robust hardware platform that is easily donned/doffed to sense myoelectric activity at the v intentional neuromotor commands into computer input. We paired this device with an infrastructure optimized to collect training data from thousands of con which allowed us to develop generic sEMG neural network decoding models that work across many people without the need for per-person calibration. Test the training set demonstrate closed-loop median performance of gesture decoding at 0.5 target acquisitions per second in a continuous navigation task, 0.9 ge

	second in a discrete gesture task, and handwriting at 17.0 adjusted words per minute. We demonstrate that input bandwidth can be further improved up to 30 sEMG decoding models to the individual, anticipating a future in which humans and machines co-adapt to provide seamless translation of human intent. To the first high-bandwidth neuromotor interface that directly leverages biosignals with performant out-of-the-box generalization across people.
language	en
urldate	2024-04-03
author	CTRL-labs at Reality Labs} and Sussillo, David and Kaifosh, Patrick and Reardon, Thomas
month	feb, year = {2024
annote	[TLDR] This work developed a highly-sensitive and robust hardware platform that is easily donned/doffed to sense myoelectric activity at the wrist and tran neuromotor commands into computer input, and is the first high-bandwidth neuromotor interface that directly leverages biosignals with performant out-of-th across people.
file	Version soumise:C\:\Users\paul-\Zotero\storage\YXXCAPH2\CTRL-labs at Reality Labs et al. - 2024 - A generic noninvasive neuromotor interface for hum
phdthesis	**peyre_sonification_2022**
type	These de doctorat
title	Sonification du mouvement pour la rééducation après une lésion cérébrale acquise : conception et évaluations de dispositifs
copyright	Licence Etalab
shorttitle	Sonification du mouvement pour la rééducation après une lésion cérébrale acquise
url	https://theses.fr/2022SORUS414
abstract	Première cause de handicap acquis de l'adulte, les lésions cérébrales acquises non dégénératives induisent de multiples troubles affectant les sphères sensitiv et psycho-sociales. La chronicisation des déficits résiduels est à l'origine d'une perte d'autonomie dans la réalisation des activités de vie quotidienne. L'acco rééducation favorise la récupération fonctionnelle grâce au recours à des méthodes, techniques et outils complémentaires. Ainsi, la poursuite de la rééducatio supervisée lors de la phase chronique est désormais encouragée. Avec le développement des technologies pour la santé, de nouvelles modalités d'accompagn actuellement l'objet d'investigations. L'émergence d'outils interactifs de sonification du mouvement permettant de disposer en temps réel d'informations so avec les mouvements réalisés constituent une voie prometteuse pour la rééducation. Cependant, des questionnements relatifs à l'orientation des choix de con particulièrement concernant les caractéristiques des retours sonores et les modalités d'interactions gestes-sons, sont actuellement au cœur des réflexions. Fac l'objectif principal de cette thèse interdisciplinaire, santé-arts-sciences, était de développer un dispositif de sonification du mouvement pour la rééducation e de patients présentant des séquelles motrices après une lésion cérébrale acquise. Dans cette perspective, le premier objectif était d'évaluer l'effet de différent sonores (caractéristiques sonores et modalités d'interactions gestes-sons) sur deux tâches gestuelles : un mouvement d'antépulsion (extension du coude), et u avec des participants de différents profils. Le deuxième objectif était de définir les critères de conception et de sélectionner les solutions adaptées pour la cré sonification du mouvement répondant aux caractéristiques et besoins de patients présentant des séquelles motrices au membre supérieur suite à une lésion cé perspective d'un usage en autonomie supervisée. Le troisième objectif était d'initier une évaluation du dispositif conçu, afin de disposer d'éléments formels une étude clinique. Les travaux menés ont notamment permis de confirmer l'effet de la présence de retours sonores interactifs lors de l'exécution de gestes e prise en considération des modalités d'interaction gestes-sons. Le processus de co-conception centré utilisateurs mis en œuvre avec des experts de plusieurs la création d'un dispositif de sonification du mouvement mobile novateur, fonctionnel, flexible (personnalisable), adapté à une situation de rééducation en au Peu onéreux, le dispositif a été dupliqué en 10 exemplaires. Les premiers résultats des évaluations réalisées auprès de thérapeutes sont très encourageants. C ainsi des perspectives d'évaluation clinique à grande échelle.
urldate	2024-04-03

school	Sorbonne université
author	Peyre, Iseline
collaborator	Bevilacqua, Frédéric and Pradat-Diehl, Pascale
month	dec, year = {2022
keywords	616.8, Acquired brain injuries, Autonomie, Autonomy, Lésion cérébrale acquise, Movement sonification, Musicothérapie, Personnes atteintes de lésions cér Rééducation, Rééducation motrice, Rehabilitation, Sonification, Sonification du mouvement
mastersthesis	**montazerin_deep_2023**
title	Deep {Learning} {Methods} for {Hand} {Gesture} {Recognition} via {High}-{Density} {Surface} {Electromyogram} ({HD}-{sEMG}) {Signals}
copyright	term_access
url	https://spectrum.library.concordia.ca/id/eprint/993031/
abstract	Hand Gesture Recognition (HGR) using surface Electromyogram (sEMG) signals can be considered as one of the most important technologies in making ef Interface (HMI) systems. In particular, sEMG-based hand gesture has been a topic of growing interest for development of assistive systems to improve the q individuals suffering from amputated limbs. Generally speaking, myoelectric prosthetic devices work by classifying existing patterns of the collected sEMG synthesizing intended gestures. While conventional myoelectric control systems, e.g., on/off control or direct-proportional, have potential advantages, challe Degree of Freedom (DoF) due to crosstalk have resulted in the emergence of data-driven solutions. More specifically, to improve efficiency, intuitiveness, a performance of hand prosthetic systems, several Artificial Intelligence (AI) algorithms ranging from conventional Machine Learning (ML) models to highly Neural Network (DNN) architectures have been designed for sEMG-based hand gesture recognition in myoelectric prosthetic devices. In this thesis, we, firs review on hand gesture recognition methods and elaborate on the recently proposed Deep Learning/Machine Learning (DL/ML) models in the literature. Th Density sEMG (HD-sEMG) dataset is introduced and the rationales behind our main focus on this particular type of sEMG dataset are explained. We, then, Transformer (ViT)-based model for gesture recognition with HD-sEMG signals and evaluate its performance under different conditions such as variable wir electrode channels, and model's complexity. We compare its performance with that of two conventional ML and one DL algorithm that are typically adopte Furthermore, we introduce another capability of our proposed framework for instantaneous training, which is its ability to classify hand gestures based on a sEMG dataset. Following that, we introduce the idea of integrating the macroscopic and microscopic neural drive information obtained from HD-sEMG dat based framework for gesture recognition, which outperforms a standalone ViT architecture in terms of classification accuracy. Here, microscopic neural driv called Motor Unit Spike Trains) refers to the neural commands sent by the brain and spinal cord to individual muscle fibers and are extracted from HD-sEM Source Separation (BSP) algorithms. Finally, we design an alternative and novel hand gesture recognition model based on the less-explored topic of Spiking (SNN), which performs spatio-temporal gesture recognition in an event-based fashion. As opposed to the classical DNN architectures, SNNs are of the capa brain's cognitive function by using biologically inspired models of neurons and synapses. Therefore, they are more biologically explainable and computatio
language	en
urldate	2024-04-03
school	Concordia University
author	Montazerin, Mansoorehsadat
month	aug, year = {2023
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\EYVJPP6V\Montazerin - 2023 - Deep Learning Methods for Hand Gesture Recognition.pdf:application/pdf;S \Zotero\storage\6ZXWYY6E\993031.html:text/html

article	visconti_technical_2018
title	Technical {Features} and {Functionalities} of {Myo} {Armband}: {An} {Overview} on {Related} {Literature} and {Advanced} {Applications} of {Myoe {Mainly} {Focused} on {Arm} {Prostheses}
volume	11
shorttitle	Technical {Features} and {Functionalities} of {Myo} {Armband}
doi	10.21307/ijssis-2018-005
abstract	Technological advances in manufacturing smart high-performances electronic devices, increasingly available at lower costs, nowadays allow to improve use many application fields. In this work, the human-machine interaction obtained by using a next generation device (Myo armband) is analyzed and discussed, to healthcare applications such as upper-limb prostheses. An overview on application fields of the Myo armband and on the latest research works related to i applications is presented; subsequently, the technical features and functionalities of this device are examined. Myo armband is a wearable device provided w myographic electrodes, a 9-axes Inertial Measurement Unit and a transmission module. It sends the data related to the detected signals, via Bluetooth Low E other electronic devices which process them and act accordingly, depending on how they are programmed (in order to drive actuators or perform other speci to the prosthetic field, Myo armband allows to overcome many issues related to the existing prostheses, representing a complete electronic platform that dete main signals related to forearm activity (muscles activation and forearm movements in the three-dimensional space) and sends these data to the connected de several typologies of prostheses are available on the market; they can be mainly distinguished into low-cost prostheses, which are light and compact but allo of movements, and high-end prostheses, which are much more complex and featured by high dexterity, but also heavy, bulky, difficult to control and very ex Myo armband is an optimum candidate for prosthetic application (and many others) and offers an excellent low-cost solution for obtaining a reliable, easy to
journal	International Journal on Smart Sensing and Intelligent Systems
author	Visconti, Paolo and Gaetani, Federico and Zappatore, Giovanni and Primiceri, Patrizio
month	jun, year = {2018
pages	1--25
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\H7DAVXE7\Visconti et al. - 2018 - Technical Features and Functionalities of Myo Armb.pdf:application/pdf
article	**montazerin_transformer-based_2023**
title	Transformer-based hand gesture recognition from instantaneous to fused neural decomposition of high-density {EMG} signals
volume	13
copyright	2023 The Author(s)
issn	2045-2322
url	https://www.nature.com/articles/s41598-023-36490-w
doi	10.1038/s41598-023-36490-w
abstract	Designing efficient and labor-saving prosthetic hands requires powerful hand gesture recognition algorithms that can achieve high accuracy with limited cor this context, the paper proposes a Compact Transformer-based Hand Gesture Recognition framework referred to as \$\$\text {CT-HGR}\$\$, which employs network to conduct hand gesture recognition using high-density surface EMG (HD-sEMG) signals. Taking advantage of the attention mechanism, which is i transformer architectures, our proposed \$\$\text {CT-HGR}\$\$framework overcomes major constraints associated with most of the existing deep learning complexity; requiring feature engineering; inability to consider both temporal and spatial information of HD-sEMG signals, and requiring a large number of

	attention mechanism in the proposed model identifies similarities among different data segments with a greater capacity for parallel computations and addre limitation problems while dealing with inputs of large sequence lengths. \$\$\text {CT-HGR}\$\$can be trained from scratch without any need for transfer le simultaneously extract both temporal and spatial features of HD-sEMG data. Additionally, the \$\$\text {CT-HGR}\$\$framework can perform instantaneou sEMG image spatially composed from HD-sEMG signals. A variant of the \$\$\text {CT-HGR}\$\$is also designed to incorporate microscopic neural drive of Motor Unit Spike Trains (MUSTs) extracted from HD-sEMG signals using Blind Source Separation (BSS). This variant is combined with its baseline ver architecture to evaluate potentials of fusing macroscopic and microscopic neural drive information. The utilized HD-sEMG dataset involves 128 electrodes t related to 65 isometric hand gestures of 20 subjects. The proposed \$\$\text {CT-HGR}\$\$framework is applied to 31.25, 62.5, 125, 250 ms window sizes o dataset utilizing 32, 64, 128 electrode channels. Our results are obtained via 5-fold cross-validation by first applying the proposed framework on the dataset separately and then, averaging the accuracies among all the subjects. The average accuracy over all the participants using 32 electrodes and a window size o which gradually increases till reaching 91.98\% for 128 electrodes and a window size of 250 ms. The \$\$\text {CT-HGR}\$\$achieves accuracy of 89.13\% recognition based on a single frame of HD-sEMG image. The proposed model is statistically compared with a 3D Convolutional Neural Network (CNN) and of Support Vector Machine (SVM) and Linear Discriminant Analysis (LDA) models. The accuracy results for each of the above-mentioned models are paire recall, F1 score, required memory, and train/test times. The results corroborate effectiveness of the proposed \$\$\text {CT-HGR}\$\$framework compared t
language	en
number	1
urldate	2024-04-03
journal	Scientific Reports
author	Montazerin, Mansooreh and Rahimian, Elahe and Naderkhani, Farnoosh and Atashzar, S. Farokh and Yanushkevich, Svetlana and Mohammadi, Arash
month	jul, year = {2023
note	Publisher: Nature Publishing Group
keywords	Computational models, Data processing
pages	11000
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\VIXYHYL7\Montazerin et al. - 2023 - Transformer-based hand gesture recognition from in.pdf:application/p
article	**malesevic_database_2021**
title	A database of high-density surface electromyogram signals comprising 65 isometric hand gestures
volume	8
copyright	2021 The Author(s)
issn	2052-4463
url	https://www.nature.com/articles/s41597-021-00843-9
doi	10.1038/s41597-021-00843-9

abstract	Control of contemporary, multi-joint prosthetic hands is commonly realized by using electromyographic signals from the muscles remaining after amputatio Although this principle is trying to imitate the natural control structure where muscles control the joints of the hand, in practice, myoelectric control provides functions to an amputee using a dexterous prosthesis. This study aims to provide an annotated database of high-density surface electromyographic signals to designing robust and versatile electromyographic control interfaces for prosthetic hands. The electromyographic signals were recorded using 128 channels w grids positioned on the forearms of 20 able-bodied volunteers. The participants performed 65 different hand gestures in an isometric manner. The hand move timed using an automated recording protocol which also synchronously recorded the electromyographic signals and hand joint forces. To assess the quality o several quantitative assessments were performed, such as frequency content analysis, channel crosstalk, and the detection of poor skin-electrode contacts.
language	en
number	1
urldate	2024-04-03
journal	Scientific Data
author	Malešević, Nebojša and Olsson, Alexander and Sager, Paulina and Andersson, Elin and Cipriani, Christian and Controzzi, Marco and Björkman, Anders and
month	feb, year = {2021
note	Publisher: Nature Publishing Group
keywords	Biomedical engineering, Electromyography – EMG, Neurology
pages	63
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\S3I4W9PG\Malešević et al. - 2021 - A database of high-density surface electromyogram .pdf:application/pdf
article	**liu_practical_2023**
title	A {Practical} {System} for 3-D {Hand} {Pose} {Tracking} {Using} {EMG} {Wearables} {With} {Applications} to {Prosthetics} and {User} {Interfaces}
volume	10
issn	2327-4662
url	https://ieeexplore.ieee.org/abstract/document/9956792
doi	10.1109/JIOT.2022.3223600
abstract	Ubiquitous finger motion tracking enables a number of exciting applications in augmented reality, sports analytics, rehabilitation-healthcare, haptics, etc. Th NeuroPose, a system that shows the feasibility of 3-D finger motion tracking using a platform of wearable electromyography (EMG) sensors. EMG sensors potential from muscles due to finger activation, thus offering rich information for fine-grained finger motion sensing. However, converting the sensor inform poses is non trivial since signals from multiple fingers superimpose at the sensor in complex patterns. Toward solving this problem, NeuroPose fuses inform constraints of finger motion with machine learning architectures on recurrent neural networks (RNNs), encoder–decoder networks, and ResNets to extract 3- noisy EMG data. The generated motion pattern is temporally smooth as well as anatomically consistent. Furthermore, a transfer learning algorithm is leverag pretrained model on one user to a new user with minimal training overhead. A systematic study with 12 users demonstrates a median error of 6.24° and a 90 tracking 3-D finger joint angles. The accuracy is robust to natural variation in sensor mounting positions as well as changes in wrist positions of the user. In validates the feasibility of mirrored bilateral training approach with applications in prosthetic devices. Finally, NeuroPose is comprehensively evaluated on b smartphones with a processing latency of 0.019 s and low energy overhead.
number	4

urldate	2024-04-03
journal	IEEE Internet of Things Journal
author	Liu, Yilin and Zhang, Shijia and Gowda, Mahanth
month	feb, year = {2023
note	Conference Name: IEEE Internet of Things Journal
keywords	Accessibility, Cameras, Electromyography, electromyography (EMG), human centered computing, Piezoelectric transducers, representation learning, Senso displays, Tracking, Wrist
pages	3407--3427
file	IEEE Xplore Abstract Record:C\:\Users\paul-\Zotero\storage\C2F8IPXK\9956792.html:text/html
article	**tian_rhythmic_2020**
title	Rhythmic {Auditory} {Stimulation} as an {Adjuvant} {Therapy} {Improved} {Post}-stroke {Motor} {Functions} of the {Upper} {Extremity}: A {Randon {Pilot} {Study}
volume	14
issn	1662-453X
shorttitle	Rhythmic {Auditory} {Stimulation} as an {Adjuvant} {Therapy} {Improved} {Post}-stroke {Motor} {Functions} of the {Upper} {Extremity}
url	https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2020.00649/full
doi	10.3389/fnins.2020.00649
abstract	To explore whether rhythmic auditory stimulation (RAS) could improve motor functions of post-stroke hemiparetic upper extremity. A prospective, randomized controlled, assessor-blinded pilot study. Thirty stroke patients were randomly distributed into the RAS group (n = 15) and the control group (n = 15). Both groups received regular therapies. The RA additional 30 min of RAS training, while the control group received additional 30 min of regular therapies for 5 days per week for 4 weeks. The Fugl-Meyer Extremity (FMA-UE), Wolf Motor Function Test (WMFT), and Barthel Index (BI) were used. The co-activation interval and co-contraction index were calc electromyography (sEMG) recordings on the affected biceps and triceps during elbow flexion and extension. Assessments were performed before and after t Significant improvements in motor functions were observed within both groups (p \< 0.05 in the FMA-UE, WMFT, and BI, respectively), as well as betwee treatments (higher scores in the RAS group, all p \< 0.05 except for p = 0.052 in the FMA-UE; group × time interaction, all p \< 0.05). Statistical significanc activation interval between groups after the treatments (lower in the RAS group; p = 0.022 during elbow extension; p = 0.001 during elbow flexion; group × 0.05 only during elbow extension). No statistical significance was found in the co-contraction index between groups; an inversed pattern of changes was obs supported by relatively higher increments in the triceps recruitments to the biceps. Using RAS in task-oriented exercises was effective in moderating co-contraction, facilitating task-oriented movements of the hemiparetic upper extremity, a among those who had emerging isolated joint movements. The effects were evident on sEMG possibly by adjusting the balance of recruitments between the antagonist.

	The study was registered at the Chinese Clinical Trial Registry (No. 1900026665).
language	English
urldate	2024-04-03
journal	Frontiers in Neuroscience
author	Tian, Rujin and Zhang, Bei and Zhu, Yulian
month	jun, year = {2020
note	Publisher: Frontiers
keywords	rhythmic auditory stimulation, Stroke, Motor function, surface EMG, Upper limb
file	Texte intégral:C\:\Users\paul-\Zotero\storage\ZPGDXM96\Tian et al. - 2020 - Rhythmic Auditory Stimulation as an Adjuvant Thera.pdf:application/pdf
article	**braun_janzen_rhythm_2021**
title	Rhythm and {Music}-{Based} {Interventions} in {Motor} {Rehabilitation}: {Current} {Evidence} and {Future} {Perspectives}
volume	15
issn	1662-5161
shorttitle	Rhythm and {Music}-{Based} {Interventions} in {Motor} {Rehabilitation}
doi	10.3389/fnhum.2021.789467
abstract	Research in basic and clinical neuroscience of music conducted over the past decades has begun to uncover music's high potential as a tool for rehabilitation understanding of how music engages parallel brain networks underpinning sensory and motor processes, arousal, reward, and affective regulation, have laid foundation for the development of theory-driven music interventions that have been systematically tested in clinical settings. Of particular significance in the rehabilitation is the notion that musical rhythms can entrain movement patterns in patients with movement-related disorders, serving as a continuous time re regulate movement timing and pace. To date, a significant number of clinical and experimental studies have tested the application of rhythm- and music-bas improve motor functions following central nervous injury and/or degeneration. The goal of this review is to appraise the current state of knowledge on the ef and rhythm to modulate movement spatiotemporal patterns and restore motor function. By organizing and providing a critical appraisal of a large body of re provide a revised framework for future research on the effectiveness of rhythm- and music-based interventions to restore and (re)train motor function.
language	eng
journal	Frontiers in Human Neuroscience
author	Braun Janzen, Thenille and Koshimori, Yuko and Richard, Nicole M. and Thaut, Michael H.
year	2021
pmid	35111007
pmcid	PMC8801707
keywords	rehabilitation, gait, auditory-motor entrainment, movement, music-based interventions, Neurologic Music Therapy (NMT), Rhythmic Auditory Stimulation,

pages	789467
file	Texte intégral:C\:\Users\paul-\Zotero\storage\CG9TFS99\Braun Janzen et al. - 2021 - Rhythm and Music-Based Interventions in Motor Reha.pdf:applicatio
article	**baniqued_braincomputer_2021**
title	Brain–computer interface robotics for hand rehabilitation after stroke: a systematic review
volume	18
issn	1743-0003
shorttitle	Brain–computer interface robotics for hand rehabilitation after stroke
url	https://doi.org/10.1186/s12984-021-00820-8
doi	10.1186/s12984-021-00820-8
abstract	Hand rehabilitation is core to helping stroke survivors regain activities of daily living. Recent studies have suggested that the use of electroencephalography- interfaces (BCI) can promote this process. Here, we report the first systematic examination of the literature on the use of BCI-robot systems for the rehabilita skills associated with hand movement and profile these systems from a technical and clinical perspective.
number	1
urldate	2024-04-03
journal	Journal of NeuroEngineering and Rehabilitation
author	Baniqued, Paul Dominick E. and Stanyer, Emily C. and Awais, Muhammad and Alazmani, Ali and Jackson, Andrew E. and Mon-Williams, Mark A. and M Raymond J.
month	jan, year = {2021
keywords	Stroke, Rehabilitation, Brain–computer interface, EEG, Motor imagery, Robotics
pages	15
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\ZHMMYY2J\Baniqued et al. - 2021 - Brain–computer interface robotics for hand rehabil.pdf:application /pdf;Snapshot:C\:\Users\paul-\Zotero\storage\T3ANCFKP\s12984-021-00820-8.html:text/html
article	**wang_effectiveness_2019**
title	Effectiveness of interventions to improve hand motor function in individuals with moderate to severe stroke: a systematic review protocol
volume	9
copyright	© Author(s) (or their employer(s)) 2019. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ.. This is distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adap non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any change the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.
issn	2044-6055, 2044-6055
shorttitle	Effectiveness of interventions to improve hand motor function in individuals with moderate to severe stroke

url	https://bmjopen.bmj.com/content/9/9/e032413
doi	10.1136/bmjopen-2019-032413
abstract	Introduction The human hand is extremely involved in our daily lives. However, the rehabilitation of hand function after stroke can be rather difficult due to structure and function, as well as neural basis that supports hand function. Specifically, in individuals with moderate to severe impairment following a stroke effective treatments that recover hand function in this population is limited, and thus has never been reviewed. With the progress of rehabilitation science an results from more and more clinical trials are now available, thereby justifying conducting a systematic review. Methods and analysis This systematic review with the methodology recommended by the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols and the Cochrane handbook for s interventions. Electronic searches will be carried out in the PubMed, CINAHL, Physiotherapy Evidence Database and Cochrane Library databases, along wi the reference lists from included studies and published systematic reviews. The date range parameters used in searching all databases is between January 199 Randomised controlled trials (RCTs) published in English, with the primary outcome focusing on hand motor function, will be included. Two reviewers will titles, abstracts and full texts, perform the evaluation of the risk bias and extract all data independently. The risk of bias of the included RCTs will be evaluat Collaboration's tool. A qualitative synthesis will be provided in text and table, to summarise the main results of the selected publications. A meta-analysis w there is sufficient homogeneity across outcomes. The quality of the included publications will be evaluated by the Grading of Recommendations Assessmen Evaluation system from the Cochrane Handbook for Systematic Reviews of Interventions. Ethics and dissemination No ethical approval is needed, and the r will be disseminated via peer-reviewed publications and conference presentations. Trial registration number CRD42019128285.
language	en
number	9
urldate	2024-04-03
journal	BMJ Open
author	Wang, Hewei and Arceo, Ray and Chen, Shugeng and Ding, Li and Jia, Jie and Yao, Jun
month	sep, year = {2019
pmid	31562163
note	Publisher: British Medical Journal Publishing Group Section: Rehabilitation medicine
keywords	rehabilitation, stroke, hand, motor function, review protocol
pages	e032413
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\YSEEWFA7\Wang et al. - 2019 - Effectiveness of interventions to improve hand mot.pdf:application/pdf
article	**huang_effectiveness_2021**
title	The {Effectiveness} of {Music} {Therapy} on {Hand} {Function} in {Patients} {With} {Stroke}: A {Systematic} {Review} of {Randomized} {Controlled
volume	12
issn	1664-2295
shorttitle	The {Effectiveness} of {Music} {Therapy} on {Hand} {Function} in {Patients} {With} {Stroke}
url	https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2021.641023/full
doi	10.3389/fneur.2021.641023

abstract	Objective: This study aims to evaluate the efficacy of music-supported therapy for stroke patients' hand function. Methods: The databases used included Cumulative Index to Nursing and Allied Health Literature (CINAHL), MEDLINE, PubMed, Embase, Music Index, a Studies published between January 2010 and August 2020 were included. The searching key terms included “music-supported therapy,” “music therapy,” “h dysfunction,” “stroke,” “ischemic,” and “hemorrhagic.” Randomized controlled trials or controlled trials involving adults who have hand function problems included in this study. The methodological quality and risk of bias of the included studies were rated by two independent assessors under the guidance of Co risk of bias tool. Results: Twelve studies that met the inclusion criteria were included in this study. Totally, the data included 598 stroke patients (345 male, 253 female) with months to 3 years, and the mean age of the participants were 61.09 years old. Based on the Cochrane risk of bias tool, study quality ranged from three to sev Compared with the control group, outcomes including hand strength, range of joint motion, dexterity of hands, arm function, and quality of life were signific music-supported therapy. Five studies reported improved dexterity of hands, and one study reported the improvement of range of motion and strength of pat supported the therapy has positive effects on patients' hand function and improving their quality of life after the therapy. The therapy ranged over a period of average duration of 30 min/session and an average of three times per week. Conclusion: Based on the results, music-supported therapy could be a useful treatment for improving hand function and activities of daily living in patients w for patients within 6 months after stroke. However, the low certainty of evidence downgrades our confidence to practice in hospital. More and more random and larger sample sizes are required for a deeper review.
language	English
urldate	2024-04-03
journal	Frontiers in Neurology
author	Huang, Wen-Hao and Dou, Zu-Lin and Jin, Hui-Min and Cui, Ying and Li, Xin and Zeng, Qing
month	may, year = {2021
note	Publisher: Frontiers
keywords	hand function, Music supported therapy, RCT - randomized controlled trial, Stroke - Diagnosis, Systematic review
file	Texte intégral:C\:\Users\paul-\Zotero\storage\R5A66UM4\Huang et al. - 2021 - The Effectiveness of Music Therapy on Hand Functio.pdf:application/pdf
article	**grau-sanchez_enriched_2021**
title	Enriched {Music}-supported {Therapy} for chronic stroke patients: a study protocol of a randomised controlled trial
volume	21
issn	1471-2377
shorttitle	Enriched {Music}-supported {Therapy} for chronic stroke patients
doi	10.1186/s12883-020-02019-1

abstract	BACKGROUND: Residual motor deficits of the upper limb in patients with chronic stroke are common and have a negative impact on autonomy, participat Music-Supported Therapy (MST) is an effective intervention to enhance motor and cognitive function, emotional well-being and quality of life in chronic st adapted the original MST training protocol to a home-based intervention, which incorporates increased training intensity and variability, group sessions, and learning to promote autonomy and motivation. METHODS: A randomised controlled trial will be conducted to test the effectiveness of this enriched MST (improving motor functions, cognition, emotional well-being and quality of life of chronic stroke patients when compared to a program of home-based exerci Repetitive Arm Supplementary Program (GRASP). Sixty stroke patients will be recruited and randomly allocated to an eMST group (n = 30) or a control GR group (n = 30). Patients will be evaluated before and after a 10-week intervention, as well as at 3-month follow-up. The primary outcome of the study is the paretic upper limb measured with the Action Research Arm Test. Secondary outcomes include other motor and cognitive functions, emotional well-being an measures as well as self-regulation and self-efficacy outcomes. DISCUSSION: We hypothesize that patients treated with eMST will show larger improveme cognitive functions, emotional well-being and quality of life than patients treated with a home-based GRASP intervention. TRIAL REGISTRATION: The tr at ClinicalTrials.gov and identified as NCT04507542 on 8 August 2020.
language	eng
number	1
journal	BMC neurology
author	Grau-Sánchez, Jennifer and Segura, Emma and Sanchez-Pinsach, David and Raghavan, Preeti and Münte, Thomas F. and Palumbo, Anna Marie and Turry, and Särkämö, Teppo and Cerquides, Jesus and Arcos, Josep Lluis and Rodríguez-Fornells, Antoni
month	jan, year = {2021
pmid	33435919
pmcid	PMC7801568
keywords	Stroke, Rehabilitation, Female, Humans, Male, Middle Aged, Music therapy, Music Therapy, Music-supported therapy, Randomized Controlled Trials as To Function, Stroke Rehabilitation, Upper Extremity
pages	19
file	Texte intégral:C\:\Users\paul-\Zotero\storage\LLQKNH6S\Grau-Sánchez et al. - 2021 - Enriched Music-supported Therapy for chronic strok.pdf:application
article	**amengual_sensorimotor_2013**
title	Sensorimotor plasticity after music-supported therapy in chronic stroke patients revealed by transcranial magnetic stimulation
volume	8
issn	1932-6203
doi	10.1371/journal.pone.0061883
abstract	BACKGROUND: Several recently developed therapies targeting motor disabilities in stroke sufferers have shown to be more effective than standard neuror approaches. In this context, several basic studies demonstrated that music training produces rapid neuroplastic changes in motor-related brain areas. Music-s been recently developed as a new motor rehabilitation intervention. METHODS AND RESULTS: In order to explore the plasticity effects of music-supporte therapeutic intervention was applied to twenty chronic stroke patients. Before and after the music-supported therapy, transcranial magnetic stimulation was a assessment of excitability changes in the motor cortex and a 3D movement analyzer was used for the assessment of motor performance parameters such as v and smoothness in a set of diadochokinetic movement tasks. Our results suggest that the music-supported therapy produces changes in cortical plasticity lead of the subjects' motor performance. CONCLUSION: Our findings represent the first evidence of the neurophysiological changes induced by this therapy in c

	and their link with the amelioration of motor performance. Further studies are needed to confirm our observations.
language	eng
number	4
journal	PloS One
author	Amengual, Julià L. and Rojo, Nuria and Veciana de Las Heras, Misericordia and Marco-Pallarés, Josep and Grau-Sánchez, Jennifer and Schneider, Sabine a Juncadella, Montserrat and Montero, Jordi and Mohammadi, Bahram and Rubio, Francisco and Rueda, Nohora and Duarte, Esther and Grau, Carles and Alt Münte, Thomas F. and Rodríguez-Fornells, Antoni
year	2013
pmid	23613966
pmcid	PMC3629163
keywords	Stroke, Humans, Middle Aged, Music Therapy, Adult, Aged, Analysis of Variance, Biomechanical Phenomena, Case-Control Studies, Chronic Disease, Der Hand, Motor Activity, Motor Cortex, Movement, Neuronal Plasticity, Transcranial Magnetic Stimulation
pages	e61883
file	Texte intégral:C\:\Users\paul-\Zotero\storage\5GWNUZRC\Amengual et al. - 2013 - Sensorimotor plasticity after music-supported ther.pdf:application/pdf
article	**street_home-based_2018**
title	Home-based neurologic music therapy for arm hemiparesis following stroke: results from a pilot, feasibility randomized controlled trial
volume	32
issn	0269-2155
shorttitle	Home-based neurologic music therapy for arm hemiparesis following stroke
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5751852/
doi	10.1177/0269215517717060
abstract	Objective: To assess the feasibility of a randomized controlled trial to evaluate music therapy as a home-based intervention for arm hemiparesis in stroke. D randomized controlled trial, with cross-over design. Randomization by statistician using computer-generated, random numbers concealed in opaque envelop Participants' homes across Cambridgeshire, UK. Subjects: Eleven people with stroke and arm hemiparesis, 3–60 months post stroke, following discharge fro rehabilitation. Interventions: Each participant engaged in therapeutic instrumental music performance in 12 individual clinical contacts, twice weekly for six Feasibility was estimated by recruitment from three community stroke teams over a 12-month period, attrition rates, completion of treatment and successful Structured interviews were conducted pre and post intervention to establish participant tolerance and preference. Action Research Arm Test and Nine-hole P collected at weeks 1, 6, 9, 15 and 18, pre and post intervention by a blinded assessor. Results: A total of 11 of 14 invited participants were recruited (interver n = 5). In total, 10 completed treatment and data collection. Conclusion: It cannot be concluded whether a larger trial would be feasible due to unavailable da of eligible patients screened. Adherence to treatment, retention and interview responses might suggest that the intervention was motivating for participants. T ClinicalTrials.gov identifier NCT 02310438.
number	1
urldate	2024-04-03

journal	Clinical Rehabilitation
author	Street, Alexander J and Magee, Wendy L and Bateman, Andrew and Parker, Michael and Odell-Miller, Helen and Fachner, Jorg
month	jan, year = {2018
pmid	28643570
pmcid	PMC5751852
pages	18--28
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\2HACVVYT\Street et al. - 2018 - Home-based neurologic music therapy for arm hemipa.pdf
article	**grau-sanchez_music-supported_2018**
series	12th {World} {Congress} of the {International} {Society} of {Physical} and {Rehabilitation} {Medicine}. {Paris}. 8-12 {July} 2018
title	Music-supported therapy in the rehabilitation of subacute stroke patients: A randomized controlled trial
volume	61
issn	1877-0657
shorttitle	Music-supported therapy in the rehabilitation of subacute stroke patients
url	https://www.sciencedirect.com/science/article/pii/S1877065718305128
doi	10.1016/j.rehab.2018.05.438
abstract	Introduction/Background Music-supported therapy (MST) has been developed as a tool in neurorehabilitation to restore hemiparesis of the upper extremity a the ability of music training to promote brain plasticity. However, the effects of MST have not been appropriately contrasted with conventional therapy. The test the effectiveness of adding MST to a standard rehabilitation program in subacute stroke patients. Material and method A two-arm, parallel-group contro where 40 patients in the first 6 months from the stroke onset were randomly allocated to conventional treatment (CT-group, n=20) or MST treatment (MST- addition to the rehabilitation program. Before and after 4 weeks of treatment, patients were evaluated on four domains: motor functions, cognitive functions, life, and brain plasticity assessed with functional Magnetic Resonance Imaging (fMRI). A follow-up at 3 months was conducted to examine the evolution of Results Both groups significantly improved their motor function, and no differences between groups were found. The only difference between groups was ol domain for quality of life. Importantly, an association was encountered between the capacity to experience pleasure from music activities and the motor imp group. A pattern of intrahemispheric reorganization in motor-related areas was observed in both groups, being more prominent in the MST-group. Conclusio treatment showed no superiority to conventional therapies for motor recovery with a similar pattern of plastic changes. Importantly, patient's intrinsic motiva musical activities was associated to better motor improvement. Trial registration NCT02208219.
urldate	2024-04-03
journal	Annals of Physical and Rehabilitation Medicine
author	Grau-Sánchez, J. and Duarte, E. and Ramos-Escobar, N. and Sierpowska, J. and Rueda, N. and Redon, S. and Veciana de las Heras, M. and Pedro, J. and Sä Rodriguez-Fornells, A.
month	jul, year = {2018
keywords	Stroke, Music therapy, Randomized controlled trial
pages	e191

file	ScienceDirect Snapshot:C\:\Users\paul-\Zotero\storage\X3HEUUSH\S1877065718305128.html:text/html
article	**dery_playing_2023**
title	Playing with your ears: {Audio}-motor skill learning is sensitive to the lateral relationship between trained hand and ear
volume	26
issn	2589-0042
shorttitle	Playing with your ears
doi	10.1016/j.isci.2023.107720
abstract	A salient feature of motor and sensory circuits in the brain is their contralateral hemispheric bias-a feature that might play a role in integration and learning c In the current behavioral study, we examined whether the lateral configuration between sound-producing hand and feedback-receiving ear affects performan audio-motor skill. Right-handed participants (n = 117) trained to play a piano sequence using their right or left hand while auditory feedback was presented r right or left ear. Participants receiving auditory feedback to the contralateral ear during training performed better than participants receiving ipsilateral feedb training hand). Furthermore, in the Left-Hand training groups, the contralateral training advantage persisted in a generalization task. Our results demonstrate learning is sensitive to the lateral configuration between motor and sensory circuits and suggest that integration of neural activity across hemispheres facilita
language	eng
number	10
journal	iScience
author	Dery, Hadar and Buaron, Batel and Mazinter, Roni and Lavi, Shalev and Mukamel, Roy
month	oct, year = {2023
pmid	37674982
pmcid	PMC10477063
keywords	Behavioral neuroscience, Cognitive neuroscience, Sensory neuroscience
pages	107720
file	Texte intégral:C\:\Users\paul-\Zotero\storage\VJGCYI8S\Dery et al. - 2023 - Playing with your ears Audio-motor skill learning.pdf:application/pdf
article	**willett_high-performance_2023**
title	A high-performance speech neuroprosthesis
volume	620
copyright	2023 The Author(s)
issn	1476-4687
url	https://www.nature.com/articles/s41586-023-06377-x
doi	10.1038/s41586-023-06377-x

abstract	Speech brain–computer interfaces (BCIs) have the potential to restore rapid communication to people with paralysis by decoding neural activity evoked by a text1,2 or sound3,4. Early demonstrations, although promising, have not yet achieved accuracies sufficiently high for communication of unconstrained sente vocabulary1–7. Here we demonstrate a speech-to-text BCI that records spiking activity from intracortical microelectrode arrays. Enabled by these high-resol study participant—who can no longer speak intelligibly owing to amyotrophic lateral sclerosis—achieved a 9.1\% word error rate on a 50-word vocabulary (than the previous state-of-the-art speech BCI2) and a 23.8\% word error rate on a 125,000-word vocabulary (the first successful demonstration, to our knowl vocabulary decoding). Our participant's attempted speech was decoded at 62 words per minute, which is 3.4 times as fast as the previous record8 and begins of natural conversation (160 words per minute9). Finally, we highlight two aspects of the neural code for speech that are encouraging for speech BCIs: spatia to speech articulators that makes accurate decoding possible from only a small region of cortex, and a detailed articulatory representation of phonemes that p paralysis. These results show a feasible path forward for restoring rapid communication to people with paralysis who can no longer speak.
language	en
number	7976
urldate	2024-04-03
journal	Nature
author	Willett, Francis R. and Kunz, Erin M. and Fan, Chaofei and Avansino, Donald T. and Wilson, Guy H. and Choi, Eun Young and Kamdar, Foram and Glasse Hochberg, Leigh R. and Druckmann, Shaul and Shenoy, Krishna V. and Henderson, Jaimie M.
month	aug, year = {2023
note	Publisher: Nature Publishing Group
keywords	Brain–machine interface, Neural decoding
pages	1031--1036
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\2E54S9N8\Willett et al. - 2023 - A high-performance speech neuroprosthesis.pdf:application/pdf
article	**guo_effects_2022**
title	Effects of auditory feedback on fine motor output and corticomuscular coherence during a unilateral finger pinch task
volume	16
issn	1662-453X
url	https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2022.896933/full
doi	10.3389/fnins.2022.896933
abstract	Auditory feedback is important to reduce movement error and improve motor performance during a precise motor task. Accurate motion guided by auditory the neural muscle transmission pathway between the sensorimotor area and the effective muscle. However, it remains unclear how neural activities and sens role in enhancing performance. The present study uses an auditory feedback system by simultaneously recording electroencephalogram (EEG), electromyog force information to measure corticomuscular coherence (CMC), neural activity, and motor performance during precise unilateral right-hand pinch by using index finger with and without auditory feedback. This study confirms three results. First, compared with no auditory feedback, auditory feedback decreases Second, compared with no auditory feedback, auditory feedback decreased the power spectrum in the beta band in the bimanual sensorimotor cortex and the ipsilateral sensorimotor cortex. Finally, CMC was computed between effector muscle of right hand and contralateral sensorimotor cortex. Analyses reveals t band significantly decreases in auditory feedback condition compared with no auditory feedback condition. The results indicate that auditory feedback decre in the alpha and beta bands and decreases corticospinal connection in the beta band during precise hand control. This study provides a new perspective on th

	feedback on behavior and brain activity and offers a new idea for designing more suitable and effective rehabilitation and training strategies to improve fine
language	English
urldate	2024-04-03
journal	Frontiers in Neuroscience
author	Guo, Jing and Liu, Tian and Wang, Jue
month	nov, year = {2022
note	Publisher: Frontiers
keywords	auditory feedback, beta band, corticomuscular coherence, power spectrum, precise motor control
file	Texte intégral:C\:\Users\paul-\Zotero\storage\9JPWC9IC\Guo et al. - 2022 - Effects of auditory feedback on fine motor output .pdf:application/pdf
article	**peyre_effect_2023**
title	Effect of sonification types in upper-limb movement: a quantitative and qualitative study in hemiparetic and healthy participants
volume	20
issn	1743-0003
shorttitle	Effect of sonification types in upper-limb movement
doi	10.1186/s12984-023-01248-y
abstract	BACKGROUND: Movement sonification, the use of real-time auditory feedback linked to movement parameters, have been proposed to support rehabilitat promising results have been reported, the effect of the type of sound used has not been studied systematically. The aim of this study was to investigate in a s of different types of sonification both quantitatively and qualitatively on patients with acquired brain lesions and healthy participants. METHODS: An exper arm sonification was developed using three different categories of sonification (direct sound modulation, musical interaction, and soundscape). Simple movi performed while sliding on a table with both arms were investigated with all participants. Quantitative analysis on the movement timing were performed cor parameters (sound condition, affected arm and dominance, sonification categories). Qualitative analysis of semi-structured interviews were also conducted, a neuropsychological evaluation of music perception. RESULTS: For both the patient and healthy groups (15 participants each), average duration for perform is significantly longer with sonification compared to the no-sound condition (p
language	eng
number	1
journal	Journal of Neuroengineering and Rehabilitation
author	Peyre, Iseline and Roby-Brami, Agnès and Segalen, Maël and Giron, Alain and Caramiaux, Baptiste and Marchand-Pauvert, Véronique and Pradat-Diehl, Pa Frédéric
month	oct, year = {2023
pmid	37798637
pmcid	PMC10552218
keywords	Movement sonification, Rehabilitation, Upper limb, Humans, Movement, Arm, Healthy Volunteers, Hemiparesis, Mixed methodology, Music

pages	136
file	Texte intégral:C\:\Users\paul-\Zotero\storage\F8YMBEBL\Peyre et al. - 2023 - Effect of sonification types in upper-limb movemen.pdf:application/pdf
article	**raghavan_music_2016**
title	Music {Upper} {Limb} {Therapy}—{Integrated}: {An} {Enriched} {Collaborative} {Approach} for {Stroke} {Rehabilitation}
volume	10
issn	1662-5161
shorttitle	Music {Upper} {Limb} {Therapy}—{Integrated}
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5053999/
doi	10.3389/fnhum.2016.00498
abstract	Stroke is a leading cause of disability worldwide. It leads to a sudden and overwhelming disruption in one's physical body, and alters the stroke survivors' s recovery requires that bodily perception, social participation and sense of self are restored; this is challenging to achieve, particularly with a single interventi synchronization of movement to external stimuli facilitates sensorimotor coupling for movement recovery, enhances emotional engagement and has positive interpersonal relationships. In this proof-of-concept study, we designed a group music-making intervention, Music Upper Limb Therapy-Integrated (MULT- physical, psychological and social domains of rehabilitation simultaneously, and investigated its effects on long-term post-stroke upper limb recovery. The s method pre-post design with 1-year follow up. Thirteen subjects completed the 45-min intervention twice a week for 6 weeks. The primary outcome was red impairment on the Fugl-Meyer Scale (FMS). Secondary outcomes included sensory impairment (two-point discrimination test), activity limitation (Modifiec well-being (WHO well-being index), and participation (Stroke Impact Scale, SIS). Repeated measures analysis of variance (ANOVA) was used to test for di and post-intervention, and 1-year follow up scores. Significant improvement was found in upper limb motor impairment, sensory impairment, activity limita immediately post-intervention that persisted at 1 year. Activities of daily living and social participation improved only from post-intervention to 1-year follo in upper limb motor impairment was more pronounced in a subset of lower functioning individuals as determined by their pre-intervention wrist range of mo subjects reported new feelings of ownership of their impaired limb, more spontaneous movement, and enhanced emotional engagement. The results suggest intervention may help stroke survivors re-create their sense of self by integrating sensorimotor, emotional and interoceptive information and facilitate long-t multiple domains of disability, even in the chronic stage post-stroke. Randomized controlled trials are warranted to confirm the efficacy of this approach. Cl Registration: National Institutes of Health, clinicaltrials.gov, NCT01586221.
urldate	2024-04-03
journal	Frontiers in Human Neuroscience
author	Raghavan, Preeti and Geller, Daniel and Guerrero, Nina and Aluru, Viswanath and Eimicke, Joseph P. and Teresi, Jeanne A. and Ogedegbe, Gbenga and Pa Alan
month	oct, year = {2016
pmid	27774059
pmcid	PMC5053999
pages	498
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\IXKSZ8WZ\Raghavan et al. - 2016 - Music Upper Limb Therapy—Integrated An Enriched
article	**sihvonen_clinical_2021**

title	Clinical and {Neural} {Predictors} of {Treatment} {Response} to {Music} {Listening} {Intervention} after {Stroke}
volume	11
issn	2076-3425
doi	10.3390/brainsci11121576
abstract	Patients with post-stroke impairments present often significant variation in response to therapeutic interventions. Recent studies have shown that daily music stroke recovery of language and memory, but reliable predictors of treatment response are unknown. Utilizing data from the music intervention arms of a sir controlled trial (RCT) on stroke patients (N = 31), we built regression models to predict the treatment response of a two-month music listening intervention verbal memory with baseline demographic, clinical and musical data as well as fMRI data from a music listening task. Clinically, greater improvement in ve language skills after the music listening intervention were predicted by the severity of the initial deficit and educational level. Neurally, greater baseline fMR vocal music listening in the left parietal cortical and medial frontal areas predicted greater treatment-induced improvement in language skills and greater bas auditory network during instrumental music listening predicted improvement in both verbal memory and language skills. Our results suggest that clinical, de neuroimaging data predicts music listening treatment response. This data could be used clinically to target music-based treatments.
language	eng
number	12
journal	Brain Sciences
author	Sihvonen, Aleksi J. and Särkämö, Teppo
month	nov, year = {2021
pmid	34942878
pmcid	PMC8699822
keywords	stroke, intervention, language, music, treatment predictor, treatment response
pages	1576
file	Texte intégral:C\:\Users\paul-\Zotero\storage\5U3ZKQZM\Sihvonen et Särkämö - 2021 - Clinical and Neural Predictors of Treatment Respon.pdf:applicatio
article	**wang_effects_2022**
title	Effects of rhythmic auditory stimulation on motor function and balance ability in stroke: A systematic review and meta-analysis of clinical randomized cont
volume	16
issn	1662-4548
shorttitle	Effects of rhythmic auditory stimulation on motor function and balance ability in stroke
doi	10.3389/fnins.2022.1043575
abstract	OBJECTIVE: Rhythmic auditory stimulation (RAS) belongs to neurologic music therapy, which has attracted clinical attention because of its efficacy in mo stroke. This study aimed to summarize the effectiveness of rhythmic auditory stimulation (RAS) for the treatment of motor function and balance ability in st systematic review and meta-analysis. METHODS: All studies were retrieved from six databases. The effects of RAS on stroke were determined using the fo motor function including step length, step cadence, velocity, Fugl-Meyer assessment (FMA); and balance ability including overall balance index (OBI) and (BBS). The risk map of bias of the quality of the studies and the meta-analysis results of the indicators was prepared using RevMan 5.2 software. RESULTS

	abstracts were retrieved. Among them, 325 duplicate studies were eliminated, and 971 studies were excluded after reading the titles and abstracts. In additio full text for further reading and screening, 47 studies were excluded. A total of 22 studies were included in the systematic review, and 18 studies were includ Assessment of quality, based on the PEDro scale, two studies had low quality, three studies had excellent quality, and the other studies had good quality; bas Collaborative Network Bias Risk Assessment Scale. A total of 15 studies specifically explained the random methods used. Meanwhile, seven studies did not sequence generation. A total of 10 studies reported that the evaluation of experimental results was blinded. In the meta-analysis, the results of motor functio (SMD = 0.99, 95\% CI (0.43, 1.55)), step length (SMD = 0.97, 95\% CI (0.74, 1.20)), and step cadence (MD = 5.16, 95\% CI (4.17, 6.14)), FMA (MD = 2.9 3.83))], were statistically significant (P < 0.01). The results of balance ability [OBI (MD = -0.51, 95\% CI (-0.86, -0.16)) and BBS (MD = 2.93, 95\% CI (1.6 statistically significant (P < 0.01). Among all the outcome indicators, three indicators were included in more than 10 studies: these are step length, step cade results showed that the two sides of the funnel chart were asymmetrical, thus these results all showed heterogeneity. The GRADEpro GDT online tool was u quality of evidence for the outcome indicators in the included studies. Five outcome indicators were included, of which three were low-quality indicators an quality indicators. CONCLUSIONS: RAS could improve gait parameters, walking function, and balance ability of individuals with stroke. However, studies indicators for balance ability of stroke patients is relatively insufficient, which also requires further research in the future. SYSTEMATIC REVIEW REGIST PROSPERO, identifier: CRD42021225102.
language	eng
journal	Frontiers in Neuroscience
author	Wang, Lei and Peng, Jin-Lin and Xiang, Wu and Huang, Yi-Jie and Chen, Ai-Lian
year	2022
pmid	36466174
pmcid	PMC9714437
keywords	rehabilitation, stroke, rhythmic auditory stimulation, motor disorders, motor fluctuations
pages	1043575
file	Texte intégral:C\:\Users\paul-\Zotero\storage\QCELRGAL\Wang et al. - 2022 - Effects of rhythmic auditory stimulation on motor .pdf:application/pdf
article	**bunketorp_kall_effects_2012**
title	The effects of a rhythm and music-based therapy program and therapeutic riding in late recovery phase following stroke: a study protocol for a three-armed trial
volume	12
issn	1471-2377
shorttitle	The effects of a rhythm and music-based therapy program and therapeutic riding in late recovery phase following stroke
url	https://doi.org/10.1186/1471-2377-12-141
doi	10.1186/1471-2377-12-141
abstract	Stroke represents one of the most costly and long-term disabling conditions in adulthood worldwide and there is a need to determine the effectiveness of reh the late phase after stroke. Limited scientific support exists for training incorporating rhythm and music as well as therapeutic riding and well-designed trials effectiveness of these treatment modalities are warranted.
number	1

urldate	2024-04-03
journal	BMC Neurology
author	Bunketorp Käll, Lina and Lundgren-Nilsson, Åsa and Blomstrand, Christian and Pekna, Marcela and Pekny, Milos and Nilsson, Michael
month	nov, year = {2012
keywords	Music Therapy, Life Satisfaction, Minimal Important Difference, Modify Rankin Scale, Stroke Survivor
pages	141
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\9H8F6XB6\Bunketorp Käll et al. - 2012 - The effects of a rhythm and music-based therapy pr.pdf:application /pdf;Snapshot:C\:\Users\paul-\Zotero\storage\X6BP65D6\1471-2377-12-141.html:text/html
article	**liu_reminiscent_2024**
title	Reminiscent music therapy combined with robot-assisted rehabilitation for elderly stroke patients: a pilot study
volume	21
issn	1743-0003
shorttitle	Reminiscent music therapy combined with robot-assisted rehabilitation for elderly stroke patients
doi	10.1186/s12984-024-01315-y
abstract	BACKGROUND: Although some studies suggest that robot-assisted technology can significantly improve upper limb function in stroke patients compared t rehabilitation training, it is still necessary to incorporate an auxiliary intervention to alleviate negative emotions, thereby alleviating the post-stroke fatigue a to actively respond to rehabilitation. However, the effect of the auxiliary intervention is unknown. OBJECTIVE: To evaluate the effect of reminiscent music robot-assisted rehabilitation in elderly patients with upper limb dysfunction. METHODS: From November 2022 to March 2023, elderly patients with upper l stroke were assigned to one of three groups, with group A receiving usual rehabilitation treatment and care plus robot-assisted rehabilitation and reminiscent B receiving usual rehabilitation treatment and care plus robot-assisted rehabilitation, and group C receiving only usual rehabilitation treatment and care. Thi this study, with 10 participants in each group. Activities of daily living, self-esteem, rehabilitation self-efficacy, positive emotion and upper limb function w and after the intervention. One-way analysis of variance, paired-sample t-test, Kruskal-Wallis H test, Wilcoxon signed rank sum test and Chi-square test wer data. RESULTS: According to the intragroup comparisons, in the three groups, all outcome measurements were significantly higher than those at baseline (a
language	eng
number	1
journal	Journal of Neuroengineering and Rehabilitation
author	Liu, Qian and Liu, Li and Liu, Zuoyan and Xu, Yang and Wang, Fang and Cheng, Hong and Hu, Xiuying
month	jan, year = {2024
pmid	38291426
pmcid	PMC10829204
keywords	Stroke, Rehabilitation, Robotics, Humans, Music therapy, Music Therapy, Recovery of Function, Stroke Rehabilitation, Upper Extremity, Aged, Music, Act Pilot Projects, Reminiscence therapy, Treatment Outcome

pages	16
file	Texte intégral:C\:\Users\paul-\Zotero\storage\2T2JJWZR\Liu et al. - 2024 - Reminiscent music therapy combined with robot-assi.pdf:application/pdf
article	**cason_bridging_2015**
title	Bridging music and speech rhythm: {Rhythmic} priming and audio–motor training affect speech perception
volume	155
issn	0001-6918
shorttitle	Bridging music and speech rhythm
url	https://www.sciencedirect.com/science/article/pii/S0001691814002650
doi	10.1016/j.actpsy.2014.12.002
abstract	Following findings that musical rhythmic priming enhances subsequent speech perception, we investigated whether rhythmic priming for spoken sentences e phonological processing – the building blocks of speech – and whether audio–motor training enhances this effect. Participants heard a metrical prime follow matching/mismatching prosodic structure), for which they performed a phoneme detection task. Behavioural (RT) data was collected from two groups: one w audio–motor training, and one who did not. We hypothesised that 1) phonological processing would be enhanced in matching conditions, and 2) audio–moto musical rhythms would enhance this effect. Indeed, providing a matching rhythmic prime context resulted in faster phoneme detection, thus revealing a cros musical rhythm on phonological processing. In addition, our results indicate that rhythmic audio–motor training enhances this priming effect. These results h implications for rhythm-based speech therapies, and suggest that metrical rhythm in music and speech may rely on shared temporal processing brain resourc
urldate	2024-04-03
journal	Acta Psychologica
author	Cason, Nia and Astésano, Corine and Schön, Daniele
month	feb, year = {2015
keywords	Music, Metre, Prosody, Rhythm, Speech
pages	43--50
file	Version soumise:C\:\Users\paul-\Zotero\storage\I4NY8L2V\Cason et al. - 2015 - Bridging music and speech rhythm Rhythmic priming.pdf:application/pdf
article	**xu_potential_2022**
title	Potential {Benefits} of {Music} {Therapy} on {Stroke} {Rehabilitation}
volume	2022
issn	1942-0900
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9217607/
doi	10.1155/2022/9386095

abstract	Stroke is associated with a high rate of disability and mortality, and survivors are usually accompanied with dysphagia, aphasia, motor dysfunction, cognitiv depression, and other complications. In the past decades, many studies have been conducted to reveal the pathogenesis and pathological mechanisms of strol treatment methods have been developed that contribute to the elevated survival rate of stroke patients. Early rehabilitation poststroke is starting to be recogn has been receiving increasing attention in order to further improve the quality of life of the patients. As an emerging method of poststroke rehabilitation, mu attenuate dysphagia and aphasia, improve cognition and motor function, alleviate negative moods, and accelerate neurological recovery in stroke patients. T summarize the recent progress that has been made using music therapy in stroke rehabilitation and is aimed at providing clinical evidence for the treatment c
urldate	2024-04-03
journal	Oxidative Medicine and Cellular Longevity
author	Xu, Chengyan and He, Zixia and Shen, Zhipeng and Huang, Fei
month	jun, year = {2022
pmid	35757506
pmcid	PMC9217607
pages	9386095
file	Texte intégral:C\:\Users\paul-\Zotero\storage\KAPE5PF2\Xu et al. - 2022 - Potential Benefits of Music Therapy on Stroke Reha.pdf:application/pdf
article	**chen_use_2016**
title	The use of augmented auditory feedback to improve arm reaching in stroke: a case series
volume	38
issn	1464-5165
shorttitle	The use of augmented auditory feedback to improve arm reaching in stroke
doi	10.3109/09638288.2015.1076530
abstract	PURPOSE: After practice, augmented feedback is the most important factor that facilitates motor learning. We assess the potential effectiveness of two type feedback on the re-learning of arm reaching in individuals with stroke: (a) real-time knowledge of performance (KP) feedback and (b) rhythmic cueing in th results (KR) feedback. METHOD: Five participants with stroke underwent short-term practice, reaching with their affected arm with KP, KR and no feedba We assessed range of motion of the upper extremity (shoulder, elbow) and trunk, mean error and variability of the performed trajectory, and movement time training. RESULTS: All participants benefitted from practice with feedback, though the effects varied across participants and feedback type. In three partici increased elbow extension and reduced compensatory trunk flexion. In four participants, KR feedback reduced movement time taken to perform the reach. C benefitted mostly from KP feedback, which increased shoulder flexion and elbow extension, and decreased compensatory trunk flexion and mean error. CO day practice with augmented auditory feedback improves reaching in individuals with stroke. This warrants further investigation with longer practice period size. IMPLICATIONS FOR REHABILITATION: After practice, augmented feedback is the second most important factor that facilitates motor learning. M auditory feedback has potential to enhance reaching abilities in individuals with stroke. Future studies are warranted to evaluate the long-term effectiveness longer training period in a larger sample size.
language	eng
number	11
journal	Disability and Rehabilitation

author	Chen, Joyce L. and Fujii, Shinya and Schlaug, Gottfried
year	2016
pmid	26314746
pmcid	PMC4769960
keywords	Stroke, Female, Humans, Male, Recovery of Function, Stroke Rehabilitation, Upper Extremity, Treatment Outcome, Exercise Therapy, Feedback, Sensory, performance, knowledge of results, Knowledge of Results, Psychological, motor learning, Psychomotor Performance, Range of Motion, Articular
pages	1115--1124
file	Version acceptée:C\:\Users\paul-\Zotero\storage\4YY5U5SP\Chen et al. - 2016 - The use of augmented auditory feedback to improve .pdf:application/pdf
article	**bianco_music_2019**
title	Music predictability and liking enhance pupil dilation and promote motor learning in non-musicians
volume	9
copyright	2019 The Author(s)
issn	2045-2322
url	https://www.nature.com/articles/s41598-019-53510-w
doi	10.1038/s41598-019-53510-w
abstract	Humans can anticipate music and derive pleasure from it. Expectations facilitate the learning of movements associated with anticipated events, and they are reward, which may further facilitate learning of the anticipated rewarding events. The present study investigates the synergistic effects of predictability and l music on arousal and motor-learning in a naïve population. Novel melodies were manipulated in their overall predictability (predictable/unpredictable) as ob model of music expectation, and ranked as high/medium/low liked based on participants' self-reports collected during an initial listening session. During thi recorded ocular pupil size as an implicit measure of listeners' arousal. During the following motor task, participants learned to play target notes of the melod (notes were of similar motor and musical complexity across melodies). Pupil dilation was greater for liked melodies, particularly when predictable. Motor pe facilitated in predictable rather than unpredictable melodies, but liked melodies were learned even in the unpredictable condition. Low-liked melodies also s mostly in participants with higher scores of task perceived competence. Taken together, these results highlight the effects of stimuli predictability on learnin however overshadowed by the effects of stimulus liking or task-related intrinsic motivation.
language	en
number	1
urldate	2024-04-03
journal	Scientific Reports
author	Bianco, R. and Gold, B. P. and Johnson, A. P. and Penhune, V. B.
month	nov, year = {2019
note	Publisher: Nature Publishing Group
keywords	Health care, Learning and memory

pages	17060
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\9N8RJMT3\Bianco et al. - 2019 - Music predictability and liking enhance pupil dila.pdf:application/pdf
article	**brown_auditorymotor_2012**
title	Auditory–motor learning influences auditory memory for music
volume	40
issn	1532-5946
url	https://doi.org/10.3758/s13421-011-0177-x
doi	10.3758/s13421-011-0177-x
abstract	In two experiments, we investigated how auditory–motor learning influences performers' memory for music. Skilled pianists learned novel melodies in four only (listening), motor only (performing without sound), strongly coupled auditory–motor (normal performance), and weakly coupled auditory–motor (perfo auditory recordings). Pianists' recognition of the learned melodies was better following auditory-only or auditory–motor (weakly coupled and strongly coup following motor-only learning, and better following strongly coupled auditory–motor learning than following auditory-only learning. Auditory and motor im modulated the learning effects: Pianists with high auditory imagery scores had better recognition following motor-only learning, suggesting that auditory im missing auditory feedback at the learning stage. Experiment 2 replicated the findings of Experiment 1 with melodies that contained greater variation in acous that were slower and less variable in tempo and intensity were remembered better following weakly coupled auditory–motor learning. These findings sugges can aid performers' auditory recognition of music beyond auditory learning alone, and that motor learning is influenced by individual abilities in mental ima acoustic features.
language	en
number	4
urldate	2024-04-03
journal	Memory \& Cognition
author	Brown, Rachel M. and Palmer, Caroline
month	may, year = {2012
keywords	Imagery, Individual differences, Music cognition, Recognition, Sequence learning
pages	567--578
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\ZDIQY2LK\Brown et Palmer - 2012 - Auditory–motor learning influences auditory memory.pdf:application/p
article	**verrienti_motivation_2023**
title	Motivation as a {Measurable} {Outcome} in {Stroke} {Rehabilitation}: A {Systematic} {Review} of the {Literature}
volume	20
copyright	http://creativecommons.org/licenses/by/3.0/
issn	1660-4601
shorttitle	Motivation as a {Measurable} {Outcome} in {Stroke} {Rehabilitation}

url	https://www.mdpi.com/1660-4601/20/5/4187
doi	10.3390/ijerph20054187
abstract	Motivated behaviours are thought to lead to enhanced performances. In the neurorehabilitation field, motivation has been demonstrated to be a link between performance, therefore playing an important role upon rehabilitation outcome determining factors. While motivation-enhancing interventions have been freq common and reliable motivation assessment strategy has not been established yet. This review aims to systematically explore and provide a comparison amo motivation assessment tools concerning stroke rehabilitation. For this purpose, a literature search (PubMed and Google Scholar) was performed, using the fo Subject Headings terms: “assessment” OR “scale” AND “motivation” AND “stroke” AND “rehabilitation”. In all, 31 randomized clinical trials and 15 clinic examined. The existing assessment tools can be grouped into two categories: the first mirroring the trade-off between patients and rehabilitation, the latter re between patients and interventions. Furthermore, we presented assessment tools which reflect participation level or apathy, as an indirect index of motivatio left to put forth a possible common motivation assessment strategy, which might provide valuable incentive to investigate in future research.
language	en
number	5
urldate	2024-04-04
journal	International Journal of Environmental Research and Public Health
author	Verrienti, Giulio and Raccagni, Cecilia and Lombardozzi, Ginevra and De Bartolo, Daniela and Iosa, Marco
month	jan, year = {2023
note	Number: 5 Publisher: Multidisciplinary Digital Publishing Institute
keywords	stroke, apathy, motivation, neurorehabilitation, participation, performance
pages	4187
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\2H7RZFU8\Verrienti et al. - 2023 - Motivation as a Measurable Outcome in Stroke Rehab.pdf:application/pdf
article	**rule_purdue_2021**
title	Purdue manual dexterity testing: A cohort study of community-dwelling elderly
volume	34
issn	1545-004X
shorttitle	Purdue manual dexterity testing
doi	10.1016/j.jht.2019.12.006
abstract	INTRODUCTION: The Purdue Pegboard Test is a manual dexterity test that requires the manipulation of pegs, washers, and collars. Our population for this on the geriatric community owing to the lack of recent available normative data. PURPOSE OF THE STUDY: The purpose of this study was to provide upd hand therapists to use in clinical practice to determine if seniors have dexterity deficits. STUDY DESIGN: This is a cohort study. METHODS: This study w convenience sample which included 128 participants. Participants were stratified into three age groups (60-69, 70-79, and 80+). Participants were asked to c Pegboard Test. Mean scores were analyzed using an independent-sample t-test and one-way analysis of variance to compare the mean scores of each designa RESULTS: A one-way analysis of variance reported statistically significant differences between the 3 age groups (F = 15.03, P
language	eng

number	1
journal	Journal of Hand Therapy: Official Journal of the American Society of Hand Therapists
author	Rule, Katherine and Ferro, Jennifer and Hoffman, Audrey and Williams, Jasmine and Golshiri, Sara and Padre, Rachelle and Avila, Jessica and Coca, Carley
year	2021
pmid	32151505
keywords	Female, Humans, Male, Aged, Hand, Assessment, Cohort Studies, Independent Living, Manual dexterity, Measurement, Motor Skills, Normative data, Purd
pages	116--120
article	**carment_manual_2018**
title	Manual {Dexterity} and {Aging}: A {Pilot} {Study} {Disentangling} {Sensorimotor} {From} {Cognitive} {Decline}
volume	9
issn	1664-2295
shorttitle	Manual {Dexterity} and {Aging}
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6215834/
doi	10.3389/fneur.2018.00910
abstract	Manual dexterity measures can be useful for early detection of age-related functional decline and for prediction of cognitive decline. However, what aspects function to assess remains unclear. Manual dexterity markers should be able to separate impairments related to cognitive decline from those related to health study, we aimed to compare manual dexterity components in patients diagnosed with cognitive decline (mean age: 84 years, N = 11) and in age comparable elderly subjects (mean age: 78 years, N = 11). In order to separate impairments due to healthy aging from deficits due to cognitive decline we also included young adults (mean age: 26 years, N = 10) and middle-aged adults (mean age: 41 years, N = 8). A comprehensive quantitative evaluation of manual dexterit three tasks: (i) visuomotor force tracking, (ii) isochronous single finger tapping with auditory cues, and (iii) visuomotor multi-finger tapping. Results showe increase in force tracking error with increasing age. Subjects with cognitive decline had increased finger tapping variability and reduced ability to select the in the multi-finger tapping task compared to cognitively intact elderly subjects. Cognitively intact elderly subjects and those with cognitive decline had prol reduced independence of finger movements compared to young adults and middle-aged adults. The findings suggest two different patterns of impaired manu to cognitive decline and another related to healthy aging. Manual dexterity tasks requiring updating of performance, in accordance with (temporal or spatial) in short-term memory, are particularly affected in cognitive decline. Conversely, tasks requiring online matching of motor output to sensory cues were affec cognitive status. Remarkably, no motor impairments were detected in patients with cognitive decline using clinical scales of hand function. The findings ma for the development of manual dexterity markers of cognitive decline.
urldate	2024-04-04
journal	Frontiers in Neurology
author	Carment, Loic and Abdellatif, Abir and Lafuente-Lafuente, Carmelo and Pariel, Sylvie and Maier, Marc A. and Belmin, Joël and Lindberg, Påvel G.
month	oct, year = {2018
pmid	30420830
pmcid	PMC6215834

pages	910
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\KCGHANDP\Carment et al. - 2018 - Manual Dexterity and Aging A Pilot Study Disentan.pd
article	**schaefer_auditory_2014**
title	Auditory rhythmic cueing in movement rehabilitation: findings and possible mechanisms
volume	369
issn	1471-2970
shorttitle	Auditory rhythmic cueing in movement rehabilitation
doi	10.1098/rstb.2013.0402
abstract	Moving to music is intuitive and spontaneous, and music is widely used to support movement, most commonly during exercise. Auditory cues are increasing rehabilitation of disordered movement, by aligning actions to sounds such as a metronome or music. Here, the effect of rhythmic auditory cueing on movem representative findings of cued movement rehabilitation are considered for several movement disorders, specifically post-stroke motor impairment, Parkinso Huntington's disease. There are multiple explanations for the efficacy of cued movement practice. Potentially relevant, non-mutually exclusive mechanisms of learning; qualitatively different motor learning owing to an auditory context; effects of increased temporal skills through rhythmic practices and motivatio rhythm. Further considerations of rehabilitation paradigm efficacy focus on specific movement disorders, intervention methods and complexity of the audito clinical interventions using rhythmic auditory cueing do not show consistently positive results, it is argued that internal mechanisms of temporal prediction a and further research may inform rehabilitation practice to increase intervention efficacy.
language	eng
number	1658
journal	Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences
author	Schaefer, Rebecca S.
month	dec, year = {2014
pmid	25385780
pmcid	PMC4240969
keywords	Humans, Motor Cortex, Music, Auditory Perception, auditory rhythm, Brain Mapping, cued movement, Cues, Exercise Movement Techniques, Magnetic Re Movement Disorders, movement rehabilitation, Periodicity
pages	20130402
file	Texte intégral:C\:\Users\paul-\Zotero\storage\CA9VQRNT\Schaefer - 2014 - Auditory rhythmic cueing in movement rehabilitatio.pdf:application/pdf
article	**roerdink_rhythm_2009**
title	Rhythm perturbations in acoustically paced treadmill walking after stroke
volume	23
issn	1545-9683
doi	10.1177/1545968309332879

abstract	BACKGROUND: In rehabilitation, acoustic rhythms are often used to improve gait after stroke. Acoustic cueing may enhance gait coordination by creating between heel strikes and metronome beats and provide a means to train the adaptability of gait coordination to environmental changes, as required in everyd OBJECTIVE: To examine the stability and adaptability of auditory-motor synchronization in acoustically paced treadmill walking in stroke patients. METH patients and 10 healthy controls walked on a treadmill at preferred speed and cadence under no metronome, single-metronome (pacing only paretic or nonpa double-metronome (pacing both footfalls) conditions. The stability of auditory-motor synchronization was quantified by the variability of the phase relation beats. In a separate session, the acoustic rhythms were perturbed and adaptations to restore auditory-motor synchronization were quantified. RESULTS: For motor synchronization was more stable for double-metronome than single-metronome conditions, with stroke patients exhibiting an overall weaker coupling metronome beats than controls. The recovery characteristics following rhythm perturbations corroborated the stability findings and further revealed that stro difficulty in accelerating their steps and instead preferred a slower-step response to restore synchronization. CONCLUSIONS: In gait rehabilitation practice, rhythms may be more effective when both footfalls are paced. In addition, rhythm perturbations during acoustically paced treadmill walking may not only be the stability of auditory-motor synchronization but also have promising implications for evaluation and training of gait adaptations in neurorehabilitation pra
language	eng
number	7
journal	Neurorehabilitation and Neural Repair
author	Roerdink, Melvyn and Lamoth, Claudine J. C. and van Kordelaar, Joost and Elich, Peter and Konijnenbelt, Manin and Kwakkel, Gert and Beek, Peter J.
month	sep, year = {2009
pmid	19307435
keywords	Stroke, Female, Humans, Male, Middle Aged, Stroke Rehabilitation, Adult, Aged, Analysis of Variance, Biomechanical Phenomena, Psychomotor Performa Stimulation, Gait, Gait Disorders, Neurologic, Walking
pages	668--678
file	Texte intégral:C\:\Users\paul-\Zotero\storage\YIQSXABK\Roerdink et al. - 2009 - Rhythm perturbations in acoustically paced treadmi.pdf:application/pdf
article	**schauer_musical_2003**
title	Musical motor feedback ({MMF}) in walking hemiparetic stroke patients: randomized trials of gait improvement
volume	17
issn	0269-2155
shorttitle	Musical motor feedback ({MMF}) in walking hemiparetic stroke patients
doi	10.1191/0269215503cr668oa
abstract	OBJECTIVE: To demonstrate the effect of rhythmical auditory stimulation in a musical context for gait therapy in hemiparetic stroke patients, when the stin measure by measure initiated by the patient's heel-strikes (musical motor feedback). Does this type of musical feedback improve walking more than a less sp DESIGN: The randomized controlled trial considered 23 registered stroke patients. Two groups were created by randomization: the control group received 1 conventional gait therapy and the test group received 15 therapy sessions with musical motor feedback. SETTING: Inpatient rehabilitation hospital. SUBJEC stroke interval was 44 days and the patients were able to walk without technical aids with a speed of approximately 0.71 m/s. MAIN OUTCOME MEASUR duration, gait symmetry, stride length and foot rollover path length (heel-on-toe-off distance). RESULT: The test group showed more mean improvement tha stride length increased by 18\% versus 0\%, symmetry deviation decreased by 58\% versus 20\%, walking speed increased by 27\% versus 4\% and rollover by 28\% versus 11\%. CONCLUSION: Musical motor feedback improves the stroke patient's walk in selected parameters more than conventional gait therap

	the patient's mind about the song and its timing may stimulate the improvement of gait even without the presence of an external pacemaker.
language	eng
number	7
journal	Clinical Rehabilitation
author	Schauer, Michael and Mauritz, Karl-Heinz
month	nov, year = {2003
pmid	14606736
keywords	Stroke, Humans, Middle Aged, Music Therapy, Stroke Rehabilitation, Treatment Outcome, Psychomotor Performance, Gait Disorders, Neurologic, Feedbac
pages	713--722
article	**thaut_kinematic_2002**
title	Kinematic optimization of spatiotemporal patterns in paretic arm training with stroke patients
volume	40
issn	0028-3932
doi	10.1016/s0028-3932(01)00141-5
abstract	The effect of rhythmic cueing on spatiotemporal control of sequential reaching movements of the paretic arm was studied in 21 hemispheric stroke patients. were studied with and without rhythmic metronome cuing in a counterbalanced design. Metronome frequencies were entrained to the naturally selected freq Results indicate statistically significant (P<0.05) improvements of spatiotemporal arm control during rhythmic entrainment. Variability of timing and reachi reduced significantly. Time series analysis of sequential movement repetitions showed an immediate reduction in variability of arm kinematics during rhyth the first two to three repetitions of each trial. Rhythm also produced significant increases in angle ranges of elbow motion (P<0.05). Analysis of acceleration the wrist joint showed significant kinematic smoothing during rhythmic cuing. The link between rhythmic sensory timing and spatiotemporal motor control a mathematical optimization model with minimization of peak acceleration as criterion. Rhythmically cued acceleration profiles fit the predicted model data (P<0.01) than the self-paced profiles. Since velocity and acceleration are mathematical derivatives of position-time trajectories, the model data suggest that e precision via temporal phase and period coupling of the motor pattern to the rhythmic time timekeeper enhances the brain's computational ability to optimall parameters across time.
language	eng
number	7
journal	Neuropsychologia
author	Thaut, M. H. and Kenyon, G. P. and Hurt, C. P. and McIntosh, G. C. and Hoemberg, V.
year	2002
pmid	11900758
keywords	Female, Humans, Male, Middle Aged, Stroke Rehabilitation, Aged, Biomechanical Phenomena, Arm, Motor Skills, Periodicity, Learning, Models, Theoreti Modalities

pages	1073--1081
article	**friedman_retraining_2014**
title	Retraining and assessing hand movement after stroke using the {MusicGlove}: comparison with conventional hand therapy and isometric grip training
volume	11
issn	1743-0003
shorttitle	Retraining and assessing hand movement after stroke using the {MusicGlove}
doi	10.1186/1743-0003-11-76
abstract	BACKGROUND: It is thought that therapy should be functional, be highly repetitive, and promote afferent input to best stimulate hand motor recovery afte struggle to access such therapy. We developed the MusicGlove, an instrumented glove that requires the user to practice gripping-like movements and thumb play a highly engaging, music-based, video game. The purpose of this study was to 1) compare the effect of training with MusicGlove to conventional hand MusicGlove training was more effective than a matched form of isometric hand movement training; and 3) determine if MusicGlove game scores predict cli METHODS: 12 chronic stroke survivors with moderate hemiparesis were randomly assigned to receive MusicGlove, isometric, and conventional hand thera design. Each subject participated in six one-hour treatment sessions three times per week for two weeks, for each training type, for a total of 18 treatment ses assessed hand impairment before and after each training type and at one-month follow-up including the Box and Blocks (B \& B) test as the primary outcom also completed the Intrinsic Motivation Inventory (IMI). RESULTS: Subjects improved hand function related to grasping small objects more after MusicGlo conventional training, as measured by the B \& B score (improvement of 3.21±3.82 vs. -0.29±2.27 blocks; P=0.010) and the 9 Hole Peg test (improvement c -0.85±1.29 pegs/minute; P=0.005). There was no significant difference between training types in the broader assessment batteries of hand function. Subjects isometric therapy than MusicGlove training, but the difference was not significant (P>0.09). Subjects sustained improvements in hand function at a one mon the MusicGlove more motivating than the other two therapies, as measured by the IMI. MusicGlove games scores correlated strongly with the B \& B score. These results support the hypothesis that hand therapy that is engaging, incorporates high numbers of repetitions of gripping and thumb-finger opposition m afferent input is a promising approach to improving an individual's ability to manipulate small objects. The MusicGlove provides a simple way to access suc
language	eng
journal	Journal of Neuroengineering and Rehabilitation
author	Friedman, Nizan and Chan, Vicky and Reinkensmeyer, Andrea N. and Beroukhim, Ariel and Zambrano, Gregory J. and Bachman, Mark and Reinkensmeye
month	apr, year = {2014
pmid	24885076
pmcid	PMC4022276
keywords	Stroke, Female, Humans, Male, Middle Aged, Stroke Rehabilitation, Adult, Aged, Hand, Movement, Exercise Therapy, Psychomotor Performance, Aged, 8 Strength, Video Games, Young Adult
pages	76
file	Texte intégral:C\:\Users\paul-\Zotero\storage\52KJBVFY\Friedman et al. - 2014 - Retraining and assessing hand movement after strok.pdf:application/pdf
article	**teremetz_novel_2015**
title	A novel method for the quantification of key components of manual dexterity after stroke
volume	12

issn	1743-0003
doi	10.1186/s12984-015-0054-0
abstract	BACKGROUND: A high degree of manual dexterity is a central feature of the human upper limb. A rich interplay of sensory and motor components in the l for independent control of fingers in terms of timing, kinematics and force. Stroke often leads to impaired hand function and decreased manual dexterity, lim living and impacting quality of life. Clinically, there is a lack of quantitative multi-dimensional measures of manual dexterity. We therefore developed the F Manipulandum (FFM), which allows quantification of key components of manual dexterity. The purpose of this study was (i) to test the feasibility of using t components of manual dexterity in hemiparetic stroke patients, (ii) to compare differences in dexterity components between stroke patients and controls, and individual profiles of dexterity components in stroke patients. METHODS: 10 stroke patients with mild-to-moderate hemiparesis and 10 healthy subjects we measures of hand function included the Action Research Arm Test and the Moberg Pick-Up Test. Four FFM tasks were used: (1) Finger Force Tracking to m (2) Sequential Finger Tapping to measure the ability to perform motor sequences, (3) Single Finger Tapping to measure timing effects, and (4) Multi-Finger ability to selectively move fingers in specified combinations (independence of finger movements). RESULTS: Most stroke patients could perform the tracki single and multi-finger tapping tasks. However, only four patients performed the sequence task. Patients showed less accurate force control, reduced tapping independence of finger movements compared to controls. Unwanted (erroneous) finger taps and overflow to non-tapping fingers were increased in patients. were not systematically related among each other, resulting in individually different profiles of deficient dexterity. Some of the FFM measures correlated wi CONCLUSIONS: Quantifying some of the key components of manual dexterity with the FFM is feasible in moderately affected hemiparetic patients. The F differences and individual profiles of deficient dexterity. The FFM is a promising tool for the measurement of key components of manual dexterity after stro improved targeting of motor rehabilitation.
language	eng
journal	Journal of Neuroengineering and Rehabilitation
author	Térémetz, Maxime and Colle, Florence and Hamdoun, Sonia and Maier, Marc A. and Lindberg, Påvel G.
month	aug, year = {2015
pmid	26233571
pmcid	PMC4522286
keywords	Stroke, Female, Humans, Male, Middle Aged, Adult, Aged, Fingers, Hand, Arm, Psychomotor Performance, Motor Skills, Paresis, Hand Strength
pages	64
file	Texte intégral:C\:\Users\paul-\Zotero\storage\FBDBJK39\Térémetz et al. - 2015 - A novel method for the quantification of key compo.pdf:application/pdf
book	**wagner_rhythm_2008**
title	The rhythm of language and speech: {Constraints}, models, metrics and applications.
shorttitle	The rhythm of language and speech
author	Wagner, Petra
month	jan, year = {2008
misc	**noauthor_heart_nodate**
title	Heart {Disease} and {Stroke} {Statistics}—2023 {Update}: A {Report} {From} the {American} {Heart} {Association} {\textbar} {Circulation}
url	https://www.ahajournals.org/doi/full/10.1161/CIR.0000000000001123

urldate	2024-04-05
article	**parker_loss_1986**
title	Loss of arm function after stroke: measurement, frequency, and recovery
volume	8
issn	0379-0797
shorttitle	Loss of arm function after stroke
doi	10.3109/03790798609166178
abstract	This study aims to establish the frequency of paralysis and other arm problems after stroke; the recovery of lost function; and to compare various tests of the per cent of the sample had no arm paralysis when first seen within 14 days. At 3 months 24 per cent of survivors had moderate or severe paralysis; 57 per ce pegs into holes within 50 seconds; 19 per cent had significant sensory disturbance; 5 per cent had shoulder pain; 8 per cent had restricted passive shoulder m cent had some paralysis of the dominant arm. Between 3 and 6 months, improvement of motor power was seen in 40 per cent of patients, and 13 per cent of function. Severity of initial paralysis was an important prognostic factor.
language	eng
number	2
journal	International Rehabilitation Medicine
author	Parker, V. M. and Wade, D. T. and Langton Hewer, R.
year	1986
pmid	3804600
keywords	Humans, Aged, Psychomotor Performance, Cerebrovascular Disorders, Contracture, Disability Evaluation, Follow-Up Studies, Hemiplegia, Pain
pages	69--73
article	**kwakkel_probability_2003**
title	Probability of regaining dexterity in the flaccid upper limb: impact of severity of paresis and time since onset in acute stroke
volume	34
issn	1524-4628
shorttitle	Probability of regaining dexterity in the flaccid upper limb
doi	10.1161/01.STR.0000087172.16305.CD
abstract	BACKGROUND AND PURPOSE: To improve the accuracy of early postonset prediction of motor recovery in the flaccid hemiplegic arm, the effects of ch over time on the accuracy of prediction were evaluated, and a prediction model for the probability of regaining dexterity at 6 months was developed. METH patients, dexterity and paresis were measured with the Action Research Arm Test, Motricity Index, and Fugl-Meyer motor evaluation. For model developme determinants were selected. Logistic regression analysis was used for prognostic factors and model development. RESULTS: At 6 months, some dexterity in found in 38\%, and complete functional recovery was seen in 11.6\% of the patients. Total anterior circulation infarcts, right hemisphere strokes, homonymo gaze deficit, visual inattention, and paresis were statistically significant related to a poor arm function. Motricity Index leg scores of at least 25 points in the

	Meyer arm scores of 11 points in the second week increasing to 19 points in the fourth week raised the probability of developing some dexterity (Action Res points) from 74\% (positive predictive value [PPV], 0.74; 95\% confidence interval [CI], 0.63 to 0.86) to 94\% (PPV, 0.83; 95\% CI, 0.76 to 0.91) at 6 mont probabilities of prediction dexterity was found after 4 weeks. CONCLUSIONS: Based on the Fugl-Meyer scores of the flaccid arm, optimal prediction of ar months can be made within 4 weeks after onset. Lack of voluntary motor control of the leg in the first week with no emergence of arm synergies at 4 weeks outcome at 6 months.
language	eng
number	9
journal	Stroke
author	Kwakkel, Gert and Kollen, Boudewijn J. and van der Grond, Jeroen and Prevo, Arie J. H.
month	sep, year = {2003
pmid	12907818
keywords	Stroke, Female, Humans, Male, Middle Aged, Recovery of Function, Stroke Rehabilitation, Upper Extremity, Aged, Treatment Outcome, Cohort Studies, M Disability Evaluation, Acute Disease, Infarction, Middle Cerebral Artery, Logistic Models, Motor Skills Disorders, Multivariate Analysis, Muscle Hypotoni Prospective Studies, Severity of Illness Index, Time Factors
pages	2181--2186
article	**thrift_global_2017**
title	Global stroke statistics
volume	12
issn	1747-4949
doi	10.1177/1747493016676285
abstract	Background Up to date data on incidence, mortality, and case-fatality for stroke are important for setting the agenda for prevention and healthcare. Aims and to update the most current incidence and mortality data on stroke available by country, and to expand the scope to case-fatality and explore how registry data complementary. Methods Data were compiled using two approaches: (1) an updated literature review building from our previous review and (2) direct acqui stroke events in the World Health Organization (WHO) mortality database for each country providing these data. To assess new and/or updated data on inci multiple databases to identify new original papers and review articles that met ideal criteria for stroke incidence studies and were published between 15 May 2016. For data on case-fatality, we searched between 1980 and 31 May 2016. We further screened reference lists and citation history of papers to identify ot from these sources. Mortality codes for ICD-8, ICD-9, and ICD-10 were extracted. Using population denominators provided for each country, we calculated mortality from stroke and mortality adjusted to the WHO world population. We used only the most recent year reported to the WHO for which both populati were available. Results Fifty-one countries had data on stroke incidence, some with data over many time periods, and some with data in more than one regio review, there were new incidence studies from 12 countries, with four meeting pre-determined quality criteria. In these four studies, the incidence of stroke, World standard population, ranged from 76 per 100,000 population per year in Australia (2009-10) up to 119 per 100,000 population per year in New Zealar latter being in those aged at least 15 years. Only in Martinique (2011-12) was the incidence of stroke greater in women than men. In countries either lacking stroke incidence, eight had national clinical registries of hospital based data. Of the 128 countries reporting mortality data to the WHO, crude mortality was (in 2003), Bulgaria, and Greece. Crude mortality and crude incidence of stroke were both positively correlated with the proportion of the population aged ≥ 6 time. Data on case-fatality were available in 42 studies in 22 countries, with large variations between regions. Conclusions In this updated review, we descri stroke incidence, case-fatality and mortality in different countries, and highlight the growing trend for national clinical registries to provide estimates in lieu

	incidence studies.
language	eng
number	1
journal	International Journal of Stroke: Official Journal of the International Stroke Society
author	Thrift, Amanda G. and Thayabaranathan, Tharshanah and Howard, George and Howard, Virginia J. and Rothwell, Peter M. and Feigin, Valery L. and Norrv Geoffrey A. and Cadilhac, Dominique A.
month	jan, year = {2017
pmid	27794138
keywords	stroke, Stroke, Humans, burden, case-fatality, epidemiology, global, Incidence, Internationality, mortality, Mortality, Registries, worldwide
pages	13--32
file	Version soumise:C\:\Users\paul-\Zotero\storage\3AME6TPW\Thrift et al. - 2017 - Global stroke statistics.pdf:application/pdf
article	**lee_improvement_2022**
title	Improvement of {Gait} in {Patients} with {Stroke} {Using} {Rhythmic} {Sensory} {Stimulation}: A {Case}-{Control} {Study}
volume	11
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2077-0383
shorttitle	Improvement of {Gait} in {Patients} with {Stroke} {Using} {Rhythmic} {Sensory} {Stimulation}
url	https://www.mdpi.com/2077-0383/11/2/425
doi	10.3390/jcm11020425
abstract	Patients with stroke suffer from impaired locomotion, exhibiting unstable walking with increased gait variability. Effects of rhythmic sensory stimulation on patients with chronic stroke are unclear. This study aims to determine the effects of rhythmic sensory stimulation on the gait of patients with chronic stroke. with stroke and twenty age- and gender-matched healthy controls walked 60 m under four conditions: normal walking with no stimulation, walking with rhy stimulation (RAS) through an earphone in the ear, walking with rhythmic somatosensory stimulation (RSS) through a haptic device on the wrist of each part with rhythmic combined stimulation (RCS: RAS + RSS). Gait performance in the stroke group significantly improved during walking with RAS, RSS, and I during normal walking ($p < 0.008$). Gait variability significantly decreased under the RAS, RSS, and RCS conditions compared to that during normal walkir Rhythmic sensory stimulation is effective in improving the gait of patients with chronic stroke, regardless of the type of rhythmic stimuli, compared to healt was greater in patients with reduced mobility, assessed by the Rivermead Mobility Index (RMI).
language	en
number	2
urldate	2024-04-05
journal	Journal of Clinical Medicine
author	Lee, Yungon and Shin, Sunghoon

month	jan, year = {2022
note	Number: 2 Publisher: Multidisciplinary Digital Publishing Institute
keywords	stroke, gait, rhythmic sensory stimulation, stimulus benefit
pages	425
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\E4VEZF4P\Lee et Shin - 2022 - Improvement of Gait in Patients with Stroke Using .pdf:application/pdf
article	**bae_development_2023**
title	Development of {Immersive} {Virtual} {Reality}-{Based} {Hand} {Rehabilitation} {System} {Using} a {Gesture}-{Controlled} {Rhythm} {Game} {Wit {Feedback}: {An} {fNIRS} {Pilot} {Study}
volume	31
issn	1558-0210
shorttitle	Development of {Immersive} {Virtual} {Reality}-{Based} {Hand} {Rehabilitation} {System} {Using} a {Gesture}-{Controlled} {Rhythm} {Game} {Wit {Feedback}
url	https://ieeexplore.ieee.org/abstract/document/10239483
doi	10.1109/TNSRE.2023.3312336
abstract	Recently, virtua reality (VR) has been widely utilized with rehabilitation to promote user engagement, which has been shown to induce brain plasticity. In th a VR-based hand rehabilitation system consisting of a personalized gesture-controlled rhythm game with vibrotactile feedback and investigated the cortical a induced by our system using functional near-infrared spectroscopy (fNIRS). Our system provides vibrotactile feedback as the user matches their hand gestur customized to their pre-recorded hand gestures. Cortical activation was measured via fNIRS during 420 seconds of alternating gameplay and rest in 11 healt stroke survivor. Regions of interest (ROI) were the prefrontal cortex (PFC), the premotor cortex \& the supplementary motor area (PMC\&SMA), the primar (SM1), and the somatosensory association cortex (SAC). The mean success rate of gesture matching among healthy subjects was 90 \% with a standard devi the success rate of the stroke survivor was 79.6 \%. The averaged cortical activation map for the 11 healthy subjects and the individual cortical activation ma survivor showed increased hemodynamic responses of oxygenated hemoglobin (HbO) during the VR-based hand rehabilitation compared to the resting cond analysis demonstrated a significant increase in HbO activation values in 19 out of 51 channels, corresponding to all ROIs except the left PFC and PMC\&SM high subject variability. The experimental results indicate that the proposed system successfully activated brain areas related to motor planning/execution, m and attention.
urldate	2024-04-05
journal	IEEE Transactions on Neural Systems and Rehabilitation Engineering
author	Bae, Sungjin and Park, Hyung-Soon
year	2023
note	Conference Name: IEEE Transactions on Neural Systems and Rehabilitation Engineering
keywords	Rhythm, Color, Functional near-infrared spectroscopy, Games, gesture recognition, hand rehabilitation systems, Stroke (medical condition), Task analysis, T feedback, virtual reality
pages	3732--3743

file	IEEE Xplore Abstract Record:C\:\Users\paul-\Zotero\storage\B99PWDBP\10239483.html:text/html;IEEE Xplore Full Text PDF:C\:\Users\paul-\Zotero\stor Park - 2023 - Development of Immersive Virtual Reality-Based Han.pdf:application/pdf
book	**beveridge_rhythmic_2018**
title	Rhythmic entrainment for hand rehabilitation using the {Leap} {Motion} controller
abstract	INTRODUCTION Millions of people around the world suffer from motor impairment or disability, yet effective, engaging, and cost-effective therapeutic sc MOTIVATION We propose a game that leverages the therapeutic aspects of music to encourage user engagement and adherence to the traditional physio-th treatment of motor impairment or disability.
author	Beveridge, Scott and Cano, Estefanía
month	sep, year = {2018
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\T3H6Z9ZH\Beveridge et Cano - 2018 - Rhythmic entrainment for hand rehabilitation using.pdf:application/pd
article	**raglio_hand_2021**
title	Hand rehabilitation with sonification techniques in the subacute stage of stroke
volume	11
issn	2045-2322
doi	10.1038/s41598-021-86627-y
abstract	After a stroke event, most survivors suffer from arm paresis, poor motor control and other disabilities that make activities of daily living difficult, severely a and personal independence. This randomized controlled trial aimed at evaluating the efficacy of a music-based sonification approach on upper limbs motor f and pain perceived during rehabilitation. The study involved 65 subacute stroke individuals during inpatient rehabilitation allocated into 2 groups which und dayweek) respectively of standard upper extremity motor rehabilitation or upper extremity treatment with sonification techniques. The Fugl-Meyer Upper E Block Test and the Modified Ashworth Scale were used to perform motor assessment and the McGill Quality of Life-it and the Numerical Pain Rating Scale and pain. The assessment was performed at baseline, after 2 weeks, at the end of treatment and at follow-up (1 month after the end of treatment). Total score Upper Extremity Scale (primary outcome measure) and hand and wrist sub scores, manual dexterity scores of the affected and unaffected limb in the Box an scores of the Numerical Pain Rating Scale (secondary outcomes measures) significantly improved in the sonification group compared to the standard of care interaction
language	eng
number	1
journal	Scientific Reports
author	Raglio, Alfredo and Panigazzi, Monica and Colombo, Roberto and Tramontano, Marco and Iosa, Marco and Mastrogiacomo, Sara and Baiardi, Paola and M Baldissarro, Eleonora and Imbriani, Chiara and Imarisio, Chiara and Eretti, Laura and Hamedani, Mehrnaz and Pistarini, Caterina and Imbriani, Marcello an and Caltagirone, Carlo
month	mar, year = {2021
pmid	33790343
pmcid	PMC8012636

keywords	Stroke, Female, Humans, Male, Middle Aged, Music Therapy, Recovery of Function, Stroke Rehabilitation, Aged, Aged, 80 and over, Quality of Life
pages	7237
file	Texte intégral:C\:\Users\paul-\Zotero\storage\RIFK62FF\Raglio et al. - 2021 - Hand rehabilitation with sonification techniques i.pdf:application/pdf
article	**huang_effectiveness_2021-1**
title	The {Effectiveness} of {Music} {Therapy} on {Hand} {Function} in {Patients} {With} {Stroke}: A {Systematic} {Review} of {Randomized} {Controlle
volume	12
issn	1664-2295
shorttitle	The {Effectiveness} of {Music} {Therapy} on {Hand} {Function} in {Patients} {With} {Stroke}
doi	10.3389/fneur.2021.641023
abstract	Objective: This study aims to evaluate the efficacy of music-supported therapy for stroke patients' hand function. Methods: The databases used included Cu Nursing and Allied Health Literature (CINAHL), MEDLINE, PubMed, Embase, Music Index, and Google Scholar. Studies published between January 201 included. The searching key terms included "music-supported therapy," "music therapy," "hand function," "hand dysfunction," "stroke," "ischemic," and "he Randomized controlled trials or controlled trials involving adults who have hand function problems caused by stroke are included in this study. The method of bias of the included studies were rated by two independent assessors under the guidance of Cochrane collaboration's risk of bias tool. Results: Twelve stu inclusion criteria were included in this study. Totally, the data included 598 stroke patients (345 male, 253 female) with recruited time from 1.7 months to 3 of the participants were 61.09 years old. Based on the Cochrane risk of bias tool, study quality ranged from three to seven out of seven points. Compared wi outcomes including hand strength, range of joint motion, dexterity of hands, arm function, and quality of life were significantly superior with music-suppor reported improved dexterity of hands, and one study reported the improvement of range of motion and strength of patients' hands, which supported the thera on patients' hand function and improving their quality of life after the therapy. The therapy ranged over a period of 4-8 weeks, with an average duration of 3 average of three times per week. Conclusion: Based on the results, music-supported therapy could be a useful treatment for improving hand function and act patients with stroke, especially for patients within 6 months after stroke. However, the low certainty of evidence downgrades our confidence to practice in h randomized controlled trials and larger sample sizes are required for a deeper review.
language	eng
journal	Frontiers in Neurology
author	Huang, Wen-Hao and Dou, Zu-Lin and Jin, Hui-Min and Cui, Ying and Li, Xin and Zeng, Qing
year	2021
pmid	34113305
pmcid	PMC8185294
keywords	hand function, music supported therapy, randomized controlled trial, stroke—diagnosis, systematic review
pages	641023
file	Texte intégral:C\:\Users\paul-\Zotero\storage\X97QHF3R\Huang et al. - 2021 - The Effectiveness of Music Therapy on Hand Functio.pdf:application/pdf
article	**raglio_active_2017**
title	Active music therapy approach for stroke patients in the post-acute rehabilitation

volume	38
issn	1590-3478
doi	10.1007/s10072-017-2827-7
abstract	Guidelines in stroke rehabilitation recommend the use of a multidisciplinary approach. Different approaches and techniques with music are used in the strok improve motor and cognitive functions but also psychological outcomes. In this randomized controlled pilot trial, relational active music therapy approaches acute phase of disease. Thirty-eight hospitalized patients with ischemic and hemorrhagic stroke were recruited and allocated in two groups. The experimenta standard of care (physiotherapy and occupational therapy daily sessions) and relational active music therapy treatments. The control group underwent the sta Motor functions and psychological aspects were assessed before and after treatments. Music therapy process was also evaluated using a specific rating scale positive trend in quality of life, functional and disability levels, and gross mobility. The experimental group showed a decrease of anxiety and, in particular, (p = 0.016). In addition, the strength of non-dominant hand (grip) significantly increased in the experimental group (p = 0.041). Music therapy assessment sh improvement over time of non-verbal and sonorous-music relationships. Future studies, including a greater number of patients and follow-up evaluations, ar promising results of this study.
language	eng
number	5
journal	Neurological Sciences: Official Journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology
author	Raglio, Alfredo and Zaliani, Alberto and Baiardi, Paola and Bossi, Daniela and Sguazzin, Cinzia and Capodaglio, Edda and Imbriani, Chiara and Gontero, C Marcello
month	may, year = {2017
pmid	28138867
keywords	Stroke, Rehabilitation, Female, Humans, Male, Middle Aged, Music therapy, Music Therapy, Stroke Rehabilitation, Aged, Analysis of Variance, Aged, 80 a Life, Brain Ischemia, Communication, Motor outcomes, Psychiatric Status Rating Scales, Psychological outcomes, Surveys and Questionnaires
pages	893--897
article	**nikmaram_musical_2019**
title	Musical {Sonification} of {Arm} {Movements} in {Stroke} {Rehabilitation} {Yields} {Limited} {Benefits}
volume	13
issn	1662-4548
doi	10.3389/fnins.2019.01378
abstract	Neurologic music therapy in rehabilitation of stroke patients has been shown to be a promising supplement to the often strenuous conventional rehabilitation this study was threefold: (i) replicate results from a previous study with a sample from one clinic (henceforth called Site 1; N = 12) using an already establis and (ii) conceptually replicate previous findings with a less costly hand-tracking system in Site 2 (N = 30), and (iii) compare both sub-studies' outcomes to e of neurologic music therapy. Stroke patients in both sites were randomly assigned to treatment or control groups and received daily training of guided seque movements additional to their standard stroke rehabilitation protocol. Treatment groups received sonification (i.e., changes in musical pitch) of their movem their affected hand up and down to reproduce a sequence of the first six notes of a C major scale. Controls received the same movement protocol, however, feedback. Sensors at the upper arm and the forearm (Xsens) or an optic sensor device (Leapmotion) allowed to measure kinematics of movements and move Behavioral measures pre and post intervention included the Fugl-Meyer assessment (FMA) and the Stroke Impact Scale (SIS) and movement data. Bayesian

	show evidence supporting an additional effect of sonification on clinical mobility assessments. However, combined movement data from both sites showed s movement smoothness for the treatment group, and an advantage for one of the two motion capturing systems. Exploratory analyses of EEG-EMG phase co movement of the paretic arm in a subset of patients suggested increases in cortico-muscular phase coherence specifically in the ipsilesional hemisphere after but not after standard rehabilitation therapy. Our findings show that musical sonification is a viable treatment supplement to current neurorehabilitation meth clinical benefits. However, given patients' enthusiasm during training and the low hardware price of one of the systems it may be considered as an add-on ho neurorehabilitation therapy.
language	eng
journal	Frontiers in Neuroscience
author	Nikmaram, Nikou and Scholz, Daniel S. and Großbach, Michael and Schmidt, Simone B. and Spogis, Jakob and Belardinelli, Paolo and Müller-Dahlhaus, F and Ziemann, Ulf and Rollnik, Jens D. and Altenmüller, Eckart
year	2019
pmid	31920526
pmcid	PMC6933006
keywords	stroke, neurorehabilitation, auditory-motor coupling, music-supported therapy, neurologic music therapy, neuroplasticity, sonification
pages	1378
file	Texte intégral:C\:\Users\paul-\Zotero\storage\L9I4ULGK\Nikmaram et al. - 2019 - Musical Sonification of Arm Movements in Stroke Re.pdf:application/pd
article	**zondervan_home-based_2016**
title	Home-based hand rehabilitation after chronic stroke: {Randomized}, controlled single-blind trial comparing the {MusicGlove} with a conventional exercise
volume	53
issn	1938-1352
shorttitle	Home-based hand rehabilitation after chronic stroke
doi	10.1682/JRRD.2015.04.0057
abstract	Individuals with chronic stroke have limited options for hand rehabilitation at home. Here, we sought to determine the feasibility and efficacy of home-based Seventeen participants with moderate hand impairment in the chronic phase of stroke were randomized to 3 wk of home-based exercise with either the Musi tabletop exercises. The primary outcome measure was the change in the Box and Blocks test score from baseline to 1 mo posttreatment. Both groups signific Box and Blocks test score, but no significant difference was found between groups. The MusicGlove group did exhibit significantly greater improvements th exercise group in motor activity log quality of movement and amount of use scores 1 mo posttherapy (p = 0.007 and p = 0.04, respectively). Participants sign their use of MusicGlove over time, completing 466 gripping movements per day on average at study end. MusicGlove therapy was not superior to conventio for the primary end point but was nevertheless feasible and led to a significantly greater increase in self-reported functional use and quality of movement of conventional home exercises. CLINICAL TRIAL REGISTRATION: ClinicalTrials.gov; "Influence of Timing on Motor Learning"; NCT01769326; https://c /show/NCT01769326.
language	eng
number	4
journal	Journal of Rehabilitation Research and Development

author	Zondervan, Daniel K. and Friedman, Nizan and Chang, Enoch and Zhao, Xing and Augsburger, Renee and Reinkensmeyer, David J. and Cramer, Steven C.
year	2016
pmid	27532880
keywords	rehabilitation, stroke, Stroke, Female, Humans, Male, Middle Aged, Stroke Rehabilitation, Adult, Aged, Hand, Motor Activity, Exercise Therapy, virtual rea controlled trial, Clothing, Cross-Over Studies, hand impairment, hand therapy, home therapy, motor activity log, music therapy, Single-Blind Method, task-
pages	457--472
file	Version soumise:C\:\Users\paul-\Zotero\storage\CL2AATUE\Zondervan et al. - 2016 - Home-based hand rehabilitation after chronic strok.pdf:application/p
article	**moumdjian_effectiveness_2017**
title	Effectiveness of music-based interventions on motricity or cognitive functioning in neurological populations: a systematic review
volume	53
issn	1973-9095
shorttitle	Effectiveness of music-based interventions on motricity or cognitive functioning in neurological populations
doi	10.23736/S1973-9087.16.04429-4
abstract	BACKGROUND: Motor and cognitive symptoms are frequent in persons with neurological disorders and often require extensive long-term rehabilitation. R music-based interventions have been introduced into neurological rehabilitation as training tools. EVIDENCE ACQUISITION: This review aims to 1) descr based intervention modalities and content which are applied in experimental studies; and 2) describe the effects of these interventions on motor and/or cogni neurological population. The databases PubMed and Web of Science were searched. Cited references of included articles where screened for potential inclus literature search up to 20th of June 2016 was conducted to include controlled trials and cohort studies that have used music-based interventions for ≥3 weeks population (in- and outpatients) targeting motor and/or cognitive symptoms. No limitations to publication date was set. EVIDENCE SYNTHESIS: Nineteen thirteen randomized controlled trials (total participants Nexp=241, Nctrl=269), four controlled trials (Nexp=59, Nctrl=53) and two cohort studies (N.=27) w studies were conducted in stroke, three in Parkinson's disease, and two in multiple sclerosis population. Modalities of music-based interventions were cluste instrument-based, listening-based, rhythm-based, and multicomponent-based music interventions. Overall, studies consistently showed that music-based inte or larger effects than conventional rehabilitation on upper limb function (N.=16; fine motricity, hand and arm capacity, finger and hand tapping velocity/vari (N.=7; gait parameters), and cognition (N.=4; verbal memory and focused attention). CONCLUSIONS: Variety of modalities using music-based interventio and grouped into four clusters. Effects of interventions demonstrate an improvement in the domains assessed. Evidence is most available for improving mot studies are warranted to investigate cognition as well as motor and cognition dysfunctions in combination. Instrument-based music interventions can improv and gross motor functions in stroke. Rhythm-based music interventions can improve gait parameters of velocity and cadence in stroke, Parkinson's disease a Cognition in the domains of verbal memory and focused attention can improve after listening-based music interventions in stroke.
language	eng
number	3
journal	European Journal of Physical and Rehabilitation Medicine
author	Moumdjian, Lousin and Sarkamo, Teppo and Leone, Carmela and Leman, Marc and Feys, Peter
month	jun, year = {2017
pmid	27879960

keywords	Humans, Music Therapy, Motor Activity, Cognition, Nervous System Diseases, Neurological Rehabilitation
pages	466--482
article	**schlaug_part_2009**
title	Part {VI} introduction: listening to and making music facilitates brain recovery processes
volume	1169
issn	1749-6632
shorttitle	Part {VI} introduction
doi	10.1111/j.1749-6632.2009.04869.x
language	eng
journal	Annals of the New York Academy of Sciences
author	Schlaug, Gottfried
month	jul, year = {2009
pmid	19673811
pmcid	PMC4430085
keywords	Humans, Music Therapy, Music, Brain, Brain Injuries
pages	372--373
file	Version acceptée:C\:\Users\paul-\Zotero\storage\5RNA4NQW\Schlaug - 2009 - Part VI introduction listening to and making musi.pdf:application/pdf
article	**altenmuller_neural_2009**
title	Neural reorganization underlies improvement in stroke-induced motor dysfunction by music-supported therapy
volume	1169
issn	1749-6632
doi	10.1111/j.1749-6632.2009.04580.x
abstract	Motor impairments are common after stroke, but efficacious therapies for these dysfunctions are scarce. By extending an earlier study on the effects of musi behavioral indices of motor function as well as electrophysiological measures were obtained before and after a series of therapy sessions to assess whether th to neural reorganization and motor recovery in patients after stroke. The study group comprised 32 stroke patients in a large rehabilitation hospital; they had motor function and no previous musical experience. Over a period of 3 weeks, these patients received 15 sessions of music-supported therapy using a manua approach. For comparison 30 additional patients received standard rehabilitation procedures. Fine as well as gross motor skills were trained by using either a electronic drum pads programmed to emit piano tones. Motor functions were assessed by an extensive test battery. In addition, we studied event-related desy /synchronization and coherences from all 62 patients performing self-paced movements of the index finger (MIDI-piano) and of the whole arm (drum pads). music-supported therapy yielded significant improvement in fine as well as gross motor skills with respect to speed, precision, and smoothness of movement data showed a more pronounced event-related desynchronization before movement onset and a more pronounced coherence in the music-supported therapy training assessment, whereas almost no differences were observed in the control group. Thus we see that music-supported therapy leads to marked improven

	after stroke and that these are accompanied by electrophysiological changes indicative of a better cortical connectivity and improved activation of the motor
language	eng
journal	Annals of the New York Academy of Sciences
author	Altenmüller, E. and Marco-Pallares, J. and Münte, T. F. and Schneider, S.
month	jul, year = {2009
pmid	19673814
keywords	Stroke, Female, Humans, Male, Middle Aged, Music Therapy, Adult, Aged, Neuronal Plasticity, Psychomotor Performance, Rehabilitation Centers
pages	395--405
article	**suh_effect_2014-1**
title	Effect of rhythmic auditory stimulation on gait and balance in hemiplegic stroke patients
volume	34
issn	1878-6448
doi	10.3233/NRE-131008
abstract	BACKGROUND: Rhythmic auditory stimulation (RAS) improves gait velocity, cadence, and stride length in hemiplegic stroke patients, yet little is known of RAS on standing balance. OBJECTIVES: To examine and compare the effectiveness on standing balance and gait parameter of two different types of gai with RAS versus gait training without RAS, was compared in two groups of hemiplegic stroke patients over a three-week period (RAS group, n = 8; control METHODS: Sixteen hemiplegic stroke patients were included in the study. Standing balance was measured by using Biosway®. Stride length, gait velocity, calculated from the number of steps and required time for a 10 m-walk. RESULTS: Pre- to post-test measures showed a significant improvement in RAS gro index (p = 0.043), mediolateral index (p = 0.006), anteroposterior index (p = 0.016), gait velocity (p = 0.012), stride length (p = 0.03) and cadence (p = 0.01 group. CONCLUSIONS: The results of this study showed that RAS was an effective therapeutic method to improve gait velocity, stride length, cadence, and hemiplegic stroke patients.
language	eng
number	1
journal	NeuroRehabilitation
author	Suh, Jee Hyun and Han, Soo Jeong and Jeon, Seo Young and Kim, Ho Jeong and Lee, Jeong Eun and Yoon, Tae Sik and Chong, Hyun Ju
year	2014
pmid	24284453
keywords	rehabilitation, stroke, gait, Rhythmic auditory stimulation, Stroke, Female, Humans, Male, Middle Aged, Stroke Rehabilitation, Aged, Acoustic Stimulation, balance, Postural Balance
pages	193--199
article	**van_wijck_making_2012**
title	Making music after stroke: using musical activities to enhance arm function

volume	1252
issn	1749-6632
shorttitle	Making music after stroke
doi	10.1111/j.1749-6632.2011.06403.x
abstract	A common long-term consequence of stroke is impaired arm function, which affects independence and quality of life in a considerable proportion of stroke s growing need for self-management strategies that enable stroke survivors to continue their recovery after rehabilitation has ceased. Interventions with high-i training and feedback are most likely to improve function. Achieving the required amount of self-practice is challenging, however. Innovative approaches ar therapies into rewarding activities that can be undertaken independently. This paper describes the key principles and development of a novel intervention tha preferred music with game technology in upper limb rehabilitation. The "tap tempo" paradigm, which uses rhythmic auditory cueing, provides repetitive upp which can be tailored to individual goals and progress (e.g., in terms of movement range and complexity), while providing sensitive quantitative feedback to acquisition and enhance self-management.
language	eng
journal	Annals of the New York Academy of Sciences
author	van Wijck, Frederike and Knox, Don and Dodds, Colin and Cassidy, Gianna and Alexander, Gillian and MacDonald, Raymond
month	apr, year = {2012
pmid	22524372
keywords	Stroke, Humans, Music Therapy, Stroke Rehabilitation, Arm, Feedback, Sensory, Video Games, Neurosciences
pages	305--311
article	**lim_therapeutic_2013**
title	The {Therapeutic} {Effect} of {Neurologic} {Music} {Therapy} and {Speech} {Language} {Therapy} in {Post}-{Stroke} {Aphasic} {Patients}
volume	37
issn	2234-0645
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3764351/
doi	10.5535/arm.2013.37.4.556
abstract	Objective To investigate the therapeutic effect of neurologic music therapy (NMT) and speech language therapy (SLT) through improvement of the aphasia stroke aphasic patients. Methods Twenty-one post-stroke, nonfluent aphasia patients who had ischemic/hemorrhagic stroke on radiologic evaluation were di SLT groups. They received NMT and SLT for 1 month. Language function was assessed by Korean version-Western Aphasia Battery before and after therap therapeutic singing and melodic intonation therapy, and SLT consisted of language-oriented therapy. Results Significant improvements were revealed in AQ after therapy in the NMT group and improvements in repetition in the SLT group of chronic stroke patients ($p<0.05$). There were significant improvements i NMT group of subacute stroke patients. However, there was no significant improvement in the SLT group of subacute stroke patients. Conclusion We concl therapies are effective treatments in the chronic stage of stroke and NMT is effective in subacute post-stroke aphasic patients.
number	4
urldate	2024-04-05

journal	Annals of Rehabilitation Medicine
author	Lim, Kil-Byung and Kim, Yong-Kyun and Lee, Hong-Jae and Yoo, Jeehyun and Hwang, Ji Youn and Kim, Jeong-Ah and Kim, Sung-Kyun
month	aug, year = {2013
pmid	24020037
pmcid	PMC3764351
pages	556--562
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\85NFIS6S\Lim et al. - 2013 - The Therapeutic Effect of Neurologic Music Therapy.pdf:appli
article	**jun_effect_2013**
title	The effect of music-movement therapy on physical and psychological states of stroke patients
volume	22
issn	1365-2702
doi	10.1111/j.1365-2702.2012.04243.x
abstract	AIMS AND OBJECTIVES: This study evaluated the effects of combined music-movement therapy on physical and psychological functioning of hospitalise BACKGROUND: Few studies have focused on music-movement therapy's effects on physical and psychological functioning of stroke patients. DESIGN: A design with pre- and post-tests was used. METHODS: A convenience sample was used: patients hospitalised for stroke and within two weeks of the onset of randomised to either an experimental group (received music-movement therapy in their wheelchairs for 60 minutes three times per week for 8 weeks) or con only routine treatment). The effect of music-movement therapy was assessed in terms of physical outcomes (range of motion, muscle strength and activities psychological outcomes (mood states, depression), measured in both groups pre- and post-test. RESULTS: The experimental group had significantly increas elbow joint flexion in physical function and improved mood state in psychological function, compared with the control group. CONCLUSIONS: Early rehal stroke patients within two weeks of the onset of stroke was effective by using music-movement therapy. It improved their mood state and increased shoulder joint flexion. RELEVANCE TO CLINICAL PRACTICE: The findings of this study suggest that rehabilitation for stroke patients should begin as early as po their hospitalisation. Nursing practice should incorporate the concept of combining music and movements to improve stroke patients' physical and psycholog from the acute phase.
language	eng
number	1-2
journal	Journal of Clinical Nursing
author	Jun, Eun-Mi and Roh, Young Hwa and Kim, Mi Ja
month	jan, year = {2013
pmid	22978325
keywords	Stroke, Female, Humans, Male, Middle Aged, Music Therapy, Stroke Rehabilitation, Adult, Movement, Activities of Daily Living, Exercise Therapy, Rang
pages	22--31
article	**friedman_musicglove_2011**

title	MusicGlove}: motivating and quantifying hand movement rehabilitation by using functional grips to play music
volume	2011
issn	2694-0604
shorttitle	MusicGlove
doi	10.1109/IEMBS.2011.6090659
abstract	People with stroke typically must perform much of their hand exercise at home without professional assistance as soon as two weeks after the stroke. Withou encouragement, individuals often lose motivation to practice using the affected hand, and this disuse contributes to further declines in hand function. We dev as a way to facilitate and motivate at home practice of hand movement. This low-cost device uses music as an interactive and motivating medium to guide h quantitatively assess hand movement recovery. It requires the user to practice functional movements, including pincer grip, key-pinch grip, and finger-thuml those movements to play different musical notes, played along to songs displayed by an interactive computer game. We report here the design of the glove a single-session experiment with 10 participants with chronic stroke. We found that the glove is well suited for use by people with an impairment level quantif Blocks score of at least around 7; that the glove can be used to obtain a measure of hand dexterity (\% of notes hit) that correlates strongly with the Box and the incorporation of music into training significantly improved both objective measures of hand motor performance and self-ratings of motivation for trainin
language	eng
journal	Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual In
author	Friedman, Nizan and Chan, Vicky and Zondervan, Danny and Bachman, Mark and Reinkensmeyer, David J.
year	2011
pmid	22254815
keywords	Stroke, Humans, Music Therapy, Stroke Rehabilitation, Hand, Movement, Equipment Design, Equipment Failure Analysis, Reproducibility of Results, Sens User-Computer Interface
pages	2359--2363
article	**gentilucci_spoken_2008**
title	Spoken {Language} and arm {Gestures} are {Controlled} by the same {Motor} {Control} {System}
volume	61
issn	1747-0218
url	https://doi.org/10.1080/17470210701625683
doi	10.1080/17470210701625683
abstract	Arm movements can influence language comprehension much as semantics can influence arm movement planning. Arm movement itself can be used as a li reviewed neurophysiological and behavioural evidence that manual gestures and vocal language share the same control system. Studies of primate premotor particular, of the so-called “mirror system”, including humans, suggest the existence of a dual hand/mouth motor command system involved in ingestion act platform on which a combined manual and vocal communication system was constructed. In humans, speech is typically accompanied by manual gesture, sp is influenced by executing or observing transitive hand actions, and manual actions play an important role in the development of speech, from the babbling s Behavioural data also show reciprocal influence between word and symbolic gestures. Neuroimaging and repetitive transcranial magnetic stimulation (rTMS system governing both speech and gesture is located in Broca's area. In general, the presented data support the hypothesis that the hand motor-control system

	order cognition.
language	en
number	6
urldate	2024-04-08
journal	Quarterly Journal of Experimental Psychology
author	Gentilucci, Maurizio and Volta, Riccardo Dalla
month	jun, year = {2008
note	Publisher: SAGE Publications
pages	944--957
article	**hill_pilot_2011**
title	A {Pilot} {Study} of {Rhythm} and {Timing} {Training} as a {Supplement} to {Occupational} {Therapy} in {Stroke} {Rehabilitation}
volume	18
issn	1074-9357
url	https://doi.org/10.1310/tsr1806-728
doi	10.1310/tsr1806-728
abstract	Introduction: Stroke is the leading cause of disability. A need exists for an effective intervention to enhance upper extremity (UE) motor abilities and activiti (ADL) performance. Objective: The objectives of this pilot study were to (1) determine the feasibility of adding Interactive Metronome (IM) to an occupatio program; and (2) determine changes in UE impairments, function, quality of life, and perceived physical performance ability and satisfaction using a combin compared with OT alone for adults with chronic stroke. Methods: This pilot study (n=10) used a 2-group (OT or IM+OT) pretest-posttest design. The interv minutes of IM + OT or OT alone, 3 days a week for 10 weeks. Outcome measures included the UE Fugl-Meyer (impairment), the Arm Motor Ability Test (Block Test (function), Stroke Impact Scale (quality of life), and the Canadian Occupational Performance Measure (perceived performance ability and satisfa feasible to add IM to OT. The IM+OT group demonstrated decreased impairment and increased quality of life. However, the OT-alone group demonstrated function, perceived physical performance ability, and satisfaction. Conclusion: These findings suggest that rhythm and timing training using the IM is a feas consider as part of therapy treatment. However, IM may fit best for prefunctional treatment, as it seemed to primarily decrease impairment. It may also serve or after treatment in order to maximize rehabilitation potential. Clinical implications and suggestions for future studies are provided.
number	6
urldate	2024-04-22
journal	Topics in Stroke Rehabilitation
author	Hill, Valerie and Dunn, Leah and Dunning, Kari and Page, Stephen J.
month	nov, year = {2011
pmid	22436310
note	Publisher: Taylor \& Francis _eprint: https://doi.org/10.1310/tsr1806-728

keywords	occupational therapy, rhythm and timing training, stroke réhabilitation
pages	728--737
article	**bobak_estimation_2018**
title	Estimation of an inter-rater intra-class correlation coefficient that overcomes common assumption violations in the assessment of health measurement scales
volume	18
issn	1471-2288
url	https://doi.org/10.1186/s12874-018-0550-6
doi	10.1186/s12874-018-0550-6
abstract	Intraclass correlation coefficients (ICC) are recommended for the assessment of the reliability of measurement scales. However, the ICC is subject to a varie assumptions such as normality and stable variance, which are rarely considered in health applications.
number	1
urldate	2024-05-31
journal	BMC Medical Research Methodology
author	Bobak, Carly A. and Barr, Paul J. and O'Malley, A. James
month	sep, year = {2018
keywords	Bayesian analysis, Hierarchical regression, ICC, Observer OPTION5, Reliability, Shared decision making, Variance function modelling
pages	93
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\A3S47QIH\Bobak et al. - 2018 - Estimation of an inter-rater intra-class correlati.pdf:application/pdf;Snapshot \Zotero\storage\RKQKA9EJ\s12874-018-0550-6.html:text/html
article	**weir_quantifying_2005**
title	Quantifying test-retest reliability using the intraclass correlation coefficient and the {SEM}
volume	19
issn	1064-8011
doi	10.1519/15184.1
abstract	Reliability, the consistency of a test or measurement, is frequently quantified in the movement sciences literature. A common metric is the intraclass correlat In addition, the SEM, which can be calculated from the ICC, is also frequently reported in reliability studies. However, there are several versions of the ICC, the movement sciences regarding which ICC to use. Further, the utility of the SEM is not fully appreciated. In this review, the basics of classic reliability the context of choosing and interpreting an ICC. The primary distinction between ICC equations is argued to be one concerning the inclusion (equations 2,1 and (equations 3,1 and 3,k) of systematic error in the denominator of the ICC equation. Inferential tests of mean differences, which are performed in the process necessary variance components for the calculation of ICC values, are useful to determine if systematic error is present. If so, the measurement schedule shou (removing trials where learning and/or fatigue effects are present) to remove systematic error, and ICC equations that only consider random error may be saf ICC values is discussed in the context of estimating the effects of measurement error on sample size, statistical power, and correlation attenuation. Finally, c

	application of the SEM are discussed. It is shown how the SEM and its variants can be used to construct confidence intervals for individual scores and to det difference needed to be exhibited for one to be confident that a true change in performance of an individual has occurred.
language	eng
number	1
journal	Journal of Strength and Conditioning Research
author	Weir, Joseph P.
month	feb, year = {2005
pmid	15705040
keywords	Humans, Reproducibility of Results, Statistics as Topic
pages	231--240
article	**chandrasekaran_does_2021**
title	Does breaking up prolonged sitting improve cognitive functions in sedentary adults? A mapping review and hypothesis formulation on the potential physiol
volume	22
issn	1471-2474
shorttitle	Does breaking up prolonged sitting improve cognitive functions in sedentary adults?
url	https://doi.org/10.1186/s12891-021-04136-5
doi	10.1186/s12891-021-04136-5
abstract	Prolonged (excessive) sitting is detrimentally associated with cardiovascular, metabolic and mental health. Moreover, prolonged sitting has been associated function, memory, attention and visuospatial skills, which are important cognitive aspects of work performance. Breaking up prolonged sitting with standing exercises at the workplace is recognized as a potential measure in improving cognition. However, preliminary evidence, primarily from acute laboratory exp formulating hypothesis on the possible mechanistic pathways. Hence, the aim of this mapping review is to gather preliminary evidence and substantiate poss mechanisms underpinning the putative effects of breaking prolonged sitting on improving cognitive function among sedentary office workers.
number	1
urldate	2024-05-31
journal	BMC Musculoskeletal Disorders
author	Chandrasekaran, Baskaran and Pesola, Arto J. and Rao, Chythra R. and Arumugam, Ashokan
month	mar, year = {2021
keywords	Brain health, Cognitive function, Executive functions, Physiology, Sitting, Workplace
pages	274
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\7CTSVFNF\Chandrasekaran et al. - 2021 - Does breaking up prolonged sitting improve cogniti.pdf:applicatio /pdf;Snapshot:C\:\Users\paul-\Zotero\storage\J8ANN4LN\s12891-021-04136-5.html:text/html

article	albulescu_give_2022
title	"{Give} me a break!" A systematic review and meta-analysis on the efficacy of micro-breaks for increasing well-being and performance
volume	17
issn	1932-6203
url	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0272460
doi	10.1371/journal.pone.0272460
abstract	Recovery activities during short breaks taken between work tasks are solutions for preventing the impairing effects of accumulated strain. No wonder then th scientific literature from various perspectives emerged on this topic. The present meta-analysis is aimed at estimating the efficacy of micro-breaks in enhanc and fatigue) and performance, as well as in which conditions and for whom are the micro-breaks most effective. We searched the existent literature on this to existing data from experimental and quasi-experimental studies. The systematic search revealed 19 records, which resulted in 22 independent study samples effects meta-analyses shown statistically significant but small effects of micro-breaks in boosting vigor (d = .36, p < .001; k = 9, n = 913), reducing fatigue (n = 803), and a non-significant effect on increasing overall performance (d = .16, p = .116; k = 15, n = 1132). Sub-groups analyses on performance types rev only for tasks with less cognitive demands. A meta-regression showed that the longer the break, the greater the boost was on performance. Overall, the data micro-breaks for well-being, while for performance, recovering from highly depleting tasks may need more than 10-minute breaks. Therefore, future studies issue.
language	en
number	8
urldate	2024-05-31
journal	PLOS ONE
author	Albulescu, Patricia and Macsinga, Irina and Rusu, Andrei and Sulea, Coralia and Bodnaru, Alexandra and Tulbure, Bogdan Tudor
month	aug, year = {2022
note	Publisher: Public Library of Science
keywords	Attention, Emotions, Employment, Fatigue, Memory, Metaanalysis, Publication ethics, Relaxation (psychology)
pages	e0272460
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\LYS4UJV9\Albulescu et al. - 2022 - Give me a break! A systematic review and meta-an.pdf:application/pdf
article	**kocaballi_understanding_2019**
title	Understanding and {Measuring} {User} {Experience} in {Conversational} {Interfaces}
volume	31
doi	10.1093/iwc/iwz015

abstract	Although various methods have been developed to evaluate conversational interfaces, there has been a lack of methods specifically focusing on evaluating u paper reviews the understandings of user experience (UX) in conversational interfaces literature and examines the six questionnaires commonly used for eva systems in order to assess the potential suitability of these questionnaires to measure different UX dimensions in that context. The method to examine the qu developing an assessment framework for main UX dimensions with relevant attributes and coding the items in the questionnaires according to the framewor (i) the understandings of UX notably differed in literature; (ii) four questionnaires included assessment items, in varying extents, to measure hedonic, aesthe dimensions of UX; (iii) while the dimension of affect was covered by two questionnaires, playfulness, motivation, and frustration dimensions were covered only. The largest coverage of UX dimensions has been provided by the Subjective Assessment of Speech System Interfaces (SASSI). We recommend using to obtain a more complete measurement of user experience or improve the assessment of a particular UX dimension. RESEARCH HIGHLIGHTS Varying u conversational interfaces literature. A UX assessment framework with UX dimensions and their relevant attributes. Descriptions of the six main questionnai conversational interfaces. A comparison of the six questionnaires based on their coverage of UX dimensions.
journal	Interacting with Computers
author	Kocaballi, A. Baki and Laranjo, Liliana and Coiera, Enrico
month	may, year = {2019
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\45B6C45E\Kocaballi et al. - 2019 - Understanding and Measuring User Experience in Con.pdf:application/pd
inproceedings	**noh_starting_2024**
address	New York, NY, USA
series	CHI} '24
title	Starting a {New} {Life} after {Crossing} the {Tumen} {River}: {How} {North} {Korean} {Defectors} {Use} {Digital} {Technology} in {Transition}
isbn	9798400703300
shorttitle	Starting a {New} {Life} after {Crossing} the {Tumen} {River}
url	https://doi.org/10.1145/3613904.3642892
doi	10.1145/3613904.3642892
abstract	In a world where digital technology is omnipresent, North Korea stands as an outlier, with most citizens uninformed about its existence. This study explores North Korean defectors as they transition to a highly digitally connected society—South Korea. Through 21 semi-structured interviews, we initially investig and challenges they encounter during their transitions. Then, we examine the role of digital technology as they adapt to the highly connected digital environ Our findings highlight that social media serves as a double-edged sword, providing the freedom to construct a desired identity while accentuating the gap be ideal selves. This empirical research offers insights into how an underrepresented population navigates the digital landscape during life transitions, shedding and ways to better address their needs.
urldate	2024-06-19
booktitle	Proceedings of the {CHI} {Conference} on {Human} {Factors} in {Computing} {Systems}
publisher	Association for Computing Machinery
author	Noh, Hayoun and Yoon, Soohyun and Jo, Hyunah and Van Kleek, Max and Kang, Younah
month	may, year = {2024
keywords	Empirical study that tells us about people, HCI for Development, Interview, Social Media/Online Communities

pages	1--26
article	**tomita_efficacy_2024**
title	Efficacy of a {Single}-{Bout} of {Auditory} {Feedback} {Training} on {Gait} {Performance} and {Kinematics} in {Healthy} {Young} {Adults}
volume	24
copyright	http://creativecommons.org/licenses/by/3.0/
issn	1424-8220
url	https://www.mdpi.com/1424-8220/24/10/3206
doi	10.3390/s24103206
abstract	This study investigated the immediate effects of auditory feedback training on gait performance and kinematics in 19 healthy young adults, focusing on bilat unilateral training. Baseline and post-training kinematic measurements, as well as the feedback training were performed on a treadmill with a constant veloc improvements were seen in step length (trained: 590.7 mm to 611.1 mm, 95\%CI [7.609, 24.373]; untrained: 591.1 mm to 628.7 mm, 95\%CI [10.698, 30.83 (trained: 13.9 mm to 16.5 mm, 95\%CI [1.284, 3.503]; untrained: 11.8 mm to 13.7 mm, 95\%CI [1.763, 3.612]), ankle dorsiflexion angle at terminal stance (deg, 95\%CI [1.092, 3.319]; untrained: 9.2 deg to 12.0 deg, 95\%CI [1.676, 3.573]), hip flexion angular velocity, (trained: −126.5 deg/s to −131.0 deg/s, 95\ untrained: −130.2 deg/s to −135.3 deg/s, 95\%CI [−10.536, −1.675]), ankle angular velocity at terminal stance (trained: −344.7 deg/s to −359.1 deg/s, 95\%C untrained: −340.3 deg/s to −376.9 deg/s, 95\%CI [−37.280, −13.166s]), and gastrocnemius EMG activity (trained: 0.60 to 0.66, 95\%CI [0.014, 0.258]; untra 95\%CI [0.049, 0.214]). These findings demonstrate the efficacy of auditory feedback training in enhancing key gait parameters, highlighting the bilateral be training.
language	en
number	10
urldate	2024-06-24
journal	Sensors
author	Tomita, Yosuke and Sekiguchi, Yoshihiro and Mayo, Nancy E.
month	jan, year = {2024
note	Number: 10 Publisher: Multidisciplinary Digital Publishing Institute
keywords	gait, feedback, kinematics
pages	3206
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\YL2PEP7C\Tomita et al. - 2024 - Efficacy of a Single-Bout of Auditory Feedback Tra.pdf:application/pdf
article	**hardy_sensitivity_2018**
title	Sensitivity of {Speech} {Output} to {Delayed} {Auditory} {Feedback} in {Primary} {Progressive} {Aphasias}
volume	9
issn	1664-2295
url	https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2018.00894/full

doi	10.3389/fneur.2018.00894
abstract	Delayed auditory feedback (DAF) is a classical paradigm for probing sensori-motor interactions in speech output and has been studied in various disorders a dysfluency and aphasia. However, little information is available concerning the effects of DAF on degenerating language networks in primary progressive ap “language-led dementias.” Here we studied two forms of speech output (reading aloud and propositional speech) under natural listening conditions (no feedb DAF at 200 ms, in a cohort of 19 patients representing all major primary progressive aphasia syndromes vs. healthy older individuals and patients with other syndromes (typical Alzheimer's disease and behavioral variant frontotemporal dementia). Healthy controls and most syndromic groups showed a quantitativ similar profile of reduced speech output rate and increased speech error rate under DAF relative to natural auditory feedback. However, there was no group d speech output rate under DAF in patients with nonfluent primary progressive aphasia and logopenic aphasia. Importantly, there was considerable individual sensitivity within syndromic groups and some patients in each group (though no healthy controls) apparently benefited from DAF, showing paradoxically in rate and/or reduced speech error rate under DAF. This work suggests that DAF may be an informative probe of pathophysiological mechanisms underpinnin aphasia: identification of “DAF responders” may open up an avenue to novel therapeutic applications.
language	English
urldate	2024-06-24
journal	Frontiers in Neurology
author	Hardy, Chris J. D. and Bond, Rebecca L. and Jaisin, Kankamol and Marshall, Charles R. and Russell, Lucy L. and Dick, Katrina and Crutch, Sebastian J. an and Warren, Jason D.
month	oct, year = {2018
note	Publisher: Frontiers
keywords	Alzheimer ' s disease, Behavioural variant frontotemporal dementia (bvFTD), delayed auditory feedback, Dementia, Frontotemporal dementia (FTD), Logop progressive aphasia, Progressive nonfluent aphasia, Semantic dementia (SD)
file	Texte intégral:C\:\Users\paul-\Zotero\storage\Z4ASFBB6\Hardy et al. - 2018 - Sensitivity of Speech Output to Delayed Auditory F.pdf:application/pdf
article	**christiner_melody_2021**
title	The {Melody} of {Speech}: {What} the {Melodic} {Perception} of {Speech} {Reveals} about {Language} {Performance} and {Musical} {Abilities}
volume	6
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2226-471X
shorttitle	The {Melody} of {Speech}
url	https://www.mdpi.com/2226-471X/6/3/132
doi	10.3390/languages6030132
abstract	Research has shown that melody not only plays a crucial role in music but also in language acquisition processes. Evidence has been provided that melody h remembering, and memorizing new language material, while relatively little is known about whether individuals who perceive speech as more melodic than the acquisition of oral languages. In this investigation, we wanted to show which impact the subjective melodic perception of speech has on the pronunciatio languages. We tested 86 participants for how melodic they perceived five unfamiliar languages, for their ability to repeat and pronounce the respective five l musical abilities, and for their short-term memory (STM). The results revealed that 59 percent of the variance in the language pronunciation tasks could be e

	predictors: the number of foreign languages spoken, short-term memory capacity, tonal aptitude, melodic singing ability, and how melodic the languages ap participants. Group comparisons showed that individuals who perceived languages as more melodic performed significantly better in all language tasks than However, even though we expected musical measures to be related to the melodic perception of foreign languages, we could only detect some correlations t musical aptitude. Overall, the findings of this investigation add a new dimension to language research, which shows that individuals who perceive natural la melodic than others also retrieve and pronounce utterances more accurately.
language	en
number	3
urldate	2024-06-24
journal	Languages
author	Christiner, Markus and Gross, Christine and Seither-Preisler, Annemarie and Schneider, Peter
month	sep, year = {2021
note	Number: 3 Publisher: Multidisciplinary Digital Publishing Institute
keywords	melodic language perception, melodic perception, melody, music perception, musical abilities, phonetic, singing ability
pages	132
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\KQJ4GTSB\Christiner et al. - 2021 - The Melody of Speech What the Melodic Perception .pdf:application/pd
article	**altenmuller_apollos_2015**
title	Apollo's gift: new aspects of neurologic music therapy
volume	217
issn	1875-7855
shorttitle	Apollo's gift
doi	10.1016/bs.pbr.2014.11.029
abstract	Music listening and music making activities are powerful tools to engage multisensory and motor networks, induce changes within these networks, and foste but functionally related brain regions with continued and life-long musical practice. These multimodal effects of music together with music's ability to tap in reward system in the brain can be used to facilitate and enhance therapeutic approaches geared toward rehabilitating and restoring neurological dysfunctions acquired or congenital brain disorder. In this article, we review plastic changes in functional networks and structural components of the brain in response to music listening and music making activities. The specific influence of music on the developing brain is emphasized and possible transfer effects on emotion processes are discussed. Furthermore, we present data on the potential of using musical tools and activities to support and facilitate neurorehabilitation. We interventions such as melodic intonation therapy and music-supported motor rehabilitation to showcase the effects of neurologic music therapies and discuss mechanisms.
language	eng
journal	Progress in Brain Research
author	Altenmüller, Eckart and Schlaug, Gottfried
year	2015

pmid	25725918
pmcid	PMC4425943
keywords	Humans, Music Therapy, Neuronal Plasticity, Music, neurorehabilitation, neurologic music therapy, Nervous System Diseases, Brain, brain plasticity, melo music-supported training
pages	237--252
file	Version acceptée:C\:\Users\paul-\Zotero\storage\NK6QTPN9\Altenmüller et Schlaug - 2015 - Apollo's gift new aspects of neurologic music the.pdf:applicat
article	**woodruff_carr_beat_2014**
title	Beat synchronization predicts neural speech encoding and reading readiness in preschoolers
volume	111
issn	0027-8424
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4210020/
doi	10.1073/pnas.1406219111
abstract	Sensitivity to fine timing cues in speech is thought to play a key role in language learning, facilitating the development of phonological processing. In fact, a synchronization, which requires fine auditory–motor synchrony, and language skills has been found in school-aged children, as well as adults. Here, we show between beat entrainment and language metrics in preschoolers and use beat synchronization ability to predict the precision of neural encoding of speech syl emergent readers. By establishing links between beat keeping, neural precision, and reading readiness, our results provide an integrated framework that offe preparative biology of reading., Temporal cues are important for discerning word boundaries and syllable segments in speech; their perception facilitates lan development. Beat synchronization and neural encoding of speech reflect precision in processing temporal cues and have been linked to reading skills. In po neural precision may contribute to rhythmic and phonological deficits. Here we establish links between beat synchronization and speech processing in childr begun to read: preschoolers who can entrain to an external beat have more faithful neural encoding of temporal modulations in speech and score higher on te skills. In summary, we propose precise neural encoding of temporal modulations as a key mechanism underlying reading acquisition. Because beat synchron at an early age, these findings may inform strategies for early detection of and intervention for language-based learning disabilities.
number	40
urldate	2024-06-27
journal	Proceedings of the National Academy of Sciences of the United States of America
author	Woodruff Carr, Kali and White-Schwoch, Travis and Tierney, Adam T. and Strait, Dana L. and Kraus, Nina
month	oct, year = {2014
pmid	25246562
pmcid	PMC4210020
pages	14559--14564
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\IBGRJFCX\Woodruff Carr et al. - 2014 - Beat synchronization predicts neural speech encod
article	**palmer_mental_1990**

title	Mental representations for musical meter
volume	16
issn	1939-1277
doi	10.1037/0096-1523.16.4.728
abstract	Investigations of the psychological representation for musical meter provided evidence for an internalized hierarchy from 3 sources: frequency distributions compositions, goodness-of-fit judgments of temporal patterns in metrical contexts, and memory confusions in discrimination judgments. The frequency with occurred in different temporal locations differentiates one meter from another and coincides with music-theoretic predictions of accent placement. Goodness events presented in metrical contexts indicated a multileveled hierarchy of relative accent strength, with finer differentiation among hierarchical levels by mu than inexperienced listeners. Memory confusions of temporal patterns in a discrimination task were characterized by the same hierarchy of inferred accent st suggest mental representations for structural regularities underlying musical meter that influence perceiving, remembering, and composing music. (PsycINF 2016 APA, all rights reserved)
number	4
journal	Journal of Experimental Psychology: Human Perception and Performance
author	Palmer, Caroline and Krumhansl, Carol L.
year	1990
note	Place: US Publisher: American Psychological Association
keywords	Music, Cognitions
pages	728--741
file	Snapshot:C\:\Users\paul-\Zotero\storage\LFQ39X8B\1991-08991-001.html:text/html;Version soumise:C\:\Users\paul-\Zotero\storage\PYH8XDC6\Palmer e Mental representations for musical meter.pdf:application/pdf
article	**moumdjian_effectiveness_2016**
title	Effectiveness of music-based interventions on motricity or cognitive functioning in neurological populations: A systematic review
volume	53
shorttitle	Effectiveness of music-based interventions on motricity or cognitive functioning in neurological populations
doi	10.23736/S1973-9087.16.04429-4
abstract	Introduction: Motor and cognitive symptoms are frequent in persons with neurological disorders and often require extensive long-term rehabilitation. Recent musicbased interventions have been introduced into neurological rehabilitation as training tools. Aim: This review aims to a) describe and define music-base modalities and content which are applied in experimental studies, and b) describe the effects of these interventions on motor and/or cognitive symptoms in th population. EVIDENCE ACQUISITION: The databases PubMed and Web of Science were searched. Cited references of included articles where screened fo systematic literature search up to 20th of June 2016 was conducted to include controlled trials and cohort studies that have used music-based interventions fo neurological population (in- and outpatients) targeting motor and/or cognitive symptoms. No limitations to publication date was set. EVIDENCE SYNTHES comprising thirteen randomised controlled trials (total participants Nexp = 241, Nctrl = 269), four controlled trials (Nexp = 59, Nctrl = 53) and two cohort st included. Fourteen studies were conducted in stroke, three in Parkinson's disease, and two in multiple sclerosis population. Modalities of musicbased interve into four groups: instrument-based, listening-based, rhythm-based, and multicomponent-based music interventions. Overall, studies consistently showed tha

	interventions had similar or larger effects than conventional rehabilitation on upper limb function (N=16; fine motricity, hand and arm capacity, finger and h /variability), mobility (N=7; gait parameters), and cognition (N = 4; verbal memory and focused attention). CONCLUSIONS: Variety of modalities using m has been identified and grouped into four clusters. Effects of interventions demonstrate an improvement in the domains assessed. Evidence is most available motricity in stroke. More studies are warranted to investigate cognition as well as motor and cognition dysfunctions in combination. Clinical implications: In interventions can improve fine motor dexterity and gross motor functions in stroke. Rhythm-based music interventions can improve gait parameters of veloc stroke, Parkinson's disease and multiple sclerosis. Cognition in the domains of verbal memory and focused attention can improve after listening-based music
journal	European journal of physical and rehabilitation medicine
author	Moumdjian, Lousin and Särkämö, Teppo and Leone, Carmela and Leman, Marc and Feys, Peter
month	nov, year = {2016
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\WI8S56T3\Moumdjian et al. - 2016 - Effectiveness of music-based interventions on motr.pdf:application/pdf
article	**schlaug_singing_2008**
title	From {Singing} to {Speaking}: {Why} {Singing} {May} {Lead} to {Recovery} of {Expressive} {Language} {Function} in {Patients} with {Broca}'s {Ap
volume	25
issn	0730-7829
shorttitle	From {Singing} to {Speaking}
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3010734/
doi	10.1525/MP.2008.25.4.315
abstract	It has been reported that patients with severely nonfluent aphasia are better at singing lyrics than speaking the same words. This observation inspired the dev Intonation Therapy (MIT), a treatment whose effects have been shown, but whose efficacy is unproven and neural correlates remain unidentified. Because o /unmask language-capable regions in the unaffected right hemisphere, MIT is particularly well suited for patients with large left-hemisphere lesions. Using t impairments and stroke size/location, we show the effects of MIT and a control intervention. Both interventions' post-treatment outcomes revealed significar propositional speech that generalized to unpracticed words and phrases; however, the MIT-treated patient's gains surpassed those of the control-treated patie associated imaging changes indicate that MIT's unique engagement of the right hemisphere, both through singing and tapping with the left hand to prime the premotor cortices for articulation, accounts for its effect over nonintoned speech therapy.
number	4
urldate	2024-06-27
journal	Music perception
author	Schlaug, Gottfried and Marchina, Sarah and Norton, Andrea
month	apr, year = {2008
pmid	21197418
pmcid	PMC3010734
pages	315--323
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\TR299QLK\Schlaug et al. - 2008 - From Singing to Speaking Why Singing May Lead to .pd

article	hillinger_correlation_2024
title	Correlation of {Self}-{Assessed} {Manual} {Dexterity} and {Actual} {Performance} in {Surgical} {Simulation} – {One} {Year} {Follow} {Up}
volume	111
issn	1365-2168
url	https://doi.org/10.1093/bjs/znae118.054
doi	10.1093/bjs/znae118.054
abstract	Simulation training has proved to be beneficial in many fields of professions including medicine and might be included already during medical school.We ev students rate their manual dexterity as good or above average and if this self-assessment (SA) correlates with the actual performance in a surgical simulation have been performed by medical students in the 3rd and 4th year for a total of 4 rounds on 2 identical Lap-Sim® surgical simulators. Three questionnaires, b simulation as well as follow up after 1 year had to be answered.Descriptive statistics stratified by career goal (surgical/ undecided/ nonsurgical) for outcome parameters are reported. Associations between surgical career goal and SA of manual dexterity with three performance parameters (tissue damage, time and estimated in linear regression models.Out of 102 participating students 41 stated surgery as their career goal, 18 a nonsurgical field, and 43 were undecided. group rated their dexterity as good or above average (86.1\% in the undecided group), 88.9\% of the nonsurgical group as below average or good. A positive tissue damage has been found statistically significant (p=0.033).One year after the simulation (n=97) 13.4\% changed their career goal from undecided to su changed from undecided to non-surgical. 92.7\% wanted to have simulation training integrated into the curriculum.A correlation of self-assessed manual dex the surgical group for tissue damage, but not for time and path length. Simulation training might help students in decision making for a surgical field and the surgeons, which could be shown in the one-year follow-up.
number	Supplement_3
urldate	2024-06-27
journal	British Journal of Surgery
author	Hillinger, S and Brunn, T and Opitz, I and Held, U and Hofer, L and Zellweger, N and Battilana, B and Werner, R
month	may, year = {2024
pages	znae118.054
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\SEA8NLWU\Hillinger et al. - 2024 - Correlation of Self-Assessed Manual Dexterity and .pdf:application /pdf;Snapshot:C\:\Users\paul-\Zotero\storage\6Y3FCLTI\7674287.html:text/html
article	**orysiak_impact_2022**
title	The {Impact} of {Protective} {Gloves} on {Manual} {Dexterity} in {Cold} {Environments}—A {Pilot} {Study}
volume	19
copyright	http://creativecommons.org/licenses/by/3.0/
issn	1660-4601
url	https://www.mdpi.com/1660-4601/19/3/1637
doi	10.3390/ijerph19031637

abstract	Our research aimed to determine the impact of two types of protective gloves. The research tested the glove performance on men exposed to a range of temp working conditions in fruit and vegetable processing. The gloves were assessed for performance within the time required to complete a specific manual task relative to the subjective thermal sensations in the male subjects. Six males participated in a total of 3 study variants: at +5 °C (with double gloves and single double gloves and single glove) and in reference conditions +20 °C (without gloves), in which they performed manual tasks. The measurement of manual ta was used to assess manual dexterity. Subjective thermal sensations were determined. Differences in the time required to complete specific tasks were observ with gloves (both at a temperature of +5 °C and −1 °C), and without gloves ($p < 0.05$). The type of protective gloves had an impact on the time needed to co and therefore may affect manual dexterity.
language	en
number	3
urldate	2024-06-27
journal	International Journal of Environmental Research and Public Health
author	Orysiak, Joanna and Młynarczyk, Magdalena and Irzmańska, Emilia
month	jan, year = {2022
note	Number: 3 Publisher: Multidisciplinary Digital Publishing Institute
keywords	cold environment, hand performance, manual dexterity performance, performance time, protective gloves
pages	1637
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\YHW9X6ST\Orysiak et al. - 2022 - The Impact of Protective Gloves on Manual Dexterit.pdf:application/pdf
article	**everard_extended_2024**
title	Extended reality to assess post-stroke manual dexterity: contrasts between the classic box and block test, immersive virtual reality with controllers, with han reality tests
volume	21
issn	1743-0003
shorttitle	Extended reality to assess post-stroke manual dexterity
url	https://doi.org/10.1186/s12984-024-01332-x
doi	10.1186/s12984-024-01332-x
abstract	Recent technological advancements present promising opportunities to enhance the frequency and objectivity of functional assessments, aligning with recen guidelines. Within this framework, we designed and adapted different manual dexterity tests in extended reality (XR), using immersive virtual reality (VR) v VR-C), immersive VR with hand-tracking (BBT-VR-HT), and mixed-reality (MD-MR).
number	1
urldate	2024-06-27
journal	Journal of NeuroEngineering and Rehabilitation

author	Everard, Gauthier and Burton, Quentin and Van de Sype, Vincent and Bibentyo, Thérèse Ntabuhashe and Auvinet, Edouard and Edwards, Martin Gareth an Sebiyo and Lejeune, Thierry
month	mar, year = {2024
keywords	Stroke, Upper Extremity, Augmented reality, Patient Outcome Assessment, Virtual reality
pages	36
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\36AINI5Z\Everard et al. - 2024 - Extended reality to assess post-stroke manual dext.pdf:application/pdf;Snaps \Zotero\storage\DN5PNI5U\s12984-024-01332-x.html:text/html
article	**gangemi_call_2022**
title	A {Call} to {Evaluate} {Manual} {Dexterity} of {Prospective} {Surgical} {Trainees}
volume	279
issn	1095-8673
doi	10.1016/j.jss.2022.06.025
abstract	INTRODUCTION: This study assesses the correlation between academic grades and gross and fine motor skills in prospective surgical trainees. METHODS Surgery Residency applicants and 32 medical students with prospective surgical interests were recruited. Manual dexterity (MD) was assessed through six ta Dexterity Test and Minnesota Manual Dexterity Test; Peg Transfer Test Fundamentals of Laparoscopic Surgery (box); Ring and Rail, Thread the Ring and S Surgical Simulator). RESULTS: Medical students with higher academic scores had longer completion times for the peg transfer test (P = 0.013). Individuals instruments and perceived themselves to have "Excellent" MD and motor coordination (MC) were more likely to score higher on the Thread the Ring test (P ,respectively). Those who perceived themselves to have "Mediocre" MD and MC performed the worst on the: O'Connor Tweezer Dexterity Test (P = 0.023) Preliminary data suggest that MD ability correlates with neither high United States Medical Licensing Examination scores nor high academic grades; howev playing a musical instrument and high self-ratings of MD/MC may be associated with better test performance.
language	eng
journal	The Journal of Surgical Research
author	Gangemi, Antonio and Shi, Kevin and Durgam, Samarth and Shah, Hemali and Havelka, George and Borhani, Martin and Radhakrishnan, Jayant
month	nov, year = {2022
pmid	35868035
keywords	Humans, Prospective Studies, Clinical Competence, General surgery residency, Internship and Residency, Laparoscopy, Minnesota, Motor skills, Prospectiv Prospective surgical trainees, Selection, Students, Medical, United States
pages	518--525
article	**miendlarzewska_how_2013**
title	How musical training affects cognitive development: rhythm, reward and other modulating variables
volume	7
issn	1662-4548

shorttitle	How musical training affects cognitive development
doi	10.3389/fnins.2013.00279
abstract	Musical training has recently gained additional interest in education as increasing neuroscientific research demonstrates its positive effects on brain develop revealed plastic changes in the brains of adult musicians but it is still unclear to what extent they are the product of intensive music training rather than of ot preexisting biological markers of musicality. In this review, we synthesize a large body of studies demonstrating that benefits of musical training extend bey aims to train and last well into adulthood. For example, children who undergo musical training have better verbal memory, second language pronunciation a and executive functions. Learning to play an instrument as a child may even predict academic performance and IQ in young adulthood. The degree of observ functional adaptation in the brain correlates with intensity and duration of practice. Importantly, the effects on cognitive development depend on the timing to sensitive periods during development, as well as on several other modulating variables. Notably, we point to motivation, reward and social context of mus are important yet neglected factors affecting the long-term benefits of musical training. Further, we introduce the notion of rhythmic entrainment and sugges mechanism supporting learning and development of executive functions. It also hones temporal processing and orienting of attention in time that may underl observed in reading and verbal memory. We conclude that musical training uniquely engenders near and far transfer effects, preparing a foundation for a ran fostering cognitive development.
language	eng
journal	Frontiers in Neuroscience
author	Miendlarzewska, Ewa A. and Trost, Wiebke J.
year	2013
pmid	24672420
pmcid	PMC3957486
keywords	brain plasticity, developmental neuroscience, music education, musical training, rhythmic entrainment
pages	279
file	Texte intégral:C\:\Users\paul-\Zotero\storage\748BSRCX\Miendlarzewska et Trost - 2013 - How musical training affects cognitive development.pdf:applica
article	**begel_dance_2022**
title	Dance {Improves} {Motor}, {Cognitive}, and {Social} {Skills} in {Children} {With} {Developmental} {Cerebellar} {Anomalies}
volume	21
issn	1473-4230
url	https://doi.org/10.1007/s12311-021-01291-2
doi	10.1007/s12311-021-01291-2
abstract	In this multiple single-cases study, we used dance to train sensorimotor synchronization (SMS), motor, and cognitive functions in children with developmen (DCA). DCA are rare dysfunctions of the cerebellum that affect motor and cognitive skills. The cerebellum plays an important role in temporal cognition, in critical for motor and cognitive development. Dancing engages the SMS neuronal circuitry, composed of the cerebellum, the basal ganglia, and the motor co hypothesized that dance has a beneficial effect on SMS skills and associated motor and cognitive functions in children with DCA. Seven children (aged 7–1 participated in a 2-month dance training protocol (3 h/week). A test–retest design protocol with multiple baselines was used to assess children's SMS skills cognitive, and social abilities. SMS skills were impaired in DCA before the training. The training led to improvements in SMS (reduced variability in paced

	executive functioning (cognitive flexibility), as well as in social skills (social cognition). The beneficial effects of the dance training were visible in all partic were maintained 2 months after the intervention. These effects are likely to be sustained by enhanced activity in SMS brain networks due to the dance trainin
language	en
number	2
urldate	2024-06-27
journal	The Cerebellum
author	Bégel, Valentin and Bachrach, Asaf and Dalla Bella, Simone and Laroche, Julien and Clément, Sylvain and Riquet, Audrey and Dellacherie, Delphine
month	apr, year = {2022
keywords	Rehabilitation, Rhythm, Cerebellar anomalies, Dance, Sensorimotor synchronization
pages	264--279
file	Version soumise:C\:\Users\paul-\Zotero\storage\IF9KX55E\Bégel et al. - 2022 - Dance Improves Motor, Cognitive, and Social Skills.pdf:application/pdf
misc	**noauthor_sensors_nodate**
title	Sensors {\textbar} {Free} {Full}-{Text} {\textbar} A {Review}: {High}-{Precision} {Angle} {Measurement} {Technologies}
url	https://www.mdpi.com/1424-8220/24/6/1755
urldate	2024-06-27
file	Sensors | Free Full-Text | A Review\: High-Precision Angle Measurement Technologies:C\:\Users\paul-\Zotero\storage\VD6R6MFY\1755.html:text/html
article	**zhan_precision_2020**
title	Precision measurement physics: physics that precision matters
volume	7
issn	2095-5138
shorttitle	Precision measurement physics
url	https://doi.org/10.1093/nsr/nwaa271
doi	10.1093/nsr/nwaa271
number	12
urldate	2024-06-27
journal	National Science Review
author	Zhan, Mingsheng and Xie, Xincheng
month	dec, year = {2020
pages	1795

file	Full Text PDF:C\:\Users\paul-\Zotero\storage\Y48RF6SZ\Zhan et Xie - 2020 - Precision measurement physics physics that precis.pdf:application/pdf;Snaps \Zotero\storage\23JV5IVD\6017311.html:text/html
article	**ireland_rhythm_2018**
title	Rhythm and {Melody} {Tasks} for {School}-{Aged} {Children} {With} and {Without} {Musical} {Training}: {Age}-{Equivalent} {Scores} and {Reliab
volume	9
issn	1664-1078
shorttitle	Rhythm and {Melody} {Tasks} for {School}-{Aged} {Children} {With} and {Without} {Musical} {Training}
url	https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2018.00426/full
doi	10.3389/fpsyg.2018.00426
abstract	Measuring musical abilities in childhood can be challenging. When music training and maturation occur simultaneously, it is difficult to separate the effects from age-based changes in cognitive and motor abilities. The goal of this study was to develop age-equivalent scores for two measures of musical ability tha with school-aged children (7–13) with and without musical training. The children's Rhythm Synchronization Task (c-RST) and the children's Melody Discri were adapted from adult tasks developed and used in our laboratories. The c-RST is a motor task in which children listen and then try to synchronize their ta woodblock rhythm while it plays twice in a row. The c-MDT is a perceptual task in which the child listens to two melodies and decides if the second was the administered these tasks to 213 children in music camps (musicians, n = 130) and science camps (non-musicians, n = 83). We also measured children's pace tapping, and phonemic discrimination as baseline motor and auditory abilities We estimated internal-consistency reliability for both tasks, and compared chi results from studies with adults. As expected, musically trained children outperformed those without music lessons, scores decreased as difficulty increased, performed the best. Using non-musicians as a reference group, we generated a set of age-based z-scores, and used them to predict task performance with add training. Years of lessons significantly predicted performance on both tasks, over and above the effect of age. We also assessed the relation between musicia tasks, baseline tasks, auditory working memory, and non-verbal reasoning. Unexpectedly, musician children outperformed non-musicians in two of three bas and c-MDT fill an important need for researchers interested in evaluating the impact of musical training in longitudinal studies, those interested in comparin different training methods, and for those assessing the impact of training on non-musical cognitive abilities such as language processing.
language	English
urldate	2024-06-27
journal	Frontiers in Psychology
author	Ireland, Kierla and Parker, Averil and Foster, Nicholas and Penhune, Virginia
month	apr, year = {2018
note	Publisher: Frontiers
keywords	age-equivalent scores, melody discrimination, musical tasks, rhythm synchronization, school-aged children
file	Texte intégral:C\:\Users\paul-\Zotero\storage\HX7TQ5VJ\Ireland et al. - 2018 - Rhythm and Melody Tasks for School-Aged Children W.pdf:application/pd
article	**bood_power_2013-1**
title	The {Power} of {Auditory}-{Motor} {Synchronization} in {Sports}: {Enhancing} {Running} {Performance} by {Coupling} {Cadence} with the {Right}
volume	8

issn	1932-6203
shorttitle	The {Power} of {Auditory}-{Motor} {Synchronization} in {Sports}
url	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0070758
doi	10.1371/journal.pone.0070758
abstract	Acoustic stimuli, like music and metronomes, are often used in sports. Adjusting movement tempo to acoustic stimuli (i.e., auditory-motor synchronization) sports performance. However, music also possesses motivational qualities that may further enhance performance. Our objective was to examine the relative motor synchronization and the motivational impact of acoustic stimuli on running performance. To this end, 19 participants ran to exhaustion on a treadmill without acoustic stimuli, 2) a metronome condition with a sequence of beeps matching participants' cadence (synchronization), and 3) a music condition wit motivational music matched to participants' cadence (synchronization+motivation). Conditions were counterbalanced and measurements were taken on sepa time to exhaustion was significantly longer with acoustic stimuli than without. Unexpectedly, however, time to exhaustion did not differ between metronom music conditions, despite differences in motivational quality. Motivational music slightly reduced perceived exertion of sub-maximal running intensity and l)maximal running intensity. The beat of the stimuli –which was most salient during the metronome condition– helped runners to maintain a consistent pace l the prescribed tempo. Thus, acoustic stimuli may have enhanced running performance because runners worked harder as a result of motivational aspects (mo motivational music) and more efficiently as a result of auditory-motor synchronization (most notable with metronome beeps). These findings imply that run music with a very prominent and consistent beat matched to the runner's cadence will likely yield optimal effects because it helps to elevate physiological ef exertion, whereas the consistent and correct cadence induced by auditory-motor synchronization helps to optimize running economy.
language	en
number	8
urldate	2024-06-27
journal	PLOS ONE
author	Bood, Robert Jan and Nijssen, Marijn and Kamp, John van der and Roerdink, Melvyn
month	aug, year = {2013
note	Publisher: Public Library of Science
keywords	Acoustics, Exercise, Heart rate, Human performance, Music perception, Psychophysics, Running, Sports
pages	e70758
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\RQUMDNS3\Bood et al. - 2013 - The Power of Auditory-Motor Synchronization in Spo.pdf:application/pdf
article	**amitay_motivation_2010**
title	Motivation and {Intelligence} {Drive} {Auditory} {Perceptual} {Learning}
volume	5
issn	1932-6203
url	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0009816
doi	10.1371/journal.pone.0009816

abstract	Background Although feedback on performance is generally thought to promote perceptual learning, the role and necessity of feedback remain unclear. We of providing varying amounts of positive feedback while listeners attempted to discriminate between three identical tones on learning frequency discriminati /Principal Findings Using this novel procedure, the feedback was meaningless and random in relation to the listeners' responses, but the amount of feedback thereof) affected learning. We found that a group of listeners who received positive feedback on 10\% of the trials improved their performance on the task (l groups provided either with excess (90\%) or with no feedback did not learn. Superimposed on these group data, however, individual listeners showed other performance. In particular, those with lower non-verbal IQ who trained in the no feedback condition performed more poorly after training. Conclusions/Sign results cannot be accounted for by learning models that ascribe an external teacher role to feedback. We suggest, instead, that feedback is used to monitor pe in relation to its perceived difficulty, and that listeners who learn without the benefit of feedback are adept at self-monitoring of performance, a trait that also performance on non-verbal IQ tests. These results show that 'perceptual' learning is strongly influenced by top-down processes of motivation and intelligenc
language	en
number	3
urldate	2024-06-27
journal	PLOS ONE
author	Amitay, Sygal and Halliday, Lorna and Taylor, Jenny and Sohoglu, Ediz and Moore, David R.
month	mar, year = {2010
note	Publisher: Public Library of Science
keywords	Learning, Analysis of variance, Control theory, Human learning, Perceptual learning, Signal processing, Statistical data, Teachers
pages	e9816
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\TKSJXU9K\Amitay et al. - 2010 - Motivation and Intelligence Drive Auditory Percept.pdf:application/pdf
article	**frikha_optimizing_2023**
title	Optimizing {Fine} {Motor} {Coordination}, {Selective} {Attention} and {Reaction} {Time} in {Children}: {Effect} of {Combined} {Accuracy} {Exercis {Activities}
volume	10
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2227-9067
shorttitle	Optimizing {Fine} {Motor} {Coordination}, {Selective} {Attention} and {Reaction} {Time} in {Children}
url	https://www.mdpi.com/2227-9067/10/5/786
doi	10.3390/children10050786
abstract	Optimizing fine motor coordination and cognitive abilities in middle childhood through new intervention programs remains one of the most essential orienta development and skills proficiency. The present study aims to identify the effect of a structure-based program intervention, combining motor accuracy exerc activities (ceramics, pottery, and creations using raw materials), on fine motor coordination, selective attention, and reaction time in middle childhood. Sixty developed male schoolchildren (age = 8.29 ± 0.74 years; body height = 130.25 ± 0.05 cm and body mass = 29.83 ± 4.68 kg; mean ± SD) volunteered to parti They were randomly assigned to a combined group intervention (MG) receiving a mixed program integrating visual art activities and motor accuracy exercis intervention (AG) receiving oriented motor accuracy exercises; and a control group (CG). Selective attention, reaction time, and fine motor coordination skil

	throw; manual dexterity: grooved pegboard test) were measured. Repeated measure ANOVA, one-way ANOVA, and Kruskal–Wallis ANOVA were perforn results showed higher changes in MG compared to AG in manual dexterity ($p < 0.001$), in motor accuracy ($p < 0.001$), and in reaction time ($p < 0.01$), but n ($p = 0.379$). In addition, higher changes were recorded in AG compared to CG in motor accuracy ($p < 0.05$), reaction time ($p < 0.05$), and in selective attenti manual dexterity ($p = 0.082$). The structured combined program best assists accuracy, manual dexterity, reaction time, and selective attention. Therefore, ins academies and teachers at schools are advised to use the combined program in the training sessions and in the non-curricular activities, respectively, to impr coordination, selective attention, and reaction time in middle childhood.
language	en
number	5
urldate	2024-06-27
journal	Children
author	Frikha, Mohamed and Alharbi, Raghad Saad
month	may, year = {2023
note	Number: 5 Publisher: Multidisciplinary Digital Publishing Institute
keywords	motor learning, ceramics, middle childhood, motor accuracy, motor coordination, pottery, raw materials, reaction time, selective attention
pages	786
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\XG8FJJ9J\Frikha et Alharbi - 2023 - Optimizing Fine Motor Coordination, Selective Atte.pdf:application/pdf
article	**noordzij_sample_2010**
title	Sample size calculations: basic principles and common pitfalls
volume	25
issn	0931-0509
shorttitle	Sample size calculations
url	https://doi.org/10.1093/ndt/gfp732
doi	10.1093/ndt/gfp732
abstract	One of the most common requests that statisticians get from investigators are sample size calculations or sample size justifications. The sample size is the nu other experimental units included in a study, and determining the sample size required to answer the research question is one of the first steps in designing a statistical textbooks describe techniques for sample size calculation, it is often difficult for investigators to decide which method to use. There are many forn can be applied for different types of data and study designs. However, all of these formulas should be used with caution since they are sensitive to errors, an selected parameters can lead to large differences in the sample size. In this paper, we discuss the basic principles of sample size calculations, the most comm reporting of these calculations.
number	5
urldate	2024-06-27
journal	Nephrology Dialysis Transplantation

author	Noordzij, Marlies and Tripepi, Giovanni and Dekker, Friedo W and Zoccali, Carmine and Tanck, Michael W and Jager, Kitty J
month	may, year = {2010
pages	1388--1393
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\CYVP9D8U\Noordzij et al. - 2010 - Sample size calculations basic principles and com.pdf:application/pdf;Sn \Zotero\storage\STUCHKNX\1842407.html:text/html
article	**mcauley_psychometric_1989**
title	Psychometric properties of the {Intrinsic} {Motivation} {Inventory} in a competitive sport setting: a confirmatory factor analysis
volume	60
issn	0270-1367
shorttitle	Psychometric properties of the {Intrinsic} {Motivation} {Inventory} in a competitive sport setting
doi	10.1080/02701367.1989.10607413
abstract	The present study was designed to assess selected psychometric properties of the Intrinsic Motivation Inventory (IMI) (Ryan, 1982), a multidimensional mea experience with regard to experimental tasks. Subjects (N = 116) competed in a basketball free-throw shooting game, following which they completed the IM computer program was employed to conduct a confirmatory factor analysis to assess the tenability of a five factor hierarchical model representing four first- dimensions and a second-order general factor representing intrinsic motivation. Indices of model acceptability tentatively suggest that the sport data adequat five factor hierarchical model. Alternative models were tested but did not result in significant improvements in the goodness-of-fit indices, suggesting the pr most accurate of the models tested. Coefficient alphas for the four dimensions and the overall scale indicated adequate reliability. The results are discussed v importance of accurate assessment of psychological constructs and the use of linear structural equations in confirming the factor structures of measures.
language	eng
number	1
journal	Research Quarterly for Exercise and Sport
author	McAuley, E. and Duncan, T. and Tammen, V. V.
month	mar, year = {1989
pmid	2489825
keywords	Female, Humans, Male, Adult, Basketball, Chi-Square Distribution, Competitive Behavior, Factor Analysis, Statistical, Motivation, Psychological Tests, Psy
pages	48--58
article	**favennec_changes_2024**
title	Changes in {Human} {Motor} {Behavior} during the {Familiarization} with a {Soft} {Back}-{Support} {Occupational} {Exoskeleton}
volume	14
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2076-3417

url	https://www.mdpi.com/2076-3417/14/3/1160
doi	10.3390/app14031160
abstract	Soft back exoskeletons are aimed at reducing musculoskeletal effort during manual handling tasks, contributing to the prevention of low back disorders like sprains or intervertebral disk problems. However, large differences in their biomechanical effects are observed in the literature. A possible explanation could of familiarization protocols with the exoskeleton. The aim of this experimental study was to characterize the familiarization process with a soft back-support exoskeleton and determine the time needed to stabilize biomechanical variables. Participants carried out 6 familiarization sessions of 1 h to the CORFOR® s Joint kinematics, postural stability, exoskeleton pressure perception, muscle activity, and performance were measured at the beginning of the first session an session during stoop and squat liftings. Results showed that back kinematics, performance, and exoskeleton pressure perception changed during the first sess after sessions 3 or 4, depending on the variable. The authors recommend a familiarization protocol for the CORFOR® soft back-exoskeleton of 4 sessions o recommendation could help CORFOR® users, for instance, in the automotive industry, the food retail industry, or the agriculture field.
language	en
number	3
urldate	2024-06-27
journal	Applied Sciences
author	Favennec, Arthur and Frère, Julien and Mornieux, Guillaume
month	jan, year = {2024
note	Number: 3 Publisher: Multidisciplinary Digital Publishing Institute
keywords	motor learning, biomechanics, exoskeleton, motion analysis, motor control
pages	1160
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\256WVPSN\Favennec et al. - 2024 - Changes in Human Motor Behavior during the Familia.pdf:application/p
incollection	**pinquart_experimental_2021**
address	Cham
title	Experimental {Studies} and {Observational} {Studies}
isbn	978-3-030-22009-9
url	https://doi.org/10.1007/978-3-030-22009-9_573
language	en
urldate	2024-06-27
booktitle	Encyclopedia of {Gerontology} and {Population} {Aging}
publisher	Springer International Publishing
author	Pinquart, Martin
editor	Gu, Danan and Dupre, Matthew E.

year	2021
doi	10.1007/978-3-030-22009-9_573
pages	1748--1756
phdthesis	**locqueville_voks_2022**
type	phdthesis
title	Voks : A {Vocal} {Instrument} {Family} {Based} on {Syllabic} {Sequencing} of {Vocal} {Samples}
shorttitle	Voks
url	https://theses.hal.science/tel-03813793
abstract	The study of Voks is presented here. Voks is a family of vocal instruments that allows for control, using one's hands, of intonation, rhythmic sequencing, an parameters of an articulated voice. This dissertation focuses more specifically on the question of rhythmic control, which is here based on the frame/content production. It relies on matching impulsions produced by a biphasic tapping gesture, with control points placed on a prerecorded voice sample. This mode o complex loop that involves both gestural production and auditory perception of rhythm, which can be modeled by the notion of the P-center, or perceptual c control of melody and vocal effort, also known as chironomy, is performed using continuous controllers such as the graphic tablet and the theremin, enablin gestures. For synthesis, two distinct vocoders, World and SuperVP, are integrated into Voks and compared. They make it possible to control melody, rythm takes as input voice samples equipped with syllabic labels; those data can be either directly fed as input or generated automatically on the fly. The Voks inst applied to music and to foreign language acquisition.
language	en
urldate	2024-06-27
school	Sorbonne Université
author	Locqueville, Grégoire
month	jun, year = {2022
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\DR7EVLZH\Locqueville - 2022 - Voks A Vocal Instrument Family Based on Syllabic.pdf:application/pdf
article	**locqueville_voks_2020**
title	Voks: {Digital} instruments for chironomic control of voice samples
volume	125
issn	0167-6393
shorttitle	Voks
url	https://www.sciencedirect.com/science/article/pii/S0167639320302788
doi	10.1016/j.specom.2020.10.002

abstract	This paper presents Voks, a new family of digital instruments that allow for real-time control and modification of pre-recorded voice signal samples. An inst is made of Voks itself, the synthesis software and a given set of chironomic (hand-driven) interfaces. Rhythm can be accurately controlled thanks to a new m syllabic control points. Timing can also be controlled with other methods, including scrubbing and playback speed variation. Pitch, vocal effort, voice tensio size, voicing ratio, aperiodicity ratio of the voice samples can be modified thanks to a real-time high-quality vocoder. Different forms of chironomic control are proposed. Pitch is controlled by continuous hand motions using a stylus on a surface (C-Voks) or a theremin (T-Voks). Other interfaces can be used as w controlled using a biphasic button. Scrubbing, playback speed and timbre related parameters can be controlled using the theremin, control surfaces or contin faders. In addition to realistic imitation of speaking or singing voices, other playing modes yield new interesting sounds. Voks participated in comparative p singing synthesis systems. It has been demonstrated in a live musical settings, using different control interfaces. In addition to musical or poetic performance performative vocal synthesis to language learning and speech reeducation are foreseen.
urldate	2024-06-27
journal	Speech Communication
author	Locqueville, Grégoire and d'Alessandro, Christophe and Delalez, Samuel and Doval, Boris and Xiao, Xiao
month	dec, year = {2020
keywords	New interfaces for musical expression, Performative synthesis, Real-time vocoder, Singing synthesis, Voice synthesis
pages	97--113
file	ScienceDirect Snapshot:C\:\Users\paul-\Zotero\storage\X959WHIW\S0167639320302788.html:text/html;Version soumise:C\:\Users\paul-\Zotero\storage\K et al. - 2020 - Voks Digital instruments for chironomic control o.pdf:application/pdf
article	**conde_task-irrelevant_2012**
title	Task-{Irrelevant} {Auditory} {Feedback} {Facilitates} {Motor} {Performance} in {Musicians}
volume	3
issn	1664-1078
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3353260/
doi	10.3389/fpsyg.2012.00146
abstract	An efficient and fast auditory–motor network is a basic resource for trained musicians due to the importance of motor anticipation of sound production in mu When playing an instrument, motor performance always goes along with the production of sounds and the integration between both modalities plays an esse of musical training. The aim of the present study was to investigate the role of task-irrelevant auditory feedback during motor performance in musicians usin task (SRTT). Our hypothesis was that musicians, due to their extensive auditory–motor practice routine during musical training, have superior performance when receiving auditory feedback during SRTT relative to musicians performing the SRTT without any auditory feedback. Behaviorally, we found that audi reinforced SRTT performance of the right hand (referring to absolute response speed) while learning capabilities remained unchanged. This finding highligh role for task-irrelevant auditory feedback in motor performance in musicians, a finding that might provide further insight into auditory–motor integration inc musical context.
urldate	2024-06-27
journal	Frontiers in Psychology
author	Conde, Virginia and Altenmüller, Eckart and Villringer, Arno and Ragert, Patrick

month	may, year = {2012
pmid	22623920
pmcid	PMC3353260
pages	146
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\95AET3TV\Conde et al. - 2012 - Task-Irrelevant Auditory Feedback Facilitates Moto.pdf:ap
article	**el_shemy_impact_2018**
title	The impact of auditory rhythmic cueing on gross motor skills in children with autism
volume	30
issn	0915-5287
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6110221/
doi	10.1589/jpts.30.1063
abstract	[Purpose] This study aimed to investigate the effect of auditory rhythmic cueing on gross motor skills in children with autism. [Participants and Methods] A children aged 8–10 years with mild to moderate autistic features participated in this study. They were randomly allocated to either the control group (n=15), specially designed physical therapy program, or the study group (n=15), which underwent the same program in addition to gait training with rhythmic audito provide rhythmic auditory stimulation, combination of a metronome beat set to the child's cadence and rhythmic cueing from the MIDI Cuebase musical pro groups received 3 sessions per week for 3 months. The Bruininks-Oseretsky Test of Motor Proficiency 2nd Edition was used to assess gross motor skills at b months of intervention. [Results] The study found statistically significant improvements in bilateral coordination, balance, running speed and agility, and str after treatment. Moreover, there were statistically significant differences between the 2 groups, with the study group showing better improvement in all outc [Conclusion] Gait training with auditory rhythmic cueing elicited a positive effect on the gross motor skills of children with autism.
number	8
urldate	2024-06-27
journal	Journal of Physical Therapy Science
author	El Shemy, Samah Attia and El-Sayed, Mohamed Salah
month	aug, year = {2018
pmid	30154601
pmcid	PMC6110221
pages	1063--1068
file	PubMed Central Full Text PDF:C\:\Users\paul-\Zotero\storage\N8CS6X3I\El Shemy et El-Sayed - 2018 - The impact of auditory rhythmic cueing on gross
article	**hofbauer_background_2024**
title	Background music varying in tempo and emotional valence differentially affects cognitive task performance: experimental within-participant comparison
issn	2520-1018

shorttitle	Background music varying in tempo and emotional valence differentially affects cognitive task performance
url	https://doi.org/10.1007/s41809-024-00144-8
doi	10.1007/s41809-024-00144-8
abstract	The degree of stimulation provided by background music appears to affect performance on cognitive tasks. Moreover, individual differences influence what beneficial or detrimental. In a within-subject design, 40 participants (Mean Age: 26.15, SD: 2.99) completed cognitive tasks (immediate and delayed recall, making) under varying (2 tempi: fast/slow×2 valences: positive/negative) musical background conditions. Further, they completed questionnaires on individ (extraversion, noise sensitivity, annoyance/distraction by background noise). Performance was assessed using analyses of variance and mixed-effect models adjusted for stimulus liking and further individual characteristics. Fast (vs. slow) tempo was associated with better immediate recall (p = .002, η2 = .08) and
language	en
urldate	2024-06-27
journal	Journal of Cultural Cognitive Science
author	Hofbauer, Lena Maria and Lachmannn, Thomas and Rodriguez, Francisca Saveria
month	jun, year = {2024
keywords	Arousal, Background music, Cognitive load, Cognitive performance
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\RBG8GAFC\Hofbauer et al. - 2024 - Background music varying in tempo and emotional va.pdf:application/p
misc	**cuciniello_identifying_2022**
title	Identifying synthetic voices qualities for conversational agents
url	http://arxiv.org/abs/2205.04149
doi	10.48550/arXiv.2205.04149
abstract	The present study aims to explore user acceptance and perceptions toward different quality levels of synthetical voices. To achieve this, four voices have bee two main factors: the quality of the voices (low vs high) and their gender (male and female). 186 volunteers were recruited and subsequently allocated into f ages respec-tively, adolescents, young adults, middle-aged and seniors. After having randomly listened to each voice, participants were asked to fill the Virt Acceptance Questionnaire (VAVAQ). Outcomes show that the two higher quality voices of Antonio and Giulia were more appreciated than the low-quality Clara by the whole sample in terms of pragmatic, hedonic and attractiveness qualities attributed to the voices. Concerning preferences towards differently ag appeared that they varied according to participants age' ranges examined. Furthermore, in terms of suitability to perform different tasks, participants conside equally adapt for healthcare and front office jobs. Antonio was also judged to be significantly more qualified to accomplish protection and security tasks, wh classified as the absolute least skilled in conducting household chores.
urldate	2024-06-27
publisher	arXiv
author	Cuciniello, M. and Amorese, T. and Cordasco, G. and Marrone, S. and Marulli, F. and Cavallo, F. and Gordeeva, O. and Carrión, Z. Callejas and Esposito, A
month	may, year = {2022
note	arXiv:2205.04149 [cs]
keywords	Computer Science - Human-Computer Interaction

file	arXiv Fulltext PDF:C\:\Users\paul-\Zotero\storage\KE5Q8IQ4\Cuciniello et al. - 2022 - Identifying synthetic voices qualities for convers.pdf:application/pd Snapshot:C\:\Users\paul-\Zotero\storage\AH32ZMEN\2205.html:text/html
inproceedings	**curtin_cognitive_2017**
address	Cham
title	Cognitive {Considerations} in {Auditory} {User} {Interfaces}: {Neuroergonomic} {Evaluation} of {Synthetic} {Speech} {Comprehension}
isbn	978-3-319-58472-0
shorttitle	Cognitive {Considerations} in {Auditory} {User} {Interfaces}
doi	10.1007/978-3-319-58472-0_9
abstract	Automated spoken language interfaces have seen a remarkable proliferation in recent years, integrating with automotive, household, industrial, and mobile p way in which we interact with our devices. While the use of an auxiliary auditory information stream has the potential to decrease interference and prevent d operation of traditional visual/mechanical interfaces, evidence from behavioral and neuroimaging studies have suggested that the brain mechanisms underlyi comprehension of synthetic speech may be different from naturally produced speech, resulting in an unnecessary additional cognitive burden.
language	en
booktitle	Engineering {Psychology} and {Cognitive} {Ergonomics}: {Performance}, {Emotion} and {Situation} {Awareness}
publisher	Springer International Publishing
author	Curtin, Adrian and Ayaz, Hasan
editor	Harris, Don
year	2017
keywords	Auditory, fNIRS, Prefrontal cortex, Synthetic speech perception
pages	106--116
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\23LM7C4C\Curtin et Ayaz - 2017 - Cognitive Considerations in Auditory User Interfac.pdf:application/pdf
article	**kim_effects_2021**
title	Effects of {User} {Interfaces} on {Take}-{Over} {Performance}: A {Review} of the {Empirical} {Evidence}
volume	12
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2078-2489
shorttitle	Effects of {User} {Interfaces} on {Take}-{Over} {Performance}
url	https://www.mdpi.com/2078-2489/12/4/162
doi	10.3390/info12040162

abstract	In automated driving, the user interface plays an essential role in guiding transitions between automated and manual driving. This literature review identifie explicitly studied the effectiveness of user interfaces in automated driving. Our main selection criterion was how the user interface (UI) affected take-over p automation levels allowing drivers to take their eyes off the road (SAE3 and SAE4). We categorized user interface (UI) factors from an automated vehicle-r perspective. Short take-over times are consistently associated with take-over requests (TORs) initiated by the auditory modality with high urgency levels. O over requests directly displayed on non-driving-related task devices and augmented reality do not affect take-over time. Additional explanations of take-over and vehicle information while driving, and take-over guiding information were found to improve situational awareness. Hence, we conclude that advanced u enhance the safety and acceptance of automated driving. Most studies showed positive effects of advanced UI, but a number of studies showed no significan studies showed negative effects of advanced UI, which may be associated with information overload. The occurrence of positive and negative results of simi different studies highlights the need for systematic UI testing across driving conditions and driver characteristics. Our findings propose future UI studies of a focusing on trust calibration and enhancing situation awareness in various scenarios.
language	en
number	4
urldate	2024-06-27
journal	Information
author	Kim, Soyeon and van Egmond, René and Happee, Riender
month	apr, year = {2021
note	Number: 4 Publisher: Multidisciplinary Digital Publishing Institute
keywords	automated vehicles, human–machine interaction, take-over request, user interface
pages	162
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\Y6GWSD2D\Kim et al. - 2021 - Effects of User Interfaces on Take-Over Performanc.pdf:application/pdf
article	**guo_empowering_2023**
title	Empowering {Hand} {Rehabilitation} with {AI}-{Powered} {Gesture} {Recognition}: A {Study} of an {sEMG}-{Based} {System}
volume	10
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2306-5354
shorttitle	Empowering {Hand} {Rehabilitation} with {AI}-{Powered} {Gesture} {Recognition}
url	https://www.mdpi.com/2306-5354/10/5/557
doi	10.3390/bioengineering10050557
abstract	Stroke is one of the most prevalent health issues that people face today, causing long-term complications such as paresis, hemiparesis, and aphasia. These co impact a patient’s physical abilities and cause financial and social hardships. In order to address these challenges, this paper presents a groundbreaking solut rehabilitation glove. This motorized glove is designed to provide comfortable and effective rehabilitation for patients with paresis. Its unique soft materials a it easy to use in clinical settings and at home. The glove can train each finger individually and all fingers together, using assistive force generated by advanc actuators controlled by sEMG signals. The glove is also durable and long-lasting, with 4–5 h of battery life. The wearable motorized glove is worn on the af assistive force during rehabilitation training. The key to this glove’s effectiveness is its ability to perform the classified hand gestures acquired from the non-

	integrating four sEMG sensors and a deep learning algorithm (the 1D-CNN algorithm and the InceptionTime algorithm). The InceptionTime algorithm class sEMG signals with an accuracy of 91.60\% and 90.09\% in the training and verification sets, respectively. The overall accuracy was 90.89\%. It showed pote developing effective hand gesture recognition systems. The classified hand gestures can be used as a control command for the motorized wearable glove pla hand, allowing it to mimic the movements of the non-affected hand. This innovative technology performs rehabilitation exercises based on the theory of mir oriented therapy. Overall, this wearable rehabilitation glove represents a significant step forward in stroke rehabilitation, offering a practical and effective so recover from stroke's physical, financial, and social impact.
language	en
number	5
urldate	2024-07-04
journal	Bioengineering
author	Guo, Kai and Orban, Mostafa and Lu, Jingxin and Al-Quraishi, Maged S. and Yang, Hongbo and Elsamanty, Mahmoud
month	may, year = {2023
note	Number: 5 Publisher: Multidisciplinary Digital Publishing Institute
keywords	flexible robot, hand rehabilitation, rope-driven, sensory stimulation, wearable device
pages	557
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\69DGGZB7\Guo et al. - 2023 - Empowering Hand Rehabilitation with AI-Powered Ges.pdf:application/pdf
article	**bonilla_progressive_2023**
title	Progressive {Rehabilitation} {Based} on {EMG} {Gesture} {Classification} and an {MPC}-{Driven} {Exoskeleton}
volume	10
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2306-5354
url	https://www.mdpi.com/2306-5354/10/7/770
doi	10.3390/bioengineering10070770
abstract	Stroke is a leading cause of disability and death worldwide, with a prevalence of 200 millions of cases worldwide. Motor disability is presented in 80\% of p physical rehabilitation plays a fundamental role for gradually recovery of mobility. In this work, we designed a robotic hand exoskeleton to support rehabilit stroke episode. The system acquires electromyographic (EMG) signals in the forearm, and automatically estimates the movement intention for five gestures. developed a predictive adaptive control of the exoskeleton to compensate for three different levels of muscle fatigue during the rehabilitation therapy exercis system could be used to assist the rehabilitation therapy of the patients by providing a repetitive, intense, and adaptive assistance.
language	en
number	7
urldate	2024-07-04
journal	Bioengineering

author	Bonilla, Daniel and Bravo, Manuela and Bonilla, Stephany P. and Iragorri, Angela M. and Mendez, Diego and Mondragon, Ivan F. and Alvarado-Rojas, Cat Julian D.
month	jul, year = {2023
note	Number: 7 Publisher: Multidisciplinary Digital Publishing Institute
keywords	rehabilitation, stroke, electromyography (EMG), exoskeleton, gesture classification, model predictive control (MPC)
pages	770
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\IMS2KKFD\Bonilla et al. - 2023 - Progressive Rehabilitation Based on EMG Gesture Cl.pdf:application/pdf
incollection	**nayak_application_2020**
title	Application of {Artificial} {Intelligence} ({AI}) in {Prosthetic} and {Orthotic} {Rehabilitation}
isbn	978-1-78984-429-0
abstract	Technological integration of Artificial Intelligence (AI) and machine learning in the Prosthetic and Orthotic industry and in the field of assistive technology the Persons with Disabilities. The concept of neural network has been used by the leading manufacturers of rehabilitation aids for simulating various anatom functions of the lost parts of the human body. The involvement of human interaction with various agents' i.e. electronic circuitry, software, robotics, etc. has impact in the rehabilitation field to develop devices like Bionic leg, mind or thought control prosthesis and exoskeletons. Application of Artificial Intelligen technology has a huge impact in achieving independent mobility and enhances the quality of life in Persons with Disabilities (PwDs).
author	Nayak, Smita and Das, Rajesh
month	oct, year = {2020
doi	10.5772/intechopen.93903
pages	1--23
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\3SSNLRWA\Nayak et Das - 2020 - Application of Artificial Intelligence (AI) in Pro.pdf:application/pdf
article	**mokri_muscle_2022**
title	Muscle force estimation from lower limb {EMG} signals using novel optimised machine learning techniques
volume	60
issn	1741-0444
doi	10.1007/s11517-021-02466-z
abstract	The main objective of this work is to establish a framework for processing and evaluating the lower limb electromyography (EMG) signals ready to be fed t We design and build a knee rehabilitation robot that works with surface EMG (sEMG) signals. In our device, the muscle forces are estimated from sEMG si machine learning techniques, i.e. support vector machine (SVM), support vector regression (SVR) and random forest (RF). In order to improve the estimatic genetic algorithm (GA) for parameter optimisation and feature extraction within the proposed methods. At the same time, a load cell and a wearable inertial (IMU) are mounted on the robot to measure the muscle force and knee joint angle, respectively. Various performance measures have been employed to asse the proposed system. Our extensive experiments and comparison with related works revealed a high estimation accuracy of 98.67\% for lower limb muscles the proposed techniques is high estimation accuracy leading to improved performance of the therapy while muscle models become especially sensitive to the the slack length. Graphical Abstract.

language	eng
number	3
journal	Medical \& Biological Engineering \& Computing
author	Mokri, Chiako and Bamdad, Mahdi and Abolghasemi, Vahid
month	mar, year = {2022
pmid	35029815
pmcid	PMC8854337
keywords	Electromyography, Humans, Algorithms, Genetic algorithm, Lower Extremity, Machine Learning, Muscles, Random forest, Rehabilitation robotics, Suppor Support Vector Machine, Support vector regression
pages	683--699
file	Texte intégral:C\:\Users\paul-\Zotero\storage\KLZFSPMR\Mokri et al. - 2022 - Muscle force estimation from lower limb EMG signal.pdf:application/pdf
article	**douglass-kirk_real-time_2023**
title	Real-time auditory feedback may reduce abnormal movements in patients with chronic stroke
volume	45
issn	1464-5165
doi	10.1080/09638288.2022.2037751
abstract	PURPOSE: The current pilot study assesses the use of real-time auditory feedback to help reduce abnormal movements during an active reaching task in pat stroke. MATERIALS AND METHODS: 20 patients with chronic stroke completed the study with full datasets (age: M = 53 SD = 14; sex: male = 75\%; tim months: M = 34, SD = 33). Patients undertook 100 repetitions of an active reaching task while listening to self-selected music which automatically muted wh was detected, determined by thresholds set by clinical therapists. A within-subject design with two conditions (with auditory feedback vs. without auditory f randomised counterbalanced order was used. The dependent variable was the duration of abnormal movement as a proportion of trial duration. RESULTS: A in the duration of abnormal movement was observed when patients received auditory feedback, F(1,18) = 9.424, p = 0.007, with a large effect size (partial η2 CONCLUSIONS: Patients with chronic stroke can make use of real-time auditory feedback to increase the proportion of time they spend in optimal moveme approach provides a motivating framework that encourages high dose with a key focus on quality of movement. Trial Registration: ISRCTN12969079 https: /ISRCTN12969079 ISRTCN trial registration REF: ISRCTN12969079IMPLICATIONS FOR REHABILITATIONMovement quality during upper limb reh targeted as part of a well-balanced rehabilitation programme.Auditory feedback is a useful tool to help patients with chronic stroke reduce compensatory mo
language	eng
number	4
journal	Disability and Rehabilitation
author	Douglass-Kirk, Pedro and Grierson, Mick and Ward, Nick S. and Brander, Fran and Kelly, Kate and Chegwidden, Will and Shivji, Dhiren and Stewart, Laur
month	feb, year = {2023
pmid	35238694

keywords	Stroke, Rehabilitation, movement, Female, Humans, Male, Middle Aged, Stroke Rehabilitation, Adult, Aged, Movement, Pilot Projects, Feedback, kinemati Dyskinesias, machine learning
pages	613--619
file	Texte intégral:C\:\Users\paul-\Zotero\storage\7MEXLVSC\Douglass-Kirk et al. - 2023 - Real-time auditory feedback may reduce abnormal mo.pdf:applicat
article	**urbina_integration_2024**
title	Integration of music-based game approaches with wearable devices for hand neurorehabilitation: a narrative review
volume	21
issn	1743-0003
shorttitle	Integration of music-based game approaches with wearable devices for hand neurorehabilitation
url	https://doi.org/10.1186/s12984-024-01379-w
doi	10.1186/s12984-024-01379-w
abstract	Restoring hand functionality is critical for fostering independence in individuals with neurological disorders. Various therapeutic approaches have emerged t function restoration, with music-based therapies demonstrating notable advantages in enhancing neuroplasticity, an integral component of neurorehabilitatio effects observed, there remains a gap in the literature regarding implementing music treatments in neurorehabilitation, such as Neurologic Music Therapy (N conjunction with emerging fields like wearable devices and game-based therapies.
number	1
urldate	2024-07-30
journal	Journal of NeuroEngineering and Rehabilitation
author	Urbina, Javier and Abarca, Victoria E. and Elias, Dante A.
month	may, year = {2024
keywords	Rehabilitation, Upper limb, Music, Game
pages	89
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\LVHZ3WFH\Urbina et al. - 2024 - Integration of music-based game approaches with we.pdf:application /pdf;Snapshot:C\:\Users\paul-\Zotero\storage\784WWN4R\s12984-024-01379-w.html:text/html
article	**bella_time_2003**
title	Time course of melody recognition: a gating paradigm study
volume	65
issn	0031-5117
shorttitle	Time course of melody recognition
doi	10.3758/bf03194831

abstract	Recognizing a well-known melody (e.g., one's national anthem) is not an all-or-none process. Instead, recognition develops progressively while the melody examine which factors govern the time course of this recognition process, the gating paradigm, initially designed to study auditory word recognition, was ad Musicians and nonmusicians were presented with segments of increasing duration of familiar and unfamiliar melodies (i.e., the first note, then the first two n notes, and so forth). Recognition was assessed after each segment either by requiring participants to provide a familiarity judgment (Experiment 1) or by ask melody that they thought had been presented (Experiment 2). In general, the more familiar the melody, the fewer the notes required for recognition. Musicia familiarity within fewer notes than did nonmusicians, whereas the reverse situation (i.e., musicians were slower than nonmusicians) occurred when a sung re However, both musicians and nonmusicians appeared to segment melodies into the same perceptual units (i.e., motives) in order to access the correct represe These results are interpreted in light of the cohort model (Marslen-Wilson, 1987), as applied to the music domain.
language	eng
number	7
journal	Perception \& Psychophysics
author	Bella, Simone Dalla and Peretz, Isabelle and Aronoff, Neil
month	oct, year = {2003
pmid	14674630
keywords	Female, Humans, Male, Adult, Music, Auditory Perception, Acoustic Stimulation, Attention, Mental Recall, Practice, Psychological, Psychoacoustics, Reac Psychology
pages	1019--1028
file	Texte intégral:C\:\Users\paul-\Zotero\storage\RKW4M77E\Bella et al. - 2003 - Time course of melody recognition a gating paradi.pdf:application/pdf
article	**tacca_wearable_2024**
title	Wearable high-density {EMG} sleeve for complex hand gesture classification and continuous joint angle estimation
volume	14
copyright	2024 The Author(s)
issn	2045-2322
url	https://www.nature.com/articles/s41598-024-64458-x
doi	10.1038/s41598-024-64458-x
abstract	High-density electromyography (HD-EMG) can provide a natural interface to enhance human–computer interaction (HCI). This study aims to demonstrate t HD-EMG forearm sleeve equipped with up to 150 electrodes to capture high-resolution muscle activity, decode complex hand gestures, and estimate contin joint angle predictions. Ten able-bodied participants performed 37 hand movements and grasps while EMG was recorded using the HD-EMG sleeve. Simult motion capture glove calculated 23 joint angles from the hand and fingers across all movements for training regression models. For classifying across the 37 algorithm was able to differentiate between sequential movements with \$\$97.3 \pm 0.3\%\$\$accuracy calculated on a 100 ms bin-by-bin basis. In a separa consisting of 19 movements randomly interspersed, decoding performance achieved an average bin-wise accuracy of \$\$92.8 \pm 0.8\%\$\$. When evaluati real-time scenarios, we found that decoders can reliably decode both movements and movement transitions, achieving an average accuracy of \$\$93.3 \pm (sequential set and \$\$88.5 \pm 0.9\%\$\$on the mixed set. Furthermore, we estimated continuous joint angles from the EMG sleeve data, achieving a \$\$R{\textasciicircum}2\$\$of \$\$0.884 \pm 0.003\$\$in the sequential set and \$\$0.750 \pm 0.008\$\$in the mixed set. Median absolute error (MAE) was all joints, with a grand average MAE of \$\$1.8 \pm 0.04{\textasciicircum}\circ\$\$and \$\$3.4 \pm 0.07{\textasciicircum}\circ\$\$for the sequential and mi

	respectively. We also assessed two algorithm modifications to address specific challenges for EMG-driven HCI applications. To minimize decoder latency, accounts for reaction time by dynamically shifting cue labels in time. To reduce training requirements, we show that pretraining models with historical data decoding performance compared with models that were not pretrained when reducing the in-session training data to only one attempt of each movement. Th combined with sophisticated machine learning algorithms, can be a powerful tool for hand gesture recognition and joint angle estimation. This technology h for applications in HCI, such as prosthetics, assistive technology, rehabilitation, and human–robot collaboration.
language	en
number	1
urldate	2024-10-28
journal	Scientific Reports
author	Tacca, Nicholas and Dunlap, Collin and Donegan, Sean P. and Hardin, James O. and Meyers, Eric and Darrow, Michael J. and Colachis IV, Samuel and Gil Friedenberg, David A.
month	aug, year = {2024
note	Publisher: Nature Publishing Group
keywords	Biomedical engineering, Computer science
pages	18564
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\TBFDQPHN\Tacca et al. - 2024 - Wearable high-density EMG sleeve for complex hand .pdf:application/pdf
article	**awad_efficacy_2024**
title	Efficacy and safety of using auditory-motor entrainment to improve walking after stroke: a multi-site randomized controlled trial of {InTandemTM}
volume	15
copyright	2024 The Author(s)
issn	2041-1723
shorttitle	Efficacy and safety of using auditory-motor entrainment to improve walking after stroke
url	https://www.nature.com/articles/s41467-024-44791-5
doi	10.1038/s41467-024-44791-5
abstract	Walking slowly after stroke reduces health and quality of life. This multi-site, prospective, interventional, 2-arm randomized controlled trial (NCT04121754 and efficacy of an autonomous neurorehabilitation system (InTandemTM) designed to use auditory-motor entrainment to improve post-stroke walking. 87 i randomized to 5-week walking interventions with InTandem or Active Control (i.e., walking without InTandem). The primary endpoints were change in wal by the 10-meter walk test pre-vs-post each 5-week intervention, and safety, measured as the frequency of adverse events (AEs). Clinical responder rates wer trial met its primary endpoints. InTandem was associated with a 2x larger increase in speed (Δ: 0.14 ± 0.03 m/s versus Δ: 0.06 ± 0.02 m/s, F(1,49) = 6.58, p = responders (40\% versus 13\%, χ2(1) ≥ 6.47, p = 0.01), and similar safety (both groups experienced the same number of AEs). The auditory-motor interventi delivered by InTandem is safe and effective in improving walking in the chronic phase of stroke.
language	en
number	1

urldate	2025-02-05
journal	Nature Communications
author	Awad, Louis N. and Jayaraman, Arun and Nolan, Karen J. and Lewek, Michael D. and Bonato, Paolo and Newman, Mark and Putrino, David and Raghavan Ryan T. and Harris, Brian A. and Parker, Danielle A. and Taylor, Sabrina R.
month	feb, year = {2024
note	Publisher: Nature Publishing Group
keywords	Stroke, Translational research
pages	1081
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\IATTURRI\Awad et al. - 2024 - Efficacy and safety of using auditory-motor entrai.pdf:application/pdf
article	**altenmuller_neural_2009-1**
title	Neural reorganization underlies improvement in stroke-induced motor dysfunction by music-supported therapy
volume	1169
issn	1749-6632
doi	10.1111/j.1749-6632.2009.04580.x
abstract	Motor impairments are common after stroke, but efficacious therapies for these dysfunctions are scarce. By extending an earlier study on the effects of musi behavioral indices of motor function as well as electrophysiological measures were obtained before and after a series of therapy sessions to assess whether th to neural reorganization and motor recovery in patients after stroke. The study group comprised 32 stroke patients in a large rehabilitation hospital; they had motor function and no previous musical experience. Over a period of 3 weeks, these patients received 15 sessions of music-supported therapy using a manua approach. For comparison 30 additional patients received standard rehabilitation procedures. Fine as well as gross motor skills were trained by using either a electronic drum pads programmed to emit piano tones. Motor functions were assessed by an extensive test battery. In addition, we studied event-related desy /synchronization and coherences from all 62 patients performing self-paced movements of the index finger (MIDI-piano) and of the whole arm (drum pads). music-supported therapy yielded significant improvement in fine as well as gross motor skills with respect to speed, precision, and smoothness of movement data showed a more pronounced event-related desynchronization before movement onset and a more pronounced coherence in the music-supported therapy training assessment, whereas almost no differences were observed in the control group. Thus we see that music-supported therapy leads to marked improven after stroke and that these are accompanied by electrophysiological changes indicative of a better cortical connectivity and improved activation of the motor
language	eng
journal	Annals of the New York Academy of Sciences
author	Altenmüller, E. and Marco-Pallares, J. and Münte, T. F. and Schneider, S.
month	jul, year = {2009
pmid	19673814
keywords	Stroke, Female, Humans, Male, Middle Aged, Music Therapy, Adult, Aged, Neuronal Plasticity, Psychomotor Performance, Rehabilitation Centers
pages	395--405
article	**thaut_connection_1999**

title	The connection between rhythmicity and brain function
volume	18
issn	0739-5175
doi	10.1109/51.752991
abstract	Although rhythm and music are not entirely synonymous terms, rhythm constitutes one of the most essential structural and organizational elements of music effect of music on human adaptation, the profound effect of rhythm on the motor system strongly suggests that the time structure of music is the essential ele specifically to motor behavior. Why the motor system appears so sensitive to auditory priming and timing stimulation can only be partially answered so far. function of the auditory system regarding processing of time information makes good functional sense within the constraints of auditory sensory processing. sensitivity to auditory entrainment may simply be an evolutionary useful function of taking advantage of the specific and unique aspects of auditory informa enhanced control and organization of motor behavior; e.g, in the time domain. Unlike processes in the motor system, many other physiological processes car entrained by external sensory stimuli. For example, there is probably a very good protective reason why other cyclical physiological processes (e.g., autonor heart rate) have only very limited entrainment capacity to external rhythmic cues. Some of the basic auditory-motor arousal connections may also have their evolutionary processes related to survival behavior; e.g., in fight or flight reactions. Much of the "why" in auditory-motor interactions, however, remains unl the absence of this knowledge, great care should be taken to not compensate for this lack of understanding of specific cause and effect processes by assignin descriptions to the behavior of biological and physical systems. The unraveling of the perceptual, physiological, and neuroanatomical basis of the interaction movement has been, and continues to be, a fascinating endeavor with important ramifications for the study of brain function, sensory perception, and motor most exciting findings in this research, however, may be the evidence that the interaction between auditory rhythm and physical response can be effectively therapeutic purposes in the rehabilitation of persons with movement disorders.
language	eng
number	2
journal	IEEE engineering in medicine and biology magazine: the quarterly magazine of the Engineering in Medicine \& Biology Society
author	Thaut, M. H. and Kenyon, G. P. and Schauer, M. L. and McIntosh, G. C.
year	1999
pmid	10101675
keywords	Electromyography, Humans, Music Therapy, Motor Activity, Arm, Music, Auditory Perception, Movement Disorders, Periodicity, Acoustic Stimulation, Ce Disorders, Hemiplegia, Brain, Arousal, Adaptation, Physiological, Behavior, Escape Reaction, Muscle, Skeletal, Parkinson Disease, Perception, Sensation
pages	101--108
article	**schaffert_review_2019**
title	A {Review} on the {Relationship} {Between} {Sound} and {Movement} in {Sports} and {Rehabilitation}
volume	10
issn	1664-1078
doi	10.3389/fpsyg.2019.00244

abstract	The role of auditory information on perceptual-motor processes has gained increased interest in sports and psychology research in recent years. Numerous n behavioral studies have demonstrated the close interaction between auditory and motor areas of the brain, and the importance of auditory information for mo control, and learning. In applied research, artificially produced acoustic information and real-time auditory information have been implemented in sports and improve motor performance in athletes, healthy individuals, and patients affected by neurological or movement disorders. However, this research is scattered scientific disciplines. The aim of this paper is to provide an overview about the interaction between movement and sound and review the current literature re natural movement sounds, movement sonification, and rhythmic auditory information in sports and motor rehabilitation. The focus here is threefold: firstly, of empirical studies using natural movement sounds and movement sonification in sports. Secondly, we review recent clinical and applied studies using rhyt information and sonification in rehabilitation, addressing in particular studies on Parkinson's disease and stroke. Thirdly, we summarize current evidence reg mechanisms and neural correlates underlying the processing of auditory information during movement execution and its mental representation. The current s reviewed provides evidence of the feasibility and effectiveness of the application of auditory information to improve movement execution, control, and (re)l motor rehabilitation. Findings also corroborate the critical role of auditory information in auditory-motor coupling during motor (re)learning and performanc area of clinical and applied research has a large potential that is yet to be fully explored.
language	eng
journal	Frontiers in Psychology
author	Schaffert, Nina and Janzen, Thenille Braun and Mattes, Klaus and Thaut, Michael H.
year	2019
pmid	30809175
pmcid	PMC6379478
keywords	stroke, rhythmic auditory stimulation, acoustic feedback, motor rehabilitation, movement sonification, Parkinson’s disease, sports
pages	244
file	Texte intégral:C\:\Users\paul-\Zotero\storage\SRZBD87J\Schaffert et al. - 2019 - A Review on the Relationship Between Sound and Mov.pdf:application/pd
article	**rossignol_locomotor_1996**
title	Locomotor capacities after complete and partial lesions of the spinal cord
volume	56
issn	0065-1400
doi	10.55782/ane-1996-1148
abstract	This paper first reviews some of the observations made on the locomotor capabilities of several animal species with a special emphasis on cats and including complete spinal lesions. We show that animals can perform well-coordinated walking movements of the hindlimbs when they are placed on a treadmill belt also adaptable to speed and perturbations. Cats with partial spinal lesions of the ventral and ventrolateral parts of the cord can perform voluntary quadrupeda overground or on the treadmill albeit with deficits in weight support and interlimb coordination. We also show that some drugs such as clonidine (an alpha-2 can be used to trigger locomotion in early-spinal cats and discuss the effects of various neurotransmitter systems on the expression of the locomotor pattern i partial spinal cats. It is concluded that a pharmacological approach could be used, in combination with other approaches, such as locomotor training and fun stimulation, to improve locomotor functions after spinal cord injuries in humans.
language	eng

number	1
journal	Acta Neurobiologiae Experimentalis
author	Rossignol, S. and Chau, C. and Brustein, E. and Bélanger, M. and Barbeau, H. and Drew, T.
year	1996
pmid	8787206
keywords	Humans, Animals, Cats, Locomotion, Spinal Cord Injuries
pages	449--463
misc	**noauthor_create_nodate**
title	Create {Next} {App}
url	http://localhost:3000/
urldate	2025-02-13
file	Create Next App:C\:\Users\paul-\Zotero\storage\ZDPXT2UD\localhost.html:text/html
article	**schmitt_childrens_2020**
title	Children's {Active} {Engagement} in {Public} {School} {Language} {Therapy} {Relates} to {Greater} {Gains}
volume	29
url	https://pubs.asha.org/doi/abs/10.1044/2020_AJSLP-19-00157
doi	10.1044/2020_AJSLP-19-00157
abstract	PurposeThe current study examined children's engagement as an active ingredient of language therapy in public schools and considered the potential interpla and dose on outcomes.MethodParticipants included 135 children with language impairment receiving business-as-usual therapy from 70 speech-language pa videotaped therapy sessions from each participating child were coded for children's level of engagement and time in language-focused therapy (dose).Result modeling was used for analyses; children's level of engagement (i.e., active engagement) was significantly, positively related to children's language gain and dose.ConclusionFindings suggest that children's active participation in therapy sessions is a significant component to effective language therapy and undersc further research.
number	3
urldate	2025-02-17
journal	American Journal of Speech-Language Pathology
author	Schmitt, Mary Beth
month	aug, year = {2020
note	Publisher: American Speech-Language-Hearing Association
pages	1505--1513
inproceedings	**delalez_adjusting_2017**

title	Adjusting the {Frame}: {Biphasic} {Performative} {Control} of {Speech} {Rhythm}
shorttitle	Adjusting the {Frame}
url	https://www.isca-archive.org/interspeech_2017/delalez17_interspeech.html
doi	10.21437/Interspeech.2017-396
abstract	Performative time and pitch scaling is a new research paradigm for prosodic analysis by synthesis. In this paper, a system for real-time recorded speech time the means of hands or feet gestures is designed and evaluated. Pitch is controlled with the preferred hand, using a stylus on a graphic tablet. Time is controll frames, or constriction gestures, defined by pairs of control points. The ”Arsis” corresponds to the constriction (weak beat of the syllable) and the ”Thesis” c vocalic nucleus (strong beat of the syllable). This biphasic control of rhythmic units is performed by the non-preferred hand using a button. Pitch and time sc according to these gestural controls with the help of a real-time pitch synchronous overlap-add technique (RT-PSOLA). Rhythm and pitch control accuracy prosodic imitation experiment: the task is to reproduce intonation and rhythm of various sentences. The results show that inter-vocalic durations differ on av The system appears as a new and effective tool for performative speech and singing synthesis. Consequences and applications in speech prosody research ar
language	en
urldate	2025-02-18
booktitle	Interspeech 2017
publisher	ISCA
author	Delalez, Samuel and d’Alessandro, Christophe
month	aug, year = {2017
pages	864--868
file	Delalez et d’Alessandro - 2017 - Adjusting the Frame Biphasic Performative Control.pdf:C\:\Users\paul-\Zotero\storage\JEZDZR2G\Delalez et d’Alessandr Frame Biphasic Performative Control.pdf:application/pdf
misc	**fernandez_onsets_2023**
title	Onsets and {Velocities}: {Affordable} {Real}-{Time} {Piano} {Transcription} {Using} {Convolutional} {Neural} {Networks}
shorttitle	Onsets and {Velocities}
url	http://arxiv.org/abs/2303.04485
doi	10.48550/arXiv.2303.04485
abstract	Polyphonic Piano Transcription has recently experienced substantial progress, driven by the use of sophisticated Deep Learning approaches and the introduc such as note onset, offset, velocity and pedal detection. This progress was coupled with an increased complexity and size of the proposed models, typically r components and high-resolution data. In this work we focus on onset and velocity detection, showing that a substantially smaller and simpler convolutional temporal resolution (24ms), is still competitive: our proposed ONSETS\&VELOCITIES model achieves state-of-the-art performance on the MAESTRO dat (F1=96.78\%) and sets a good novel baseline for onset+velocity (F1=94.50\%), while having {\textasciitilde}3.1M parameters and maintaining real-time cap commodity hardware. We provide open-source code to reproduce our results and a real-time demo with a pretrained model.
urldate	2025-02-27
publisher	arXiv

author	Fernandez, Andres
month	jun, year = {2023
note	arXiv:2303.04485 [cs]
keywords	Computer Science - Sound, Electrical Engineering and Systems Science - Audio and Speech Processing
annote	Comment: Accepted at EUSIPCO 2023
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\LKQ79QLD\Fernandez - 2023 - Onsets and Velocities Affordable Real-Time Piano .pdf:application/pdf;Snap \Zotero\storage\D24SF86G\2303.html:text/html
misc	**wang_furelise_2024**
title	F{ürElise}: {Capturing} and {Physically} {Synthesizing} {Hand} {Motions} of {Piano} {Performance}
shorttitle	F{ürElise}
url	http://arxiv.org/abs/2410.05791
doi	10.48550/arXiv.2410.05791
abstract	Piano playing requires agile, precise, and coordinated hand control that stretches the limits of dexterity. Hand motion models with the sophistication to accu playing have a wide range of applications in character animation, embodied AI, biomechanics, and VR/AR. In this paper, we construct a first-of-its-kind larg contains approximately 10 hours of 3D hand motion and audio from 15 elite-level pianists playing 153 pieces of classical music. To capture natural perform markerless setup in which motions are reconstructed from multi-view videos using state-of-the-art pose estimation models. The motion data is further refine using the high-resolution MIDI key-pressing data obtained from sensors in a specialized Yamaha Disklavier piano. Leveraging the collected dataset, we dev can synthesize physically-plausible hand motions for musical scores outside of the dataset. Our approach employs a combination of imitation learning and re obtain policies for physics-based bimanual control involving the interaction between hands and piano keys. To solve the sampling efficiency problem with t we use a diffusion model to generate natural reference motions, which provide high-level trajectory and fingering (finger order and placement) information. reference motion alone does not provide sufficient accuracy for piano performance modeling. We then further augmented the data by using musical similarit motions from the captured dataset to boost the precision of the RL policy. With the proposed method, our model generates natural, dexterous motions that g outside the training dataset.
urldate	2025-02-27
publisher	arXiv
author	Wang, Ruocheng and Xu, Pei and Shi, Haochen and Schumann, Elizabeth and Liu, C. Karen
month	oct, year = {2024
note	arXiv:2410.05791 [cs]
keywords	Computer Science - Sound, Electrical Engineering and Systems Science - Audio and Speech Processing, Computer Science - Artificial Intelligence, Comput
annote	Comment: SIGGRAPH Asia 2024. Project page: https://for-elise.github.io/
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\UJCDEPHG\Wang et al. - 2024 - FürElise Capturing and Physically Synthesizing Ha.pdf:application/pdf;Sna \Zotero\storage\N4L9GRHP\2410.html:text/html
misc	**gan_pianomotion10m_2025**

title	PianoMotion10M}: {Dataset} and {Benchmark} for {Hand} {Motion} {Generation} in {Piano} {Performance
shorttitle	PianoMotion10M
url	http://arxiv.org/abs/2406.09326
doi	10.48550/arXiv.2406.09326
abstract	Recently, artificial intelligence techniques for education have been received increasing attentions, while it still remains an open problem to design the effecti instructing systems. Although key presses can be directly derived from sheet music, the transitional movements among key presses require more extensive g performance. In this work, we construct a piano-hand motion generation benchmark to guide hand movements and fingerings for piano playing. To this end, dataset, PianoMotion10M, consisting of 116 hours of piano playing videos from a bird's-eye view with 10 million annotated hand poses. We also introduce a model that generates hand motions from piano audios through a position predictor and a position-guided gesture generator. Furthermore, a series of evaluatio to assess the performance of the baseline model, including motion similarity, smoothness, positional accuracy of left and right hands, and overall fidelity of Despite that piano key presses with respect to music scores or audios are already accessible, PianoMotion10M aims to provide guidance on piano fingering f The source code and dataset can be accessed at https://github.com/agnJason/PianoMotion10M.
urldate	2025-03-03
publisher	arXiv
author	Gan, Qijun and Wang, Song and Wu, Shengtao and Zhu, Jianke
month	feb, year = {2025
note	arXiv:2406.09326 [cs]
keywords	Computer Science - Sound, Electrical Engineering and Systems Science - Audio and Speech Processing, Computer Science - Artificial Intelligence, Comput Vision and Pattern Recognition, Computer Science - Multimedia
annote	Comment: ICLR 2025 Spotlight
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\RFSG9ID8\Gan et al. - 2025 - PianoMotion10M Dataset and Benchmark for Hand Mot.pdf:application/pdf;Sn \Zotero\storage\RIC4EZSP\2406.html:text/html
misc	**pavllo_3d_2019**
title	3D} human pose estimation in video with temporal convolutions and semi-supervised training
url	http://arxiv.org/abs/1811.11742
doi	10.48550/arXiv.1811.11742
abstract	In this work, we demonstrate that 3D poses in video can be effectively estimated with a fully convolutional model based on dilated temporal convolutions ov also introduce back-projection, a simple and effective semi-supervised training method that leverages unlabeled video data. We start with predicted 2D keyp video, then estimate 3D poses and finally back-project to the input 2D keypoints. In the supervised setting, our fully-convolutional model outperforms the pr the literature by 6 mm mean per-joint position error on Human3.6M, corresponding to an error reduction of 11\%, and the model also shows significant impr HumanEva-I. Moreover, experiments with back-projection show that it comfortably outperforms previous state-of-the-art results in semi-supervised settings scarce. Code and models are available at https://github.com/facebookresearch/VideoPose3D
urldate	2025-03-06
publisher	arXiv

author	Pavllo, Dario and Feichtenhofer, Christoph and Grangier, David and Auli, Michael
month	mar, year = {2019
note	arXiv:1811.11742 [cs]
keywords	Computer Science - Computer Vision and Pattern Recognition
annote	Comment: CVPR 2019
file	Preprint PDF:C\:\Users\paul-\Zotero\storage\6EEESPLA\Pavllo et al. - 2019 - 3D human pose estimation in video with temporal co.pdf:application/pdf;Sna \Zotero\storage\FAGHQTBD\1811.html:text/html
inproceedings	**pavllo_3d_2019-1**
address	Long Beach, CA, USA
title	3D} {Human} {Pose} {Estimation} in {Video} {With} {Temporal} {Convolutions} and {Semi}-{Supervised} {Training
copyright	https://ieeexplore.ieee.org/Xplorehelp/downloads/license-information/IEEE.html
isbn	978-1-72813-293-8
url	https://ieeexplore.ieee.org/document/8954163/
doi	10.1109/CVPR.2019.00794
language	en
urldate	2025-03-12
booktitle	2019 {IEEE}/{CVF} {Conference} on {Computer} {Vision} and {Pattern} {Recognition} ({CVPR})
publisher	IEEE
author	Pavllo, Dario and Feichtenhofer, Christoph and Grangier, David and Auli, Michael
month	jun, year = {2019
pages	7745--7754
file	Pavllo et al. - 2019 - 3D Human Pose Estimation in Video With Temporal Co.pdf:C\:\Users\paul-\Zotero\storage\8CYR34WV\Pavllo et al. - 2019 - 3D Hum Video With Temporal Co.pdf:application/pdf
inproceedings	**pavllo_3d_2019-2**
address	Long Beach, CA, USA
title	3D} {Human} {Pose} {Estimation} in {Video} {With} {Temporal} {Convolutions} and {Semi}-{Supervised} {Training
copyright	https://ieeexplore.ieee.org/Xplorehelp/downloads/license-information/IEEE.html
isbn	978-1-72813-293-8
url	https://ieeexplore.ieee.org/document/8954163/
doi	10.1109/CVPR.2019.00794

language	en
urldate	2025-03-12
booktitle	2019 {IEEE}/{CVF} {Conference} on {Computer} {Vision} and {Pattern} {Recognition} ({CVPR})
publisher	IEEE
author	Pavllo, Dario and Feichtenhofer, Christoph and Grangier, David and Auli, Michael
month	jun, year = {2019
pages	7745--7754
file	Pavllo et al. - 2019 - 3D Human Pose Estimation in Video With Temporal Co.pdf:C\:\Users\paul-\Zotero\storage\PMGDSC2K\Pavllo et al. - 2019 - 3D Hun Video With Temporal Co.pdf:application/pdf
article	**furuya_surmounting_2025**
title	Surmounting the ceiling effect of motor expertise by novel sensory experience with a hand exoskeleton
volume	10
issn	2470-9476
doi	10.1126/scirobotics.adn3802
abstract	For trained individuals such as athletes and musicians, learning often plateaus after extensive training, known as the "ceiling effect." One bottleneck to overc prior physical experience with the skill to be learned. Here, we challenge this issue by exposing expert pianists to fast and complex finger movements that ca voluntarily, using a hand exoskeleton robot that can move individual fingers quickly and independently. Although the skill of moving the fingers quickly pla of piano practice, passive exposure to otherwise impossible complex finger movements generated by the exoskeleton robot at a speed faster than the pianists them to play faster. Neither a training for fast but simple finger movements nor one for slow but complex movements with the exoskeleton enhanced the ove The exoskeleton training with one hand also improved the motor skill of the untrained contralateral hand, demonstrating the intermanual transfer effect. The patterns of coordinated activities across multiple finger muscles during piano playing but not in general motor and somatosensory functions or in anatomical hand (range of motion). Patterns of the multifinger movements evoked by transcranial magnetic stimulation over the left motor cortex were also changed thr to fast and complex finger movements, which accompanied increased involvement of constituent movement elements characterizing the individuated finger demonstrate evidence that somatosensory exposure to an unexperienced motor skill allows surmounting of the ceiling effect in a task-specific but effector-in
language	eng
number	98
journal	Science Robotics
author	Furuya, Shinichi and Oku, Takanori and Nishioka, Hayato and Hirano, Masato
month	jan, year = {2025
pmid	39813311
keywords	Electromyography, Robotics, Female, Humans, Male, Adult, Fingers, Hand, Movement, Transcranial Magnetic Stimulation, Music, Range of Motion, Artic Learning, Young Adult, Equipment Design, Muscle, Skeletal, Exoskeleton Device
pages	eadn3802

article	noauthor_real-time_2025
title	Real-{Time} {Myoelectric}-{Based} {Neural}-{Drive} {Decoding} for {Concurrent} and {Continuous} {Control} of {Robotic} {Finger} {Forces} {\textb
url	https://www.researchgate.net/publication/388741945_Real-Time_Myoelectric-Based_Neural-Drive_Decoding_for_Concurrent_and_Continuous_Control_of_Robotic_Finger_Forces
doi	10.1109/THMS.2025.3532209
abstract	Request PDF {\textbar} Real-Time Myoelectric-Based Neural-Drive Decoding for Concurrent and Continuous Control of Robotic Finger Forces {\textbar} injuries, such as due to amputation, spinal cord injury, and stroke, can affect hand functions, profoundly impacting... {\textbar} Find, read and cite all the res ResearchGate
language	en
urldate	2025-05-06
journal	ResearchGate
month	feb, year = {2025
file	Snapshot:C\:\Users\paul-\Zotero\storage\IYBYQHEJ\388741945_Real-Time_Myoelectric-Based_Neural-Drive_Decoding_for_Concurrent_and_Continuou
article	**thaut_neurobiological_2015**
title	Neurobiological foundations of neurologic music therapy: rhythmic entrainment and the motor system
volume	5
issn	1664-1078
shorttitle	Neurobiological foundations of neurologic music therapy
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4344110/
doi	10.3389/fpsyg.2014.01185
abstract	Entrainment is defined by a temporal locking process in which one system's motion or signal frequency entrains the frequency of another system. This proce phenomenon that can be observed in physical (e.g., pendulum clocks) and biological systems (e.g., fire flies). However, entrainment can also be observed be and motor systems. The function of rhythmic entrainment in rehabilitative training and learning was established for the first time by Thaut and colleagues in studies in the early 1990s. It was shown that the inherent periodicity of auditory rhythmic patterns could entrain movement patterns in patients with moveme review:). Physiological, kinematic, and behavioral movement analysis showed very quickly that entrainment cues not only changed the timing of movement spatial and force parameters. Mathematical models have shown that anticipatory rhythmic templates as critical time constraints can result in the complete sp dynamics of a movement over the entire movement cycle, thereby optimizing motor planning and execution. Furthermore, temporal rhythmic entrainment ha extended into applications in cognitive rehabilitation and speech and language rehabilitation, and thus become one of the major neurological mechanisms lin to brain rehabilitation. These findings provided a scientific basis for the development of neurologic music therapy.
urldate	2025-06-02
journal	Frontiers in Psychology
author	Thaut, Michael H. and McIntosh, Gerald C. and Hoemberg, Volker
month	feb, year = {2015

pmid	25774137
pmcid	PMC4344110
pages	1185
file	Texte intégral:C\:\Users\paul-\Zotero\storage\H896VRUT\Thaut et al. - 2015 - Neurobiological foundations of neurologic music th.pdf:application/pdf
article	**repp_sensorimotor_2005**
title	Sensorimotor synchronization: a review of the tapping literature
volume	12
issn	1069-9384
shorttitle	Sensorimotor synchronization
doi	10.3758/bf03206433
abstract	Sensorimotor synchronization (SMS), the rhythmic coordination of perception and action, occurs in many contexts, but most conspicuously in music perforr laboratory, it is most often studied in the form of finger tapping to a sequence of auditory stimuli. This review summarizes theories and empirical findings ol task. Its eight sections deal with the role of intention, rate limits, the negative mean asynchrony, variability, models of error correction, perturbation studies, SMS, and SMS in musical contexts. The central theoretical issue is considered to be how best to characterize the perceptual information and the internal pro people to achieve and maintain SMS. Recent research suggests that SMS is controlled jointly by two error correction processes (phase correction and period in their degrees of cognitive control and may be associated with different brain circuits. They exemplify the general distinction between subconscious mecha regulation and conscious processes involved in perceptual judgment and action planning.
language	eng
number	6
journal	Psychonomic Bulletin \& Review
author	Repp, Bruno H.
month	dec, year = {2005
pmid	16615317
keywords	Humans, Music, Psychomotor Performance, Periodicity, Time Factors
pages	969--992
file	Texte intégral:C\:\Users\paul-\Zotero\storage\SZ7NSWCL\Repp - 2005 - Sensorimotor synchronization a review of the tapp.pdf:application/pdf
article	**bella_why_2013**
title	Why {Movement} {Is} {Captured} by {Music}, but {Less} by {Speech}: {Role} of {Temporal} {Regularity}
volume	8
issn	1932-6203
shorttitle	Why {Movement} {Is} {Captured} by {Music}, but {Less} by {Speech}

url	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0071945
doi	10.1371/journal.pone.0071945
abstract	Music has a pervasive tendency to rhythmically engage our body. In contrast, synchronization with speech is rare. Music's superiority over speech in drivin results from isochrony of musical beats, as opposed to irregular speech stresses. Moreover, the presence of regular patterns of embedded periodicities (i.e., m making music particularly conducive to movement. We investigated these possibilities by asking participants to synchronize with isochronous auditory stim and speech distractors were presented at one of various phase relationships with respect to the target. In Exp. 1, familiar musical excerpts and fragments of c used as distractors. The stimuli were manipulated in terms of beat/stress isochrony and average pitch to achieve maximum comparability. In Exp. 2, the distr known songs performed with lyrics, on a reiterated syllable, and spoken lyrics, all having the same meter. Music perturbed synchronization with the target st fragments. However, music superiority over speech disappeared when distractors shared isochrony and the same meter. Music's peculiar and regular tempor be the main factor fostering tight coupling between sound and movement.
language	en
number	8
urldate	2025-06-02
journal	PLOS ONE
author	Bella, Simone Dalla and Białuńska, Anita and Sowiński, Jakub
month	aug, year = {2013
note	Publisher: Public Library of Science
keywords	Music perception, Speech, Bioacoustics, Linguistics, Musculoskeletal mechanics, Pitch perception, Speech signal processing, Syllables
pages	e71945
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\AC56MCIA\Bella et al. - 2013 - Why Movement Is Captured by Music, but Less by Spe.pdf:application/pdf
article	**hayden_clinical_2025**
title	Clinical validation of a novel hand dexterity measurement device
volume	4
issn	2767-3170
url	https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000744
doi	10.1371/journal.pdig.0000744
abstract	The lack of sensitive objective outcome measures for hand dexterity is a barrier for clinical assessment of neurological conditions and has negatively affecte we clinically validate a new method for measuring hand dexterity, a novel hand worn sensor that digitises the Finger Tapping Test. The device was assessed healthy controls and 51 people with Amyotrophic Lateral Sclerosis (ALS) and compared against rating scales and traditional measures (Nine Hole Peg test a 14 features were extracted from the device and using a logistic regression algorithm, a 0-100 dexterity performance score was generated for each participant age/sex differences. The device returned objective ratings of a participant's hand dexterity (dominant, non-dominant and overall score). The average overall score in all healthy participants was 88 ± 17 (mean ± standard deviation). The overall dexterity score was statistically significantly worse in participants with healthy subset: 80 ± 20, ALS: 45 ± 32, p-value < 0.0001). The device also had a higher completion rate, (94\% dominant hand) compared to the traditional m dominant hand). This test and scoring system have been validated and the regression model was developed using a framework that is potentially applicable t

	condition. This device could act as an objective outcome measure in clinical trials and may be useful in improving patient care.
language	en
number	3
urldate	2025-06-17
journal	PLOS Digital Health
author	Hayden, Conor D. and Murphy, Bruce P. and Gilsenan, Deirdre and Nasseroleslami, Bahman and Hardiman, Orla and Murray, Deirdre
month	mar, year = {2025
note	Publisher: Public Library of Science
keywords	Fingers, Amyotrophic lateral sclerosis, Hand strength, Hands, Health informatics, Medical devices and equipment, Questionnaires, Thumbs
pages	e0000744
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\SX2THFRX\Hayden et al. - 2025 - Clinical validation of a novel hand dexterity meas.pdf:application/pdf
article	**peyre_effect_2023-1**
title	Effect of sonification types in upper-limb movement: a quantitative and qualitative study in hemiparetic and healthy participants
volume	20
issn	1743-0003
shorttitle	Effect of sonification types in upper-limb movement
url	https://doi.org/10.1186/s12984-023-01248-y
doi	10.1186/s12984-023-01248-y
abstract	Movement sonification, the use of real-time auditory feedback linked to movement parameters, have been proposed to support rehabilitation. Nevertheless, i have been reported, the effect of the type of sound used has not been studied systematically. The aim of this study was to investigate in a single session the e of sonification both quantitatively and qualitatively on patients with acquired brain lesions and healthy participants.
number	1
urldate	2025-06-17
journal	Journal of NeuroEngineering and Rehabilitation
author	Peyre, Iseline and Roby-Brami, Agnès and Segalen, Maël and Giron, Alain and Caramiaux, Baptiste and Marchand-Pauvert, Véronique and Pradat-Diehl, Pa Frédéric
month	oct, year = {2023
keywords	Movement sonification, Rehabilitation, Upper limb, Hemiparesis, Mixed methodology
pages	136

file	Full Text PDF:C\:\Users\paul-\Zotero\storage\KW8F35P4\Peyre et al. - 2023 - Effect of sonification types in upper-limb movemen.pdf:application/pdf;Snap \Zotero\storage\2827PCG2\s12984-023-01248-y.html:text/html
article	**wodu_stroke_2024**
title	Stroke {Survivors}' {Interaction} {With} {Hand} {Rehabilitation} {Devices}: {Observational} {Study}
volume	9
copyright	Unless stated otherwise, all articles are open-access distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/li permits unrestricted use, distribution, and reproduction in any medium, provided the original work ("first published in the Journal of Medical Internet Resea with original URL and bibliographic citation information. The complete bibliographic information, a link to the original publication on http://www.jmir.org/ copyright and license information must be included.
shorttitle	Stroke {Survivors}' {Interaction} {With} {Hand} {Rehabilitation} {Devices}
url	https://biomedeng.jmir.org/2024/1/e54159
doi	10.2196/54159
abstract	Background: The hand is crucial for carrying out activities of daily living as well as social interaction. Functional use of the upper limb is affected in up to 5 survivors 3 to 6 months after stroke. Rehabilitation can help restore function, and several rehabilitation devices have been designed to improve hand functio these devices is compromised in people with more severe loss of function. Objective: In this study, we aimed to observe stroke survivors with poor hand fun range of commonly used hand rehabilitation devices. Methods: Participants were engaged in an 8-week rehabilitation intervention at a technology-enriched r participants spent 50-60 minutes of the 2-hour session in the upper limb section at least twice a week. Each participant communicated their rehabilitation goa Research Arm Test (ARAT) was used to measure and categorize hand function as poor (scores of 0-9), moderate (scores of 10-56), or good (score of 57). Pa observed during their interactions with 3 hand-based rehabilitation devices that focused on hand rehabilitation: the GripAble, NeuroBall, and Semi-Circular Observations of device interactions were recorded for each session. Results: A total of 29 participants were included in this study, of whom 10 (34\%) had p (59\%) had moderate hand function, and 2 (7\%) had good hand function. There were no differences in the age and years after stroke among participants wit and those with moderate (P=.06 and P=.09, respectively) and good (P=.37 and P=.99, respectively) hand function. Regarding the ability of the 10 participant function to interact with the 3 hand-based rehabilitation devices, 2 (20\%) participants with an ARAT score greater than 0 were able to interact with the devi (80\%) who had an ARAT score of 0 could not. Their inability to interact with these devices was clinically examined, and the reason was determined to be a presence of (1) muscle tone or stiffness or (2) muscle weakness. Conclusions: Not all stroke survivors with impairments in their hands can make use of curre rehabilitation technologies. Those with an ARAT score of 0 cannot actively interact with hand rehabilitation devices, as they cannot carry out the hand move such interaction. The design of devices for hand rehabilitation should consider the accessibility needs of those with poor hand function.
language	EN
number	1
urldate	2025-06-17
journal	JMIR Biomedical Engineering
author	Wodu, Chioma Obinuchi and Sweeney, Gillian and Slachetka, Milena and Kerr, Andrew
month	jun, year = {2024
note	Company: JMIR Biomedical Engineering Distributor: JMIR Biomedical Engineering Institution: JMIR Biomedical Engineering Label: JMIR Biomedical En JMIR Publications Inc., Toronto, Canada

pages	e54159
file	Snapshot:C\:\Users\paul-\Zotero\storage\A2WJPQAH\e54159.html:text/html;Texte intégral:C\:\Users\paul-\Zotero\storage\LW42DLAQ\Wodu et al. - 2024 Interaction With Hand Rehabilita.pdf:application/pdf
article	**mackay_comparative_2025**
title	Comparative {Structured} {Observation}
volume	32
issn	1073-0516
url	https://dl.acm.org/doi/10.1145/3711838
doi	10.1145/3711838
abstract	Although HCI researchers often generate and compare new design concepts, they lack an established method for rigorously conducting qualitative assessme characterize Comparative Structured Observation as a qualitative research method that takes advantage of the structure of controlled experiments to generate ecologically relevant experiences with two or more design variants, often implemented as medium-fidelity prototypes. Researchers observe users and ask the reflect on each variant. We identify criteria for creating a successful Comparative Structured Observation study and illustrate variations of the method by an studies. We also examine six additional studies (three “near” and three “far”) to clarify the boundary between what should and should not be considered a C Observation. We discuss the benefits and limitations of the method and argue that gathering comparative reflections about design variants can help researche their design concepts.
number	2
urldate	2025-06-17
journal	ACM Trans. Comput.-Hum. Interact.
author	Mackay, Wendy E. and McGrenere, Joanna
month	apr, year = {2025
pages	14:1--14:27
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\RETFKHG6\Mackay et McGrenere - 2025 - Comparative Structured Observation.pdf:application/pdf
article	**marcus_acoustic_1981**
title	Acoustic determinants of perceptual center (P-center) location
volume	30
copyright	http://www.springer.com/tdm
issn	0031-5117, 1532-5962
url	http://link.springer.com/10.3758/BF03214280
doi	10.3758/BF03214280
language	en

number	3
urldate	2025-07-02
journal	Perception \& Psychophysics
author	Marcus, Stephen Michael
month	may, year = {1981
pages	247--256
file	Marcus - 1981 - Acoustic determinants of perceptual center (P-cent.pdf:C\:\Users\paul-\Zotero\storage\IBNEUYV5\Marcus - 1981 - Acoustic determinants cent.pdf:application/pdf
article	**marcus_acoustic_1981-1**
title	Acoustic determinants of perceptual center (P-center) location
volume	30
issn	0031-5117
doi	10.3758/BF03214280
abstract	Morton, Marcus, and Frankish (1976) defined "perceptual center," or "P-center," as a neutral term to describe that which is regular in a perceptually regular s sounds. This paper describes a paradigm for the determination of P-center location and the effect of various acoustic parameters on empirically determined P shown that P-center location is affected by both initial consonant duration and, secondarily, subsequent vowel and consonant duration. A simple two-parame duration of the whole stimulus is developed and gives good performance in predicting P-center location. The application of this model to continuous speech suggested that there is little value in attempting to determine any single acoustic or articulatory correlate of P-center location, or in attempting to define P-ce in time. Rather, these results indicate that P-centers are a property of the whole stimulus and reflect properties of both the production and perception of speec
number	3
journal	Perception \& Psychophysics
author	Marcus, S.M.
year	1981
pages	247--256
file	Texte intégral:C\:\Users\paul-\Zotero\storage\H2L3HVWF\Marcus - 1981 - Acoustic determinants of perceptual center (P-cent.pdf:application/pdf
article	**pompino-marschall_psychoacoustic_1989**
title	On the psychoacoustic nature of the P-center phenomenon
volume	17
issn	0095-4470
url	https://www.sciencedirect.com/science/article/pii/S0095447019304280
doi	10.1016/S0095-4470(19)30428-0

abstract	It seems almost generally accepted that the P-center, defined as the psychological moment of occurrence of a syllable, is only determined by the duration of consonantal onset and of its rhyme, independently of one another and independent of their phonetic categorization. Summing up the results of a number of o experiments—already described elsewhere—it will be shown that in contrast to this view (1) the effect of prevocalic consonant duration is not independent o vice versa and that both effects are non-linear, (2) the syllable's rhyme does not affect its P-center as a unit, but that there are also different effects of vowel duration, again interacting with one another, and (3) phonetically different syllables behave differently and that P-center shifts can also be induced by non-te elaboration of a purely psychoacoustic model—proposed elsewhere—based on the time course of the loudness within each critical band [sone/Bark] can acc effects fairly well. Furthermore, this model distinguishes between "points of onset" and "temporal centers of gravity", which both seem to be psychologicall experimental tasks involving subjectively regular rhythm. Finally the P-center phenomenon is discussed in connection with the predictions of this perceptual current models of articulatory timing.
number	3
urldate	2025-07-02
journal	Journal of Phonetics
author	Pompino-Marschall, Bernd
month	jul, year = {1989
pages	175--192
file	ScienceDirect Snapshot:C\:\Users\paul-\Zotero\storage\CHYT3PGV\S0095447019304280.html:text/html
article	**tacca_identifying_2024**
title	Identifying alterations in hand movement coordination from chronic stroke survivors using a wearable high-density {EMG} sleeve
volume	21
issn	1741-2552
doi	10.1088/1741-2552/ad634d
abstract	Objective.Non-invasive, high-density electromyography (HD-EMG) has emerged as a useful tool to collect a range of neurophysiological motor information demonstrated changes in EMG features that occur after stroke, which correlate with functional ability, highlighting their potential use as biomarkers. Howev have largely explored these EMG features in isolation with individual electrodes to assess gross movements, limiting their potential clinical utility. This stud function of stroke survivors by combining interpretable features extracted from a wearable HD-EMG forearm sleeve.Approach.Here, able-bodied (N= 7) an subjects (N= 7) performed 12 functional hand and wrist movements while HD-EMG was recorded using a wearable sleeve. A variety of HD-EMG features, decomposed to assess alterations in motor coordination.Main Results.Stroke subjects, on average, had higher co-contraction and reduced muscle coupling w their hand and actuate their thumb. Additionally, muscle synergies decomposed in the stroke population were relatively preserved, with a large spatial overla matched synergies. Alterations in synergy composition demonstrated reduced coupling between digit extensors and muscles that actuate the thumb, as well a activity in the stroke group. Average synergy activations during movements revealed differences in coordination, highlighting overactivation of antagonist m compensatory strategies. When combining co-contraction and muscle synergy features, the first principal component was strongly correlated with upper-ext sub-score of stroke participants (R2= 0.86). Principal component embeddings of individual features revealed interpretable measures of motor coordination a alterations.Significance.These results demonstrate the feasibility of predicting motor function through features decomposed from a wearable HD-EMG sleev leveraged to improve stroke research and clinical care.
language	eng

number	4
journal	Journal of Neural Engineering
author	Tacca, Nicholas and Baumgart, Ian and Schlink, Bryan R. and Kamath, Ashwini and Dunlap, Collin and Darrow, Michael J. and Colachis Iv, Samuel and Pu Branch, Joshua and Wengerd, Lauren and Friedenberg, David A. and Meyers, Eric C.
month	aug, year = {2024
pmid	39008975
keywords	stroke, Stroke, Electromyography, Female, Humans, Male, Middle Aged, Adult, Aged, Chronic Disease, Hand, Movement, Psychomotor Performance, Mus biomarker, co-contraction, electromyography, HD-EMG, muscle synergies, Survivors, Wearable Electronic Devices, wearables
article	**noauthor_pdf_nodate**
title	({PDF}) {Perceptual} centers (P-{Centers})
issn	0033-295X
url	https://www.researchgate.net/publication/232497075_Perceptual_centers_P-Centers
abstract	PDF {\textbar} Words presented with regular acoustic onsets are not perceptually regular. The requirements for perceived regularity were investigated, and t read and cite all the research you need on ResearchGate
language	en
urldate	2025-07-09
journal	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\4U44L2ZP\232497075_Perceptual_centers_P-Centers.html:text/html
article	**scott_point_1998**
title	The point of P-centres
volume	61
issn	1430-2772
url	https://doi.org/10.1007/PL00008162
doi	10.1007/PL00008162
abstract	P-centres are the subjective moments of occurrence of acoustic stimuli and capture properties of regularity and synchrony in production and perception. Two described which compare the effects of onset and offset amplitude variation and stimulus duration on P-centre location. The discussion is extended to consid in cross modal temporal phenomena.
language	en
number	1
urldate	2025-07-09
journal	Psychological Research

author	Scott, Sophie K.
month	mar, year = {1998
keywords	Auditory Perception, Psychoacoustics, Acoustic Stimulus, Amplitude Variation, Auditory System, Capture Property, Evoked potentials, Phonology and Pho Perception, Stimulus Duration, Temporal Phenomenon
pages	4--11
inproceedings	**noauthor_pdf_nodate-1**
title	({PDF}) {Performance} {Limits} for {Envelope} based {Automatic} {Syllable} {Segmentation}
url	https://www.researchgate.net/publication/224062516_Performance_Limits_for_Envelope_based_Automatic_Syllable_Segmentation
doi	10.1049/cp:20060489
abstract	PDF {\textbar} In this paper the upper performance limits of automatic syllable segmentation algorithms using single or multiple frequency band envelopes read and cite all the research you need on ResearchGate
language	en
urldate	2025-07-09
booktitle	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\K8K2GEUD\224062516_Performance_Limits_for_Envelope_based_Automatic_Syllable_Segmentation.html:text/ acceptée:C\:\Users\paul-\Zotero\storage\XNAJMGT9\(PDF) Performance Limits for Envelope based Automa.pdf:application/pdf
inproceedings	**noauthor_pdf_nodate-2**
title	({PDF}) {Performance} {Limits} for {Envelope} based {Automatic} {Syllable} {Segmentation}
url	https://www.researchgate.net/publication/224062516_Performance_Limits_for_Envelope_based_Automatic_Syllable_Segmentation
doi	10.1049/cp:20060489
abstract	PDF {\textbar} In this paper the upper performance limits of automatic syllable segmentation algorithms using single or multiple frequency band envelopes read and cite all the research you need on ResearchGate
language	en
urldate	2025-07-09
booktitle	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\JKI7T5Y6\224062516_Performance_Limits_for_Envelope_based_Automatic_Syllable_Segmentation.html:text/htı acceptée:C\:\Users\paul-\Zotero\storage\WYE4LDCW\(PDF) Performance Limits for Envelope based Automa.pdf:application/pdf
misc	**noauthor_pdf_nodate-3**
title	({PDF}) {Adjusting} the {Frame}: {Biphasic} {Performative} {Control} of {Speech} {Rhythm}
url	https://www.researchgate.net/publication/319185126_Adjusting_the_Frame_Biphasic_Performative_Control_of_Speech_Rhythm?utm_source=chatgpt.com
urldate	2025-07-09

file	(PDF) Adjusting the Frame\: Biphasic Performative Control of Speech Rhythm:C\:\Users\paul-\Zotero\storage\97BUYB8D\319185126_Adjusting_the_Frame_Biphasic_Performative_Control_of_Speech_Rhythm.html:text/html
article	**rabah_novel_2022**
title	A novel tablet-based application for assessment of manual dexterity and its components: a reliability and validity study in healthy subjects
volume	19
issn	1743-0003
shorttitle	A novel tablet-based application for assessment of manual dexterity and its components
doi	10.1186/s12984-022-01011-9
abstract	BACKGROUND: We developed five tablet-based tasks (applications) to measure multiple components of manual dexterity. AIM: to test reliability and vali dexterity measures in healthy participants. METHODS: Tasks included: (1) Finger recognition to assess mental rotation capacity. The subject taps with the f virtual hand in three orientations (reaction time, correct trials). (2) Rhythm tapping to evaluate timing of finger movements performed with, and subsequentl cue (inter-stimulus interval). (3) Multi-finger tapping to assess independent finger movements (reaction time, correct trials, unwanted finger movements). (4 assess production and memorization of visually cued finger sequences (successful taps). (5) Line-tracking to assess movement speed and accuracy while trac moving line on the screen with the fingertip (duration, error). To study inter-rater reliability, 34 healthy subjects (mean age 35 years) performed the tablet ta raters. Relative reliability (Intra-class correlation, ICC) and absolute reliability (Standard error of measurement, SEM) were established. Task validity was e subjects (mean age 49 years, range: 20-78 years) by correlating tablet measures with age, clinical dexterity assessments (time taken to pick-up objects in Bo and Moberg Pick Up Test, MPUT) and with measures obtained using a finger force-sensor device. RESULTS: Most timing measures showed excellent relia reliability was found for correct trials across tasks, and reliability was poor for unwanted movements. Inter-session learning occurred in some measures. Age and more variable reaction times in finger recognition, less correct trials in multi-finger tapping, and slower line-tracking. Reaction times correlated with tho finger force-sensor device. No significant correlations between tablet measures and BBT or MPUT were found. Inter-task correlation among tablet-derived r CONCLUSIONS: Most tablet-based dexterity measures showed good-to-excellent reliability (ICC ≥ 0.60) except for unwanted movements during multi-fin decline in performance and association with finger force-sensor measures support validity of tablet measures. Tablet-based components of dexterity compler clinical dexterity assessments. Future work is required to establish measurement properties in patients with neurological and psychiatric disorders.
language	eng
number	1
journal	Journal of Neuroengineering and Rehabilitation
author	Rabah, Ayah and Le Boterff, Quentin and Carment, Loïc and Bendjemaa, Narjes and Térémetz, Maxime and Dupin, Lucile and Cuenca, Macarena and Mas, Marie-Odile and Maier, Marc A. and Lindberg, Påvel G.
month	mar, year = {2022
pmid	35331273
pmcid	PMC8953393
keywords	Stroke, Humans, Middle Aged, Upper Extremity, Adult, Hand, Healthy Volunteers, Manual dexterity, Measurement, Reproducibility of Results, Reliability, medicine, Inter-rater reliability, Tablet application, Validity
pages	35

file	Texte intégral:C\:\Users\paul-\Zotero\storage\JQD4554Y\Rabah et al. - 2022 - A novel tablet-based application for assessment of.pdf:application/pdf
misc	**noauthor_efficacy_nodate**
title	Efficacy of interactive manual dexterity training after stroke: a pilot single-blinded randomized controlled trial {\textbar} {Journal} of {NeuroEngineering} {\textbar} {Full} {Text}
url	https://jneuroengrehab.biomedcentral.com/articles/10.1186/s12984-023-01213-9
urldate	2025-07-16
file	Efficacy of interactive manual dexterity training after stroke\: a pilot single-blinded randomized controlled trial | Journal of NeuroEngineering and Rehabilit Text:C\:\Users\paul-\Zotero\storage\YK7BHEF7\s12984-023-01213-9.html:text/html
article	**ozernov-palchik_musical_2018**
title	Musical rhythm and reading development: does beat processing matter?
issn	1749-6632
shorttitle	Musical rhythm and reading development
doi	10.1111/nyas.13853
abstract	There is mounting evidence for links between musical rhythm processing and reading-related cognitive skills, such as phonological awareness. This may be speech are rhythmic: both involve processing complex sound sequences with systematic patterns of timing, accent, and grouping. Yet, there is a salient diffe and speech rhythm: musical rhythm is often beat-based (based on an underlying grid of equal time intervals), while speech rhythm is not. Thus, the role of b the reading-rhythm relationship is not clear. Is there is a distinct relation between beat-based processing mechanisms and reading-related language skills, or link entirely due to shared mechanisms for processing nonbeat-based aspects of temporal structure? We discuss recent evidence for a distinct link between b and early reading abilities in young children, and suggest experimental designs that would allow one to further methodically investigate this relationship. We based processing taps into a listener's ability to use rich contextual regularities to form predictions, a skill important for reading development.
language	eng
journal	Annals of the New York Academy of Sciences
author	Ozernov-Palchik, Ola and Patel, Aniruddh D.
month	may, year = {2018
pmid	29781084
keywords	beat-based structure, phonological awareness, reading development, rhythm processing, temporal regularity
article	**parrell_spatiotemporal_2014**
title	Spatiotemporal coupling between speech and manual motor actions
volume	42
issn	0095-4470
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3900246/
doi	10.1016/j.wocn.2013.11.002

abstract	Much evidence has been found for pervasive links between the manual and speech motor systems, including evidence from infant development, deictic poin tapping and speaking tasks. We expand on the last of these paradigms to look at intra- and cross-modal effects of emphatic stress, as well as the effects of co absence of explicit rhythm. In this study, subjects repeatedly tapped their finger and synchronously repeated a single spoken syllable. On each trial, subjects stress on one finger tap or one spoken syllable. Results show that both movement duration and magnitude are affected by emphatic stress regardless of whet same domain (e.g., effects on the oral articulators when a spoken repetition is stressed) or across domains (e.g., effects on the oral articulators when a tap is s size of the effects differs between intra-and cross-domain emphases, the implementation of stress affects both motor domains, indicating a tight connection. seen even in the absence of stress, though it is highlighted under stress. The results of this study support the idea that implementation of prosody is not doma general aspects of the motor system.
urldate	2025-07-16
journal	Journal of phonetics
author	Parrell, Benjamin and Goldstein, Louis and Lee, Sungbok and Byrd, Dani
month	jan, year = {2014
pmid	24465063
pmcid	PMC3900246
pages	1--11
file	Version acceptée:C\:\Users\paul-\Zotero\storage\8MM8ZJYF\Parrell et al. - 2014 - Spatiotemporal coupling between speech and manual .pdf:application/pd
article	**singer_feeling_nodate**
title	Feeling the {Beat}: {Temporal} {Predictability} is {Associated} with {Ongoing} {Changes} in {Music}-{Induced} {Pleasantness}
volume	6
issn	2514-4820
shorttitle	Feeling the {Beat}
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10348017/
doi	10.5334/joc.286
abstract	Music is a complex phenomenon that elicits a range of emotional responses, influenced by numerous variables, such as rhythm, melody and harmony. One i music is listeners' ability to predict its continuation as it unfolds – an inherent attribute hypothesized to contribute to our emotional response to music. In thi this link by examining the relationship between temporal predictability – the ability to predict the timing of the next event – and the ongoing changes in mus pleasantness. Temporal predictability was operationalized as the degree to which taps of 20 musically trained participants, who tapped to the beat along thre contrastive musical pieces, were aligned. We then examined the degree to which this measure could explain the ongoing emotional experience, as reflected i of arousal and valence, in a separate group of 40 participants that listened to these pieces. Our findings reveal a positive correlation between fluctuations in r temporal predictability, even when controlling for a set of other musical features, in four out of five musical sections. The only exception being a lyrical slov findings were further supported by a large online database of annotated musical emotions (n = 1780 songs), where a consistent and robust correlation betwee an automatically extracted feature of pulse clarity was demonstrated. Overall, our findings shed light on the significance of temporal predictability as a contr hedonic experience of music, especially within the tempo range of salient beat perception.
number	1

urldate	2025-07-16
journal	Journal of Cognition
author	Singer, Neomi and Jacoby, Nori and Hendler, Talma and Granot, Roni
pmid	37457107
pmcid	PMC10348017
pages	34
file	Texte intégral:C\:\Users\paul-\Zotero\storage\DRT5BTAL\Singer et al. - Feeling the Beat Temporal Predictability is Assoc.pdf:application/pdf
article	**fiveash_processing_2021**
title	Processing rhythm in speech and music: {Shared} mechanisms and implications for developmental speech and language disorders
volume	35
issn	1931-1559
shorttitle	Processing rhythm in speech and music
doi	10.1037/neu0000766
abstract	Objective: Music and speech are complex signals containing regularities in how they unfold in time. Similarities between music and speech/language in tern features, rhythmic structure, and hierarchical structure have led to a large body of literature suggesting connections between the two domains. However, the mechanisms behind this connection remain to be elucidated. Method: In this theoretical review article, we synthesize previous research and present a framev shared neural mechanisms for music and speech rhythm processing. We outline structural similarities of rhythmic signals in music and speech, synthesize pr speech rhythm theories, discuss impaired timing in developmental speech and language disorders, and discuss music rhythm training as an additional, potent therapeutic tool to enhance speech/language processing in these disorders. Results: We propose the processing rhythm in speech and music (PRISM) framev three underlying mechanisms that appear to be shared across music and speech/language processing: Precise auditory processing, synchronization/entrainme to external stimuli, and sensorimotor coupling. The goal of this framework is to inform directions for future research that integrate cognitive and biological e relationships between rhythm processing in music and speech. Conclusion: The current framework can be used as a basis to investigate potential links betwe deficits in developmental disorders, impairments in the proposed mechanisms, and pathology-specific deficits which can be targeted in treatment and trainin therapy outcomes. On these grounds, we propose future research directions and discuss implications of our framework. (PsycInfo Database Record (c) 2022 reserved)
number	8
journal	Neuropsychology
author	Fiveash, Anna and Bedoin, Nathalie and Gordon, Reyna L. and Tillmann, Barbara
year	2021
note	Place: US Publisher: American Psychological Association
keywords	Music, Rhythm, Training, Developmental Disabilities, Language Disorders, Oral Communication, Speech Development, Speech Rhythm, Speech Therapy
pages	771--791

file	Snapshot:C\:\Users\paul-\Zotero\storage\Z4TDNB87\doiLanding.html:text/html;Texte intégral:C\:\Users\paul-\Zotero\storage\NIZG37Q5\Fiveash et al. - 20 in speech and music Shared mech.pdf:application/pdf
article	**luo_studying_2023**
title	Studying rhythm processing in speech through the lens of auditory-motor synchronization
volume	17
issn	1662-453X
url	https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2023.1146298/full
doi	10.3389/fnins.2023.1146298
abstract	Continuous speech is organized into a hierarchy of rhythms. Accurate processing of this rhythmic hierarchy through the interactions of auditory and motor s to speech perception and production. In this mini-review, we aim to evaluate the implementation of behavioral auditory-motor synchronization paradigms w processing in speech. First, we present an overview of the classic finger-tapping paradigm and its application in revealing differences in auditory-motor sync the typical and clinical populations. Next, we highlight key findings on rhythm hierarchy processing in speech and non-speech stimuli from finger-tapping st we discuss the potential caveats of the finger-tapping paradigm and propose the speech-speech synchronization (SSS) task to be a promising tool for future s seek to raise interest in developing new methods to shed light on the neural mechanisms of speech processing.
language	English
urldate	2025-07-22
journal	Frontiers in Neuroscience
author	Luo, Lu and Lu, Lingxi
month	mar, year = {2023
note	Publisher: Frontiers
keywords	Rhythm, Speech, Auditory-motor synchronization, Finger-tapping task, speech-to-speech synchronization task
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\SC932VFC\Luo et Lu - 2023 - Studying rhythm processing in speech through the l.pdf:application/pdf
article	**falk_it_2016**
title	It is better*when*expected: aligning speech and motor rhythms enhances verbal processing
volume	31
issn	2327-3798, 2327-3801
shorttitle	It is better*when*expected
url	http://www.tandfonline.com/doi/full/10.1080/23273798.2016.1144892
doi	10.1080/23273798.2016.1144892

abstract	Rhythm is a powerful way to shape the processing of complex sounds such as speech or music by generating temporal expectancies in the listener. Here, we multisensory expectancies generated by aligning speech and motor rhythms may enhance verbal processing. Participants listened to rhythmically regular Ge detected word changes occurring on stressed or unstressed syllables. Participants were cued to produce finger taps simultaneously with the auditory speech r were aligned or misaligned with stressed syllables. Detection of word changes was facilitated when manual movements were temporally aligned with the au Moreover, motor alignment enhanced sensitivity to detect changes on stressed syllables compared to a perceptual control condition. Thus, rhythmic speech s concurrent movement in multisensory contexts has beneficial effects on verbal processing. This finding lends support to models of expectancydriven speech
language	en
number	5
urldate	2025-07-22
journal	Language, Cognition and Neuroscience
author	Falk, Simone and Dalla Bella, Simone
month	may, year = {2016
note	Publisher: Informa UK Limited
pages	699--708
file	Falk et Dalla Bella - 2016 - It is betterwhenexpected aligning speech a.pdf:C\:\Users\paul-\Zotero\storage\5APKZXVG\Falk et Dalla Bella - 2016 - It is bett aligning speech a.pdf:application/pdf
article	**noauthor_pdf_nodate-4**
title	({PDF}) {It} is better when expected: {Aligning} speech and motor rhythms enhances verbal processing
issn	2327-3798
shorttitle	({PDF}) {It} is better when expected
url	https://www.researchgate.net/publication/292140690_It_is_better_when_expected_Aligning_speech_and_motor_rhythms_enhances_verbal_processing
abstract	PDF {\textbar} Rhythm is a powerful way to shape the processing of complex sounds such as speech or music by generating temporal expectancies in the lis Find, read and cite all the research you need on ResearchGate
language	en
urldate	2025-07-22
journal	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\F9YYBUE7\292140690_It_is_better_when_expected_Aligning_speech_and_motor_rhythms_enhances_verbal_pro
article	**noauthor_pdf_nodate-5**
title	({PDF}) {Tapping} into linguistic rhythm
url	https://www.researchgate.net/publication/351951671_Tapping_into_linguistic_rhythm
doi	10.5334/labphon.248

abstract	PDF {\textbar} Rhythmic properties of speech and language have been a matter of long-standing debates, with both traditional production and perception stu read and cite all the research you need on ResearchGate
language	en
urldate	2025-07-22
journal	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\H928AMGZ\351951671_Tapping_into_linguistic_rhythm.html:text/html;Texte intégral:C\:\Users\paul-\Zotero\sto Tapping into linguistic rhythm.pdf:application/pdf
article	**poeppel_speech_2020**
title	Speech rhythms and their neural foundations
volume	21
copyright	2020 Springer Nature Limited
issn	1471-0048
url	https://www.nature.com/articles/s41583-020-0304-4
doi	10.1038/s41583-020-0304-4
abstract	The recognition of spoken language has typically been studied by focusing on either words or their constituent elements (for example, low-level features or recently, the 'temporal mesoscale' of speech has been explored, specifically regularities in the envelope of the acoustic signal that correlate with syllabic inf central role in production and perception processes. The temporal structure of speech at this scale is remarkably stable across languages, with a preferred ran 8 Hz. Importantly, this rhythmicity is required by the processes underlying the construction of intelligible speech. A lot of current work focuses on audio-mo speech, highlighting behavioural and neural evidence that demonstrates how properties of perceptual and motor systems, and their relation, can underlie the rhythms. The data invite the hypothesis that the speech motor cortex is best modelled as a neural oscillator, a conjecture that aligns well with current propos fundamental role of neural oscillations in perception and cognition. The findings also show motor theories (of speech) in a different light, placing new mech accounts of the action–perception interface.
language	en
number	6
urldate	2025-07-29
journal	Nature Reviews Neuroscience
author	Poeppel, David and Assaneo, M. Florencia
month	jun, year = {2020
note	Publisher: Nature Publishing Group
keywords	Cortex, Language
pages	322--334
inproceedings	**kim_overview_2022**

address	Prague, Czech Republic
title	An {Overview} of {Automatic} {Piano} {Performance} {Assessment} within the {Music} {Education} {Context}:
isbn	978-989-758-562-3
shorttitle	An {Overview} of {Automatic} {Piano} {Performance} {Assessment} within the {Music} {Education} {Context}
url	https://www.scitepress.org/DigitalLibrary/Link.aspx?doi=10.5220/0011137600003182
doi	10.5220/0011137600003182
language	en
urldate	2025-09-10
booktitle	Proceedings of the 14th {International} {Conference} on {Computer} {Supported} {Education}
publisher	SCITEPRESS - Science and Technology Publications
author	Kim, Hyon and Ramoneda, Pedro and Miron, Marius and Serra, Xavier
year	2022
pages	465--474
file	Kim et al. - 2022 - An Overview of Automatic Piano Performance Assessm.pdf:C\:\Users\paul-\Zotero\storage\GVGY76TI\Kim et al. - 2022 - An Overview Performance Assessm.pdf:application/pdf
inproceedings	**yuksel_learn_2016**
address	San Jose California USA
title	Learn {Piano} with {BACh}: {An} {Adaptive} {Learning} {Interface} that {Adjusts} {Task} {Difficulty} {Based} on {Brain} {State}
isbn	978-1-4503-3362-7
shorttitle	Learn {Piano} with {BACh}
url	https://dl.acm.org/doi/10.1145/2858036.2858388
doi	10.1145/2858036.2858388
abstract	We present Brain Automated Chorales (BACh), an adaptive brain-computer system that dynamically increases the levels of difficulty in a musical learning t cognitive workload measured by functional near-infrared spectroscopy. As users' cognitive workload fell below a certain threshold, suggesting that they had and could handle more cognitive information, BACh automatically increased the difficulty of the learning task. We found that learners played with significar and speed in the brain-based adaptive task compared to our control condition. Participant feedback indicated that they felt they learned better with BACh and of the level changes. The underlying premise of BACh can be applied to learning situations where a task can be broken down into increasing levels of difficu
language	en
urldate	2025-09-10
booktitle	Proceedings of the 2016 {CHI} {Conference} on {Human} {Factors} in {Computing} {Systems}
publisher	ACM

author	Yuksel, Beste F. and Oleson, Kurt B. and Harrison, Lane and Peck, Evan M. and Afergan, Daniel and Chang, Remco and Jacob, Robert Jk
month	may, year = {2016
pages	5372--5384
file	Yuksel et al. - 2016 - Learn Piano with BACh An Adaptive Learning Interf.pdf:C\:\Users\paul-\Zotero\storage\GXUXPT9M\Yuksel et al. - 2016 - Learn Pia Adaptive Learning Interf.pdf:application/pdf
article	**wilson_feedback_2023**
title	Feedback in augmented and virtual reality piano tutoring systems: a mini review
volume	4
issn	2673-4192
shorttitle	Feedback in augmented and virtual reality piano tutoring systems
url	https://www.frontiersin.org/journals/virtual-reality/articles/10.3389/frvir.2023.1207397/full
doi	10.3389/frvir.2023.1207397
abstract	Researchers in music education are exploring the use of virtual reality (VR) and augmented reality (AR) to support piano instruction. Beginner piano studen infrequent lessons, which they practice on their own. This lack of instructor feedback creates opportunities for students to develop improper technique. Curr AR and VR to guide solo practice use moving shapes to help students to identify what notes to play. Improvements in commercial AR/VR technology will b more detailed real-time feedback.
language	English
urldate	2025-09-10
journal	Frontiers in Virtual Reality
author	Wilson, Katie and Pfeiffer, Phillip E.
month	jun, year = {2023
note	Publisher: Frontiers
keywords	Music, virtual reality, augmented reality, Feedback systems, Mixed reality, piano tutoring
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\J5WXUCUA\Wilson et Pfeiffer - 2023 - Feedback in augmented and virtual reality piano tu.pdf:application/p
inproceedings	**rogers_pino_2014**
address	New York, NY, USA
series	ITS} '14
title	P.I.A.N.O.: {Faster} {Piano} {Learning} with {Interactive} {Projection}
isbn	978-1-4503-2587-5
shorttitle	P.I.A.N.O.

url	https://dl.acm.org/doi/10.1145/2669485.2669514
doi	10.1145/2669485.2669514
abstract	Learning to play the piano is a prolonged challenge for novices. It requires them to learn sheet music notation and its mapping to respective piano keys, toge details. Smooth playing further requires correct finger postures. The result is a slow learning progress, often causing frustration and strain. To overcome thes P.I.A.N.O., a piano learning system with interactive projection that facilitates a fast learning process. Note information in form of an enhanced piano roll not projected onto the instrument and allows mapping of notes to piano keys without prior sight-reading skills. Three learning modes support the natural learnin feedback and performance evaluation. We report the results of two user studies, which show that P.I.A.N.O. supports faster learning, requires significantly le provides better user experience, and increases perceived musical quality compared to sheet music notation and non-projected piano roll notation.
urldate	2025-09-10
booktitle	Proceedings of the {Ninth} {ACM} {International} {Conference} on {Interactive} {Tabletops} and {Surfaces}
publisher	Association for Computing Machinery
author	Rogers, Katja and Röhlig, Amrei and Weing, Matthias and Gugenheimer, Jan and Könings, Bastian and Klepsch, Melina and Schaub, Florian and Rukzio, E and Weber, Michael
month	nov, year = {2014
pages	149--158
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\WIAMT7CH\Rogers et al. - 2014 - P.I.A.N.O. Faster Piano Learning with Interactive.pdf:application/pdf
article	**gemicioglu_passive_2024**
title	Passive {Haptic} {Rehearsal} for {Augmented} {Piano} {Learning} in the {Wild}
volume	8
url	https://dl.acm.org/doi/10.1145/3699748
doi	10.1145/3699748
abstract	Passive haptic learning (PHL) is a method for training motor skills via intensive repetition of haptic stimuli while a user is focused on other tasks. For the pr PHL to music education, we propose passive haptic rehearsal (PHR) where PHL is combined with deliberate active practice. We designed a piano teaching s haptic gloves compatible with daily wear, a Casio keyboard with light-up keys, and an online learning portal that enables users to track performance, choose with the gloves and keyboard. We conducted a longitudinal two-week study in the wild, where 36 participants with musical experience learned to play two p without PHR. For 20 participants with complete and valid data, we found that PHR boosted the learning rate for the matching accuracy by 49.7\% but did no effect on learning the notes' rhythm. Participants across all skill levels in the study require approximately two days less to reach mastery on the songs practic We also confirmed that PHR boosts recall between active practice sessions. We hope that our results and system will enable the deployment of PHL beyond
number	4
urldate	2025-09-10
journal	Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.
author	Gemicioglu, Tan and Hopper, Elijah and Dwivedi, Brahmi and Kulkarni, Richa and Bhandarkar, Asha and Rajan, Priyanka and Eng, Nathan and Ramanujar Charles and Gilliland, Scott M. and Mason, Celeste and Seim, Caitlyn and Starner, Thad

month	nov, year = {2024
pages	187:1--187:26
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\XUGB5S4G\Gemicioglu et al. - 2024 - Passive Haptic Rehearsal for Augmented Piano Learn.pdf:application/
inproceedings	**tamaki_possessedhand_2011**
address	New York, NY, USA
series	CHI} '11
title	PossessedHand}: techniques for controlling human hands using electrical muscles stimuli
isbn	978-1-4503-0228-9
shorttitle	PossessedHand
url	https://dl.acm.org/doi/10.1145/1978942.1979018
doi	10.1145/1978942.1979018
abstract	If a device can control human hands, the device can be useful for HCI and tangible application's output. To aid the controlling of finger movement, we prese device with a forearm belt that can inform when and which fingers should be moved. PossessedHand controls the user's fingers by applying electrical stimul around the forearm. Each muscle is stimulated via 28 electrode pads. Muscles at different depths in the forearm can be selected for simulation by varying the PossessedHand can automatically calibrate the system for individuals. The automatic calibration system estimates relations between each electrode pad, stim muscle movement. Experiments show that PossessedHand can control the motion of 16 joints in the hand. Further, we also discuss an application based on t playing a musical instrument.
urldate	2025-09-10
booktitle	Proceedings of the {SIGCHI} {Conference} on {Human} {Factors} in {Computing} {Systems}
publisher	Association for Computing Machinery
author	Tamaki, Emi and Miyaki, Takashi and Rekimoto, Jun
month	may, year = {2011
pages	543--552
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\JSEIHA53\Tamaki et al. - 2011 - PossessedHand techniques for controlling human ha.pdf:application/pdf
inproceedings	**noauthor_pdf_nodate-6**
title	({PDF}) {MirrorFugue}: {Communicating} hand gesture in remote piano collaboration
shorttitle	({PDF}) {MirrorFugue}
url	https://www.researchgate.net/publication/221308550_MirrorFugue_Communicating_hand_gesture_in_remote_piano_collaboration
doi	10.1145/1935701.1935705
abstract	PDF {\textbar} Playing a musical instrument involves a complex set of continuous gestures, both to play the notes and to convey expression. To learn an... { cite all the research you need on ResearchGate

language	en
urldate	2025-09-10
booktitle	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\APPL9MJT\221308550_MirrorFugue_Communicating_hand_gesture_in_remote_piano_collaboration.html:text/ht
inproceedings	**noauthor_remirrorfugue_2025**
title	ReMirrorFugue}: {Examining} the {Emotional} {Experience} of {Presence} and ({Illusory}) {Communications} {Across} {Time} {\textbar} {Request} {
shorttitle	ReMirrorFugue
url	https://www.researchgate.net/publication/391270594_ReMirrorFugue_Examining_the_Emotional_Experience_of_Presence_and_Illusory_Communications
doi	10.1145/3706598.3713328
abstract	Request PDF {\textbar} On Apr 25, 2025, Xiao Xiao and others published ReMirrorFugue: Examining the Emotional Experience of Presence and (Illusory) Across Time {\textbar} Find, read and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
booktitle	ResearchGate
month	aug, year = {2025
file	Snapshot:C\:\Users\paul-\Zotero\storage\4XJFLLF4\391270594_ReMirrorFugue_Examining_the_Emotional_Experience_of_Presence_and_Illusory_Comn
misc	**noauthor_pdf_nodate-7**
title	({PDF}) {Optimizing} piano practice with a utility-based scaffold
url	https://www.researchgate.net/publication/353068240_Optimizing_piano_practice_with_a_utility-based_scaffold
abstract	PDF {\textbar} A typical part of learning to play the piano is the progression through a series of practice units that focus on individual dimensions of the... { cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
journal	ResearchGate
doi	10.48550/arXiv.2106.12937
file	Snapshot:C\:\Users\paul-\Zotero\storage\5JC8V5NT\353068240_Optimizing_piano_practice_with_a_utility-based_scaffold.html:text/html
misc	**heyen_data-driven_2022**
title	Data-{Driven} {Visual} {Reflection} on {Music} {Instrument} {Practice}
url	http://arxiv.org/abs/2203.13320
doi	10.48550/arXiv.2203.13320

abstract	We propose a data-driven approach to music instrument practice that allows studying patterns and long-term trends through visualization. Inspired by life lo tracking, we imagine musicians to record their practice sessions over the span of months or years. The resulting data in the form of MIDI or audio recording sporadically to track progress and guide decisions. Toward this vision, we started exploring various visualization designs together with a group of nine guita with data and feedback over the course of three months.
urldate	2025-09-10
publisher	arXiv
author	Heyen, Frank and Ngo, Quynh Quang and Kurzhals, Kuno and Sedlmair, Michael
month	mar, year = {2022
note	arXiv:2203.13320 [cs]
keywords	Computer Science - Human-Computer Interaction, Computer Science - Graphics
annote	Comment: CHI 2022 IMI Workshop https://teamdarmstadt.de/imi/
file	Preprint PDF:C\:\Users\paul-\Zotero\storage\KWKCEGEU\Heyen et al. - 2022 - Data-Driven Visual Reflection on Music Instrument .pdf:application/pdf;S \Zotero\storage\WWSQYP3Y\2203.html:text/html
inproceedings	**noauthor_eyepiano_2025**
title	EyePiano}: {Leveraging} {Gaze} {For} {Reflective} {Piano} {Learning} {\textbar} {Request} {PDF
shorttitle	EyePiano
url	https://www.researchgate.net/publication/372243663_EyePiano_Leveraging_Gaze_For_Reflective_Piano_Learning
doi	10.1145/3563657.3596065
abstract	Request PDF {\textbar} On Jul 10, 2023, Jakob Karolus and others published EyePiano: Leveraging Gaze For Reflective Piano Learning {\textbar} Find, re research you need on ResearchGate
language	en
urldate	2025-09-10
booktitle	ResearchGate
month	aug, year = {2025
file	Snapshot:C\:\Users\paul-\Zotero\storage\HKSBBH7M\372243663_EyePiano_Leveraging_Gaze_For_Reflective_Piano_Learning.html:text/html;Texte intég \Zotero\storage\8FZXBNML\2025 - EyePiano Leveraging Gaze For Reflective Piano Lea.pdf:application/pdf
misc	**noauthor_pdf_nodate-8**
title	({PDF}) {Teach} {Me} {How} to {ImproVISe}: {Co}-{Designing} an {Augmented} {Piano} {Training} {System} for {Improvisation}
shorttitle	({PDF}) {Teach} {Me} {How} to {ImproVISe}
url	https://www.researchgate.net/publication/378005943_Teach_Me_How_to_ImproVISe_Co-Designing_an_Augmented_Piano_Training_System_for_Improv

abstract	PDF {\textbar} Improvisation is a vital but often neglected aspect of traditional piano teaching. Challenges such as difficulty in assessment and subjectivity. and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
journal	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\IEIE8TBF\378005943_Teach_Me_How_to_ImproVISe_Co-Designing_an_Augmented_Piano_Training_System_f /html
misc	**moringen_optimizing_2021**
title	Optimizing piano practice with a utility-based scaffold
url	http://arxiv.org/abs/2106.12937
doi	10.48550/arXiv.2106.12937
abstract	A typical part of learning to play the piano is the progression through a series of practice units that focus on individual dimensions of the skill, such as hand posture, or correct timing. Ideally, a focus on a particular practice method should be made in a way to maximize the learner's progress in learning to play the each learn differently, and because there are many choices for possible piano practice tasks and methods, the set of practice tasks should be dynamically ada learner. However, having a human teacher guide individual practice is not always feasible since it is time consuming, expensive, and not always available. In optimize in the space of practice methods, the so-called practice modes. The proposed optimization process takes into account the skills of the individual lea learning. In this work we present a modeling framework to guide the human learner through the learning process by choosing practice modes that have the h (i.e., improvement in piano playing skill). To this end, we propose a human learner utility model based on a Gaussian process, and exemplify the model train for practice scaffolding on an example of simulated human learners.
urldate	2025-09-10
publisher	arXiv
author	Moringen, Alexandra and Rüttgers, Sören and Zintgraf, Luisa and Friedman, Jason and Ritter, Helge
month	jun, year = {2021
note	arXiv:2106.12937 [cs]
keywords	Computer Science - Human-Computer Interaction, Computer Science - Artificial Intelligence
file	Preprint PDF:C\:\Users\paul-\Zotero\storage\QYCCUK8N\Moringen et al. - 2021 - Optimizing piano practice with a utility-based sca.pdf:application/pdf;Sn \Zotero\storage\5ZYHR7GL\2106.html:text/html
article	**mackay_comparative_2025-1**
title	Comparative {Structured} {Observation}
volume	32
issn	1073-0516, 1557-7325
url	https://dl.acm.org/doi/10.1145/3711838

doi	10.1145/3711838
abstract	Although HCI researchers often generate and compare new design concepts, they lack an established method for rigorously conducting qualitative assessme characterize Comparative Structured Observation as a qualitative research method that takes advantage of the structure of controlled experiments to generate ecologically relevant experiences with two or more design variants, often implemented as medium-fidelity prototypes. Researchers observe users and ask the reflect on each variant. We identify criteria for creating a successful Comparative Structured Observation study and illustrate variations of the method by an studies. We also examine six additional studies (three “near” and three “far”) to clarify the boundary between what should and should not be considered a C Observation . We discuss the benefits and limitations of the method and argue that gathering comparative reflections about design variants can help research their design concepts.
language	en
number	2
urldate	2025-09-10
journal	ACM Transactions on Computer-Human Interaction
author	Mackay, Wendy E. and McGrenere, Joanna
month	apr, year = {2025
pages	1--27
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\8YATZYTA\Mackay et McGrenere - 2025 - Comparative Structured Observation.pdf:application/pdf
inproceedings	**noauthor_pdf_nodate-9**
title	({PDF}) {When} second wave {HCI} meets third wave challenges
url	https://www.researchgate.net/publication/221247920_When_second_wave_HCI_meets_third_wave_challenges
doi	10.1145/1182475.1182476
abstract	PDF {\textbar} This paper surveys the current status of second generation HCI theory, faced with the challenges brought to HCI by the so-called third wave. read and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
booktitle	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\D4B27ZIF\221247920_When_second_wave_HCI_meets_third_wave_challenges.html:text/html;Version soumise: \Zotero\storage\QAIULZXM\(PDF) When second wave HCI meets third wave challe.pdf:application/pdf
article	**ma_empowering_2025**
title	Empowering music education with technology: a bibliometric perspective
volume	12
copyright	2025 The Author(s)
issn	2662-9992

shorttitle	Empowering music education with technology
url	https://www.nature.com/articles/s41599-025-04616-2
doi	10.1057/s41599-025-04616-2
abstract	As technology becomes an integral part of educational content and methodology, its research significance continues to grow, particularly in the relationship technology. The primary aim of this study is to quantify and analyze academic research outcomes concerning the use of technology in music education. The drawn from the WoS core database, encompassing academic achievements from 1991 to 2024. Various bibliometric software tools and three major laws wer analysis, examining publication distribution, relevant journals and authors, research countries, keywords, and current and future research themes. Presently, focused on four themes: technology integration and interaction, adaptive learning and creative teaching methods, educational frameworks and performance, inclusion of children and adolescents. Looking ahead, the two frontier hot topics in this field are remote and online education, and innovation in higher educ models. This study aims to contribute to the comprehensive bibliometric analysis literature on the use of technology in music education.
language	en
number	1
urldate	2025-09-10
journal	Humanities and Social Sciences Communications
author	Ma, Yidi and Wang, Chengliang
month	mar, year = {2025
note	Publisher: Palgrave
keywords	Development studies, Education
pages	345
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\ZGIYUZ9C\Ma et Wang - 2025 - Empowering music education with technology a bibl.pdf:application/pdf
article	**noauthor_pdf_2025**
title	({PDF}) {Technology} {Use} and {Attitudes} in {Music} {Learning}
url	https://www.researchgate.net/publication/333507541_Technology_Use_and_Attitudes_in_Music_Learning
doi	10.3389/fict.2019.00011
abstract	PDF {\textbar} While the expansion of technologies into the music education classroom has been studied in great depth, there is a lack of published literatur read and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
journal	ResearchGate
month	aug, year = {2025
file	Snapshot:C\:\Users\paul-\Zotero\storage\S8LERSNI\333507541_Technology_Use_and_Attitudes_in_Music_Learning.html:text/html;Texte intégral:C\:\Use \Zotero\storage\H8WMAU5K\2025 - (PDF) Technology Use and Attitudes in Music Learni.pdf:application/pdf

article	deja_survey_2022
title	A {Survey} of {Augmented} {Piano} {Prototypes}: {Has} {Augmentation} {Improved} {Learning} {Experiences}?
volume	6
shorttitle	A {Survey} of {Augmented} {Piano} {Prototypes}
url	https://dl.acm.org/doi/10.1145/3567719
doi	10.1145/3567719
abstract	Humans have been developing and playing musical instruments for millennia. With technological advancements, instruments were becoming ever more sop decades computer-supported innovations have also been introduced in hardware design, usability, and aesthetics. One of the most commonly digitally augme piano. Besides electronic keyboards, several prototypes augmenting pianos with different projections providing various levels of interactivity on and around implemented in order to support piano players. However, it is still unclear whether these solutions support the learning process. In this paper, we present a sy augmented piano prototypes focusing on instrument learning based on the four themes derived from interviews with piano experts to understand better the p piano. These themes are (i) synchronised movement and body posture, (ii) sight-reading, (iii) ensuring motivation, and (iv) encouraging improvisation. We f are saturated on the synchronisation themes, and there are opportunities for sight-reading, motivation, and improvisation themes. We conclude by presenting augmenting piano systems towards enriching the piano learning experience as well as on possible directions to expand knowledge in the area.
number	ISS
urldate	2025-09-10
journal	Proc. ACM Hum.-Comput. Interact.
author	Deja, Jordan Aiko and Mayer, Sven and Čopič Pucihar, Klen and Kljun, Matjaž
month	nov, year = {2022
pages	566:226--566:253
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\4NXU93ZI\Deja et al. - 2022 - A Survey of Augmented Piano Prototypes Has Augmen.pdf:application/pdf
misc	**noauthor_game_nodate**
title	Game of {Tones}: {Learning} to {Play} {Songs} on a {Piano} {Using} {Projected} {Instructions} and {Games}
shorttitle	Game of {Tones}
url	https://www.researchgate.net/publication/266655881_Game_of_Tones_Learning_to_Play_Songs_on_a_Piano_Using_Projected_Instructions_and_Games
abstract	Download Citation {\textbar} Game of Tones: Learning to Play Songs on a Piano Using Projected Instructions and Games {\textbar} Learning to play a mus the piano requires a substantial amount of practice and perseverance in learning to read and play... {\textbar} Find, read and cite all the research you need on
language	en
urldate	2025-09-10
journal	ResearchGate
doi	10.1145/2559206.2574799

file	Snapshot:C\:\Users\paul-\Zotero\storage\BDFFD7ZQ\266655881_Game_of_Tones_Learning_to_Play_Songs_on_a_Piano_Using_Projected_Instructions_a /html
article	**noauthor_enhanced_2025**
title	Enhanced {Learning} of {Jazz} {Chords} with a {Projector} {Based} {Piano} {Keyboard} {Augmentation}
url	https://www.researchgate.net/publication/337488453_Enhanced_Learning_of_Jazz_Chords_with_a_Projector_Based_Piano_Keyboard_Augmentation
doi	10.1007/978-3-030-35343-8_21
abstract	Download Citation {\textbar} Enhanced Learning of Jazz Chords with a Projector Based Piano Keyboard Augmentation {\textbar} Learning jazz piano is co difficult. Most people cannot afford private piano tuition and there are many freely available video... {\textbar} Find, read and cite all the research you need
language	en
urldate	2025-09-10
journal	ResearchGate
month	aug, year = {2025
file	Snapshot:C\:\Users\paul-\Zotero\storage\S6TWX7PV\337488453_Enhanced_Learning_of_Jazz_Chords_with_a_Projector_Based_Piano_Keyboard_Augme
inproceedings	**noauthor_music_nodate**
title	Music education using augmented reality with a head mounted display {\textbar} {Request} {PDF}
url	https://www.researchgate.net/publication/262347530_Music_education_using_augmented_reality_with_a_head_mounted_display
abstract	Request PDF {\textbar} Music education using augmented reality with a head mounted display {\textbar} Traditional music education places a large emphas practice. Studies have shown that individual practice is frequently not very... {\textbar} Find, read and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
booktitle	ResearchGate
file	Snapshot:C\:\Users\paul-\Zotero\storage\MX8BD3CP\262347530_Music_education_using_augmented_reality_with_a_head_mounted_display.html:text/htm
inproceedings	**birhanu_keynvision_2017**
address	New York, NY, USA
series	CHI} {PLAY} '17 {Extended} {Abstracts
title	KeynVision}: {Exploring} {Piano} {Pedagogy} in {Mixed} {Reality
isbn	978-1-4503-5111-9
shorttitle	KeynVision
url	https://dl.acm.org/doi/10.1145/3130859.3131336
doi	10.1145/3130859.3131336

abstract	Integrating new technology such as Augmented Reality into an established learning context is not a simple process. We explore the use of mixed reality for p focusing on notation literacy, and playtest a series of prototypes with a panel of practitioners in piano teaching. Our piano training game prototypes called K use the Microsoft Hololens to playfully introduce octave scales, chords, and arpeggios to beginners. Based on the results, the augmented contextualization o an effective tool for aiding students in the process of learning to play piano.
urldate	2025-09-10
booktitle	Extended {Abstracts} {Publication} of the {Annual} {Symposium} on {Computer}-{Human} {Interaction} in {Play}
publisher	Association for Computing Machinery
author	Birhanu, Amare and Rank, Stefan
month	oct, year = {2017
pages	299--304
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\IU5WS2SL\Birhanu et Rank - 2017 - KeynVision Exploring Piano Pedagogy in Mixed Real.pdf:application/p
article	**noauthor_arpiano_2025**
title	ARPiano} {Efficient} {Music} {Learning} {Using} {Augmented} {Reality}: {First} {International} {Conference}, {ICITL} 2018, {Portoroz}, {Slovenia} 2018, {Proceedings} {\textbar} {Request} {PDF
shorttitle	ARPiano} {Efficient} {Music} {Learning} {Using} {Augmented} {Reality
url	https://www.researchgate.net/publication /327218357_ARPiano_Efficient_Music_Learning_Using_Augmented_Reality_First_International_Conference_ICITL_2018_Portoroz_Slovenia_August_2
doi	10.1007/978-3-319-99737-7_1
abstract	Request PDF {\textbar} ARPiano Efficient Music Learning Using Augmented Reality: First International Conference, ICITL 2018, Portoroz, Slovenia, Aug Proceedings {\textbar} ARPiano uses a MIDI keyboard and a multifunction knob to create a novel mixed reality experience that supports visual music learni Find, read and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
journal	ResearchGate
month	aug, year = {2025
file	Snapshot:C\:\Users\paul-\Zotero\storage\II4MZZ94\327218357_ARPiano_Efficient_Music_Learning_Using_Augmented_Reality_First_International_Conf
article	**hackl_holokeys_nodate**
title	HoloKeys} - {An} {Augmented} {Reality} {Application} for {Learning} the {Piano
abstract	This paper describes the design and the implementation approach of a piano training application. HoloKeys is an Augmented Reality tool which is capable t to be played on a real piano. Musical pieces are loaded as MIDI files, interpreted and can be displayed in two different ways. This prototype provides many extension which can make it a powerful teaching tool.
language	en

author	Hackl, Dominik and Anthes, Christoph
file	Hackl et Anthes - HoloKeys - An Augmented Reality Application for Le.pdf:C\:\Users\paul-\Zotero\storage\SDFHMIYN\Hackl et Anthes - HoloKeys - An Application for Le.pdf:application/pdf
inproceedings	**molloy_mixed_2019**
title	Mixed {Reality} {Piano} {Tutor}: A {Gamified} {Piano} {Practice} {Environment}
shorttitle	Mixed {Reality} {Piano} {Tutor}
url	https://ieeexplore.ieee.org/document/8706474
doi	10.23919/ELINFOCOM.2019.8706474
abstract	Learning to play piano has many benefits ranging from relieving stress to stimulating neural connections and language skills. However, the learning process intrinsic motivation results in a high drop-out rate of novice piano students. Gamification has been proposed as a technique to overcome lack of intrinsic mo successfully applied in education and health. In this paper we present a novel gamification technique for piano tutoring. The method is integrated with an AR reduce cognitive load, and a continuous feedback system, to make the learning process less strenuous and more motivating. Our results show that users cons intuitive, interesting, and fun, and that users had a greater note playing accuracy than with Synthesia, a popular commercial piano learning system.
urldate	2025-09-10
booktitle	2019 {International} {Conference} on {Electronics}, {Information}, and {Communication} ({ICEIC})
author	Molloy, Will and Huang, Edward and Wünsche, Burkhard C.
month	jan, year = {2019
keywords	Three-dimensional displays, Rhythm, Games, Virtual reality, augmented reality, gamification, Keyboards, piano learning, piano tutor, Tools, visual feedbac
pages	1--7
file	Snapshot:C\:\Users\paul-\Zotero\storage\85HI5QGV\8706474.html:text/html
article	**noauthor_pdf_2025-1**
title	({PDF}) {FunPianoAR}: A {Novel} {AR} {Application} for {Piano} {Learning} {Considering} {Paired} {Play} {Based} on {Multi}-{Marker} {Trackin
issn	1742-6596
shorttitle	({PDF}) {FunPianoAR}
url	https://www.researchgate.net/publication/333461385_FunPianoAR_A_Novel_AR_Application_for_Piano_Learning_Considering_Paired_Play_Based_on_
abstract	PDF {\textbar} This paper describes the design and implementation of a novel augmented reality application, referred as FunPianoAR, that aiming to activat read and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
journal	ResearchGate
month	aug, year = {2025

inproceedings	stanbury_holokeys_2021
address	New York, NY, USA
series	VRST} '21
title	HoloKeys}: {Interactive} {Piano} {Education} {Using} {Augmented} {Reality} and {IoT
isbn	978-1-4503-9092-7
shorttitle	HoloKeys
url	https://dl.acm.org/doi/10.1145/3489849.3489921
doi	10.1145/3489849.3489921
abstract	The rise of online learning poses unique challenges in music education, where live demonstration and musical synchronization are critical for student succes HoloKeys, a music education interface which allows instructors to play remotely located pianos using an augmented reality headset and wifi-enabled microc approach allows students to receive distance education which is more direct, immersive, and comprehensive than conventional video conferencing allows fo remote students to observe live instructional demonstration on a physical keyboard in their immediate environment just as they would in traditional settings. two separate components: an augmented reality user interface and a piano playing apparatus. Our system aims to extend online music education beyond desl physical world, thereby addressing crucial obstacles encountered by educators and students transitioning into online education.
urldate	2025-09-10
booktitle	Proceedings of the 27th {ACM} {Symposium} on {Virtual} {Reality} {Software} and {Technology}
publisher	Association for Computing Machinery
author	Stanbury, Austin Jerald and Said, Ines and Kang, Hyo Jeong
month	dec, year = {2021
pages	1--3
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\JAWTQ36M\Stanbury et al. - 2021 - HoloKeys Interactive Piano Education Using Augmen.pdf:application/p
inproceedings	huang_piano_2011
title	Piano {AR}: A {Markerless} {Augmented} {Reality} {Based} {Piano} {Teaching} {System}
volume	2
shorttitle	Piano {AR}
url	https://ieeexplore.ieee.org/document/6038212
doi	10.1109/IHMSC.2011.82
abstract	This paper presents a novel markerless augmented reality based piano teaching system, which tracks the real keyboard of piano naturally. Following the virt the keyboard, beginners of piano can practice playing piano. Instead of the marker in traditional AR system, the geometrical parameters of the piano keyboa calculate the transformation matrix from keyboard coordinate to camera coordinate. We first extract all the contours of the image, from which possible cont found. Then the area of keyboard is identified by the structure of white and black keys predefined. We evaluated the system with speed and accuracy. As a r operates at 15fps, and the average error worked out by our method is about 1.97 pixels. The usefulness of our approach is reflected in an example that a user

urldate	2025-09-10
booktitle	2011 {Third} {International} {Conference} on {Intelligent} {Human}-{Machine} {Systems} and {Cybernetics}
author	Huang, Feng and Zhou, Yu and Yu, Yao and Wang, Ziqiang and Du, Sidan
month	aug, year = {2011
keywords	Cameras, augmented reality, Keyboards, Equations, Estimation, Image edge detection, image processing, machine vision, Mathematical model, Shape
pages	47--52
file	Snapshot:C\:\Users\paul-\Zotero\storage\85U4QIVJ\6038212.html:text/html
inproceedings	**fernandez_piano_2016**
title	Piano learning application with feedback provided by an {AR} virtual character
url	https://ieeexplore.ieee.org/document/7800380
doi	10.1109/GCCE.2016.7800380
abstract	This paper proposes a mobile application that uses Augmented Reality to enhance the experience of learning to play keyboard instruments. The proposed ap Cardboard to let users visualize a virtual character and instructions. The character gives the user real-time visual feedback according to their music playing p evaluated by the system. This application lets users choose among pre-defined songs that can be learned and practiced by using a simplified notation that eli using actual musical notation. Having in mind the psychological benefits of music learning, we aim to present a fun and easy way to learn music and keep th during the process.
urldate	2025-09-10
booktitle	2016 {IEEE} 5th {Global} {Conference} on {Consumer} {Electronics}
author	Fernandez, Carlos A Torres and Paliyawan, Pujana and Yin, Chu Chun and Thawonmas, Ruck
month	oct, year = {2016
keywords	Music, music, Augmented reality, Keyboards, Visualization, Animation, Augmented Reality, character animation, Computer-based instruction, Google, Inst
pages	1--2
file	Snapshot:C\:\Users\paul-\Zotero\storage\SBTWFZYU\7800380.html:text/html
misc	**noauthor_pilot_nodate**
title	Pilot {Study} of {Piano} {Learning} with {AR} {Smart} {Glasses} {Considering} {Both} {Single} and {Paired} {Play} {\textbar} {SpringerLink}
url	https://link.springer.com/chapter/10.1007/978-3-319-92037-5_39
urldate	2025-09-10
file	Pilot Study of Piano Learning with AR Smart Glasses Considering Both Single and Paired Play | SpringerLink:C\:\Users\paul-\Zotero\storage\BTZTMA6P\978-3-319-92037-5_39.html:text/html
article	**zeng_funpianoar_2019**
title	FunPianoAR}: A {Novel} {AR} {Application} for {Piano} {Learning} {Considering} {Paired} {Play} {Based} on {Multi}-{Marker} {Tracking

volume	1229
issn	1742-6596
shorttitle	FunPianoAR
url	https://dx.doi.org/10.1088/1742-6596/1229/1/012072
doi	10.1088/1742-6596/1229/1/012072
abstract	This paper describes the design and implementation of a novel augmented reality application, referred as FunPianoAR, that aiming to activate the interests o experience of piano learning. The FunPianoAR with a user-friendly interface considers the paired play to further reduce the difficulty of playing the piano fo paired play, one user plays the melody and the other plays the harmony part. The app is developed by using Android Studio 3.0 and artoolkitX 1.0, an open s reality SDK, and installed on the AR smart glasses, Epson Moverio BT300. Due to the textureless features of real piano and registration precision, the applic by using the fiducial marker tracking instead of the markerless recognition. Besides, we divided the piano keyboard into four zones and each zone used a sep tracking to solve the limited field of view (FOV) to some extent. The virtual keys can be accurately superimposed on the piano keys by using multi-marker t was also conducted to compare the effects of two types of augmented reality information superimposed, i.e., the instant way without the hints for the next no FunPianoAR with the hints for the coming notes to be played. The correctness rate was calculated and the time differences between the notes that two player same time were collected. The FunPianoAR shows more advantages over the instant way based on the evaluation results.
language	en
number	1
urldate	2025-09-10
journal	Journal of Physics: Conference Series
author	Zeng, Hong and He, Xingxi and Pan, Honghu
month	may, year = {2019
note	Publisher: IOP Publishing
pages	012072
file	IOP Full Text PDF:C\:\Users\paul-\Zotero\storage\BRPL6VNS\Zeng et al. - 2019 - FunPianoAR A Novel AR Application for Piano Learn.pdf:application/p
article	**sandler_piano_2022**
title	Piano pedaller: a measurement system for classification and visualisation of piano pedalling techniques
shorttitle	Piano pedaller
url	https://www.academia.edu/69635623/Piano_pedaller_a_measurement_system_for_classification_and_visualisation_of_piano_pedalling_techniques
abstract	This paper presents the results of a study of piano pedalling techniques on the sustain pedal using a newly designed measurement system named Piano Pedal comprised of an optical sensor mounted in the piano pedal bearing block and an
urldate	2025-09-10
author	Sandler, Mark
month	jan, year = {2022

file	Snapshot:C\:\Users\paul-\Zotero\storage\DYMW93VZ\Piano_pedaller_a_measurement_system_for_classification_and_visualisation_of_piano_pedalling_te
misc	**noauthor_pdf_nodate-10**
title	({PDF}) {Performer} {Experience} on a {Continuous} {Keyboard} {Instrument}
url	https://www.researchgate.net/publication/353581812_Performer_Experience_on_a_Continuous_Keyboard_Instrument
urldate	2025-09-10
inproceedings	**nith_dextrems_2021**
address	New York, NY, USA
series	UIST} '21
title	DextrEMS}: {Increasing} {Dexterity} in {Electrical} {Muscle} {Stimulation} by {Combining} it with {Brakes
isbn	978-1-4503-8635-7
shorttitle	DextrEMS
url	https://dl.acm.org/doi/10.1145/3472749.3474759
doi	10.1145/3472749.3474759
abstract	Electrical muscle stimulation (EMS) is an emergent technique that miniaturizes force feedback, especially popular for untethered haptic devices, such as mo AR. However, the actuation displayed by interactive systems based on EMS is coarse and imprecise. EMS systems mostly focus on inducing movements in such as legs, arms, and wrists; whereas individual finger poses, which would be required, for example, to actuate a user's fingers to fingerspell even the simp language, are not possible. The lack of dexterity in EMS stems from two fundamental limitations: (1) lack of independence: when a particular finger is actua current runs through nearby muscles, causing unwanted actuation of adjacent fingers; and, (2) unwanted oscillations: while it is relatively easy for EMS to st is very hard for EMS to stop and hold that finger at a precise angle; because, to stop a finger, virtually all EMS systems contract the opposing muscle, typica controllers (e.g., PID)—unfortunately, even with the best controller tuning, this often results in unwanted oscillations. To tackle these limitations, we propos based haptic device featuring mechanical brakes attached to each finger joint. The key idea behind dextrEMS is that while the EMS actuates the fingers, it is that stops the finger in a precise position. Moreover, it is also the brakes that allow dextrEMS to select which fingers are moved by EMS, eliminating unwan preventing adjacent fingers from moving. We implemented dextrEMS as an untethered haptic device, weighing only 68g, that actuates eight finger joints inc (metacarpophalangeal and proximal interphalangeal joints for four fingers), which we demonstrate in a wide range of haptic applications, such as assisted fir tutorial, guitar tutorial, and a VR game. Finally, in our technical evaluation, we found that dextrEMS outperformed EMS alone by doubling its independence unwanted oscillations.
urldate	2025-09-10
booktitle	The 34th {Annual} {ACM} {Symposium} on {User} {Interface} {Software} and {Technology}
publisher	Association for Computing Machinery
author	Nith, Romain and Teng, Shan-Yuan and Li, Pengyu and Tao, Yujie and Lopes, Pedro
month	oct, year = {2021
pages	414--430
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\JU2U8JJF\Nith et al. - 2021 - DextrEMS Increasing Dexterity in Electrical Muscl.pdf:application/pdf

misc	gemicioglu_passive_2022
title	Passive {Haptic} {Rehearsal} for {Accelerated} {Piano} {Skill} {Acquisition}
url	http://arxiv.org/abs/2203.12749
doi	10.48550/arXiv.2203.12749
abstract	Passive haptic learning (PHL) uses vibrotactile stimulation to train piano songs using repetition, even when the recipient of stimulation is focused on other ta the benefits of playing piano cannot be acquired without actively playing the instrument. In this position paper, we posit that passive haptic rehearsal, where assisted by separate sessions of passive stimulation, is of greater everyday use than solely PHL. We propose a study to examine the effects of passive haptic piano learners and consider how to incorporate passive rehearsal into everyday practice.
urldate	2025-09-10
publisher	arXiv
author	Gemicioglu, Tan and Teuscher, Noah and Dwivedi, Brahmi and Park, Soobin and Miller, Emerson and Mason, Celeste and Seim, Caitlyn and Starner, Thad
month	apr, year = {2022
note	arXiv:2203.12749 [cs]
keywords	Computer Science - Human-Computer Interaction
annote	Comment: 4 pages, 3 figures. Accepted to the IMI 2022 workshop at the CHI Conference on Human Factors in Computing Systems (CHI '22). Updated and conference
file	Preprint PDF:C\:\Users\paul-\Zotero\storage\S65P952F\Gemicioglu et al. - 2022 - Passive Haptic Rehearsal for Accelerated Piano Ski.pdf:application/pdf;S \Zotero\storage\27Z6KMYG\2203.html:text/html
inproceedings	**lee_hapticus_2025**
address	New York, NY, USA
series	CHI} '25
title	Hapticus: {Exploring} the {Effects} of {Haptic} {Feedback} and its {Customization} on {Motor} {Skill} {Learning}: {Tactile}, {Haptic}, and {Somatosen
isbn	9798400713941
shorttitle	Hapticus
url	https://dl.acm.org/doi/10.1145/3706598.3713821
doi	10.1145/3706598.3713821
abstract	Numerous haptic devices have been proposed to support motor learning, such as a hand exoskeleton with mechanical linkages, a vibrotactile glove, and an E Stimulation (EMS) device. Understanding the impact of each type of feedback on users' learning performance and experience, as well as the effects of custo feedback each user receives, is vital to achieving both efficient and highly motivating learning. To this end, we compared learning performance and experien haptic devices for piano learning. It revealed the distinct characteristics of each device, notably, the exoskeleton was the most preferred despite certain drawl conducted a user study to evaluate the effectiveness of haptic customization, allowing participants to customize the order of haptic feedback, demonstrating improved agency and performance. These findings would benefit haptic designers by providing more efficient and optimized haptic feedback for motor learn

urldate	2025-09-10
booktitle	Proceedings of the 2025 {CHI} {Conference} on {Human} {Factors} in {Computing} {Systems}
publisher	Association for Computing Machinery
author	Lee, Kyungyeon and Yang, Daniel S and Singh, Kriti and Nishida, Jun
month	apr, year = {2025
pages	1--20
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\UVE4RPHM\Lee et al. - 2025 - Hapticus Exploring the Effects of Haptic Feedback.pdf:application/pdf
inproceedings	**choudhary_adaptive_2025**
address	New York, NY, USA
series	CHI} '25
title	Adaptive {Electrical} {Muscle} {Stimulation} {Improves} {Muscle} {Memory}
isbn	9798400713941
url	https://dl.acm.org/doi/10.1145/3706598.3713676
doi	10.1145/3706598.3713676
abstract	Electrical muscle stimulation (EMS) has been leveraged to assist in learning motor skills by actuating the user's muscles. However, existing systems provide demonstration—actuating the correct movements, regardless of the user's learning progress. Instead, we contrast two versions of a piano-tutoring system: a c that moves the participant's fingers to play the sequence of movements correctly, and a novel adaptive-EMS system that changes its guidance strategy based performance. The adaptive-EMS dynamically adjusts its guidance: (1) demonstrate by playing the entire sequence when errors are frequent; (2) correct by li and actuating the correct one when errors are moderate; and (3) warn by lifting incorrect fingers when errors are low. We found that adaptive-EMS improve (recall) and was preferred by participants. We believe this approach could inspire new types of physical tutoring systems that promote adaptive over static gu
urldate	2025-09-10
booktitle	Proceedings of the 2025 {CHI} {Conference} on {Human} {Factors} in {Computing} {Systems}
publisher	Association for Computing Machinery
author	Choudhary, Siya and Nith, Romain and Ho, Yun and Brooks, Jas and Guruvugari, Mithil and Lopes, Pedro
month	apr, year = {2025
pages	1--11
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\NJIANVBP\Choudhary et al. - 2025 - Adaptive Electrical Muscle Stimulation Improves Mu.pdf:application/p
inproceedings	**xiao_andante_2014**
title	Andante: {Walking} {Figures} {On} {The} {Piano} {Keyboard} {To} {Visualize} {Musical} {Motion}
copyright	Creative Commons Attribution 4.0, Open Access
shorttitle	Andante

url	https://zenodo.org/record/1178987
doi	10.5281/ZENODO.1178987
abstract	We present Andante, a representation of music as animated characters walking along the piano keyboard that appear to play the physical keys with each step music pedagogy that emphasizes expressive, full-body communication early in the learning process, Andante promotes an understanding of the music rooted advantage of walking as one of the most fundamental human rhythms. We describe three example visualizations on a preliminary prototype as well as applic examples for practice feedback, improvisation and composition. Through our project, we reflect on some high level considerations for the NIME community
urldate	2025-09-10
publisher	Zenodo
author	Xiao, Xiao and Tome, Basheer and Ishii, Hiroshi
month	jun, year = {2014
annote	[TLDR] Through the project, this work reflects on some high level considerations for the NIME community and presents Andante, a representation of music walking along the piano keyboard that appear to play the physical keys with each step.
misc	**noauthor_pdf_nodate-11**
title	({PDF}) {The} {Role} of {Tutoring} in {Problem} {Solving}
url	https://www.researchgate.net/publication/228039919_The_Role_of_Tutoring_in_Problem_Solving
urldate	2025-09-10
inproceedings	**liscio_helping_2025**
address	New York, NY, USA
series	TEI} '25
title	Helping {Popular} {Musicians} {Learn} by {Ear}: {Analyzing} {Video} {Lessons} to {Inform} the {Design} of {Memory}-{Oriented} {Human}-{Recor
isbn	9798400711978
shorttitle	Helping {Popular} {Musicians} {Learn} by {Ear}
url	https://dl.acm.org/doi/10.1145/3689050.3704953
doi	10.1145/3689050.3704953
abstract	Popular musicians often learn songs by ear, from recordings, using technology that provides control over playback events. However, we know little about th interactions, and our research goal is to develop novel and useful ones. Guided by a preliminary study of 18 YouTube videos of musicians learning by ear, w grounded-theory analysis of 28 YouTube lessons and an instructional DVD that teaches by-ear learning. Based on our findings, we characterize the sub-task learn songs from recordings, their variations, and discuss how memory plays a role in ear learning. Armed with these insights, we offer recommendations fo novel, memory-oriented human-recording interactions—facilitating active listening sessions, restricting playback to fit working memory, extracting musical memorization, and re-synthesizing the recording to play notes indefinitely—all of which are grounded in real-world observations, and studies in both neuros
urldate	2025-09-10
booktitle	Proceedings of the {Nineteenth} {International} {Conference} on {Tangible}, {Embedded}, and {Embodied} {Interaction}

publisher	Association for Computing Machinery
author	Liscio, Christopher and Brown, Daniel G.
month	mar, year = {2025
pages	1--13
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\BAKVP76A\Liscio et Brown - 2025 - Helping Popular Musicians Learn by Ear Analyzing .pdf:application/pd
article	**robinson_developing_2020**
title	Developing {Self}-{Directed} {Learners}
volume	84
issn	0002-9459
url	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7159015/
doi	10.5688/ajpe847512
abstract	Objective. To discuss the meaning of self-directed learning, challenges with implementation, and strategies to overcome obstacles in educational settings. In self-directed learning, differentiate it from similar terminology, and discuss the empirical evidence for its development and strategies for its use within highe Self-directed learning as a defined teaching pedagogy has been around since the 1960s and can be used in classroom and experiential settings. It is a term tha describe a set of skills that college graduates should possess. A self-directed learning environment is dramatically different from a lecture-based classroom v determines the goals, the assessments administered, and pacing of the course content. During the self-directed learning process, the learner sets goals, detern be assessed, defines the structure and sequence of activities and a timeline, identifies resources, and seeks out feedback. When teaching individuals who are must be taken to appropriately scaffold and structure learning to develop the underlying soft skills needed for students to be successful as self-directed learne implementing this pedagogy in a classroom setting, challenges are faced both by the learner and the educator. Faculty members should proactively plan for p during the course design process.
number	3
urldate	2025-09-10
journal	American Journal of Pharmaceutical Education
author	Robinson, Jennifer D. and Persky, Adam M.
month	mar, year = {2020
pmid	32313284
pmcid	PMC7159015
pages	847512
file	Texte intégral:C\:\Users\paul-\Zotero\storage\J78KCX63\Robinson et Persky - 2020 - Developing Self-Directed Learners.pdf:application/pdf
article	**noauthor_pdf_2025-2**
title	({PDF}) {Examining} {Self}-{Assessment} of {Student} {Teachers}' {Self}-{Directed} {Learning}: A {Case} {Study} at {Battambang} {Teacher} {Edu {Cambodia}

shorttitle	({PDF}) {Examining} {Self}-{Assessment} of {Student} {Teachers}' {Self}-{Directed} {Learning}
url	https://www.researchgate.net/publication/387523580_Examining_Self-Assessment_of_Student_Teachers'_Self-Directed_Learning_A_Case_Study_at_Battambang_Teacher_Education_College_Cambodia
doi	10.31014/aior.1993.07.04.529
abstract	PDF {\textbar} This case study, a collaborative investigation into the self-assessment of student teachers' self-directed learning in the Battambang Teacher... and cite all the research you need on ResearchGate
language	en
urldate	2025-09-10
journal	ResearchGate
month	aug, year = {2025
file	Version soumise:C\:\Users\paul-\Zotero\storage\8NDUTFZ3\2025 - (PDF) Examining Self-Assessment of Student Teacher.pdf:application/pdf
article	**kong_high-resolution_2021**
title	High-{Resolution} {Piano} {Transcription} {With} {Pedals} by {Regressing} {Onset} and {Offset} {Times}
volume	29
issn	2329-9290
url	https://dl.acm.org/doi/10.1109/TASLP.2021.3121991
doi	10.1109/TASLP.2021.3121991
abstract	Automatic music transcription (AMT) is the task of transcribing audio recordings into symbolic representations. Recently, neural network-based methods ha AMT, and have achieved state-of-the-art results. However, many previous systems only detect the onset and offset of notes frame-wise, so the transcription the frame hop size. There is a lack of research on using different strategies to encode onset and offset targets for training. In addition, previous AMT system misaligned onset and offset labels of audio recordings. Furthermore, there are limited researches on sustain pedal transcription on large-scale datasets. In this high-resolution AMT system trained by regressing precise onset and offset times of piano notes. At inference, we propose an algorithm to analytically calcul and offset times of piano notes and pedal events. We show that our AMT system is robust to the misaligned onset and offset labels compared to previous sys system achieves an onset F1 of 96.72\&\#x0025; on the MAESTRO dataset, outperforming previous onsets and frames system of 94.80\&\#x0025;. Our sys onset F1 score of 91.86\&\#x0025;, which is the first benchmark result on the MAESTRO dataset. We have released the source code and checkpoints of our \<uri\>https://github.com/bytedance/piano_transcription\</uri\>
urldate	2025-09-11
journal	IEEE/ACM Trans. Audio, Speech and Lang. Proc.
author	Kong, Qiuqiang and Li, Bochen and Song, Xuchen and Wan, Yuan and Wang, Yuxuan
month	oct, year = {2021
pages	3707--3717
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\BR55DAY3\Kong et al. - 2021 - High-Resolution Piano Transcription With Pedals by.pdf:application/pdf

article	noauthor_seeing_2025
title	Seeing {How} {It} {Sounds}: {Observation}, {Imitation}, and {Improved} {Learning} in {Piano} {Playing} {\textbar} {Request} {PDF}
issn	0737-0008
shorttitle	Seeing {How} {It} {Sounds}
url	https://www.researchgate.net/publication/309210549_Seeing_How_It_Sounds_Observation_Imitation_and_Improved_Learning_in_Piano_Playing
abstract	Request PDF {\textbar} Seeing How It Sounds: Observation, Imitation, and Improved Learning in Piano Playing {\textbar} Fifty free eprints available from http://www.tandfonline.com/eprint/qnWNq2ncMfk3isWaYyUD/full This study centers upon a piano... {\textbar} Find, read and cite all the research you nee
language	en
urldate	2025-09-11
journal	ResearchGate
month	aug, year = {2025
article	**mian_comparing_2024**
title	Comparing {Real} and {Imitative} {Practice} with {No} {Practice} during {Observational} {Learning} of {Hand} {Motor} {Skills} from {Animations}
volume	14
copyright	http://creativecommons.org/licenses/by/3.0/
issn	2227-7102
url	https://www.mdpi.com/2227-7102/14/9/949
doi	10.3390/educsci14090949
abstract	In two experiments, we compared the effects of practice (real and imitative) with no practice on the observational learning of hand motor skills from animate investigated learning to play a series of piano clips of varying complexity. Results demonstrated improved learning efficiency with imitative practice compa Experiment 2 featured a paper-folding task, and results indicated that real practice led to significantly greater learning than no practice. Furthermore, a signif found with gender and practice, where females learned best with both real and imitative practice, but males did not benefit from these interventions. Howeve females in the no practice condition. Overall, we found benefits of practice versus no practice for both tasks. However, the most effective type of practice wa task: imitative practice for piano playing, and real practice for paper folding. Task complexity and gender were also found to be moderating factors.
language	en
number	9
urldate	2025-09-11
journal	Education Sciences
author	Mian, Maliha Naushad and Beder, Hannah and Marcus, Nadine and Ayres, Paul
month	sep, year = {2024
note	Publisher: Multidisciplinary Digital Publishing Institute

keywords	cognitive load theory, embodied cognition, gender, instructional animations, observational learning, practice
pages	949
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\42RNJVNX\Mian et al. - 2024 - Comparing Real and Imitative Practice with No Prac.pdf:application/pdf
inproceedings	**xiao_andantino_2016**
address	New York, NY, USA
series	IDC} '16
title	Andantino: {Teaching} {Children} {Piano} with {Projected} {Animated} {Characters}
isbn	978-1-4503-4313-8
shorttitle	Andantino
url	https://dl.acm.org/doi/10.1145/2930674.2930689
doi	10.1145/2930674.2930689
abstract	This paper explores how multi-modal body-syntonic interactive systems may be used to teach children to play the piano beyond the typical focus on reading "surface correctness". Our work draws from Dalcroze Eurhythmics, a method of music pedagogy aimed at instilling an understanding of music rooted in the Dalcrozian process of piano learning as a five-step iterative cycle of: listen, internalize, extend, analyze, and improvise. As a case study of how digital techn this process, we present Andantino, a set of extensions of Andante, which projects musical lines as miniature light silhouettes that appear to walk on the key We discuss features of Andantino based on each stage, or step, of the iterative framework and discuss directions for future research, based on two preliminar between the ages of 7 and 13.
urldate	2025-09-11
booktitle	Proceedings of the {The} 15th {International} {Conference} on {Interaction} {Design} and {Children}
publisher	Association for Computing Machinery
author	Xiao, Xiao and Puentes, Pablo and Ackermann, Edith and Ishii, Hiroshi
month	jun, year = {2016
pages	37--45
file	Full Text PDF:C\:\Users\paul-\Zotero\storage\32HSNMUD\Xiao et al. - 2016 - Andantino Teaching Children Piano with Projected .pdf:application/pdf
article	**braun_using_2006**
title	Using thematic analysis in psychology
volume	3
issn	1478-0887
url	https://doi.org/10.1191/1478088706qp063oa
doi	10.1191/1478088706qp063oa

abstract	Thematic analysis is a poorly demarcated, rarely acknowledged, yet widely used qualitative analytic method within psychology. In this paper, we argue that and theoretically flexible approach to analysing qualitative data. We outline what thematic analysis is, locating it in relation to other qualitative analytic meth themes or patterns, and in relation to different epistemological and ontological positions. We then provide clear guidelines to those wanting to start thematic in a more deliberate and rigorous way, and consider potential pitfalls in conducting thematic analysis. Finally, we outline the disadvantages and advantages conclude by advocating thematic analysis as a useful and flexible method for qualitative research in and beyond psychology.
number	2
urldate	2025-09-11
journal	Qualitative Research in Psychology
author	Braun, Virginia and Clarke, Victoria
month	jan, year = {2006
note	Publisher: Routledge _eprint: https://doi.org/10.1191/1478088706qp063oa
keywords	epistemology, flexibility, patterns, qualitative psychology, thematic analysis
pages	77--101